

Feature Selection for Distribution Shift

Sep 13th, 2026

Abstract

Distribution shift is ubiquitous in real-world machine learning systems, frequently causing performance degradation and thus undermine the economic benefit of the deployed ML models. While many existing methodologies have been developed in regime of understanding this phenomenon from hypothesis testing, online adaptation to understanding this phenomenon from the feature level is a relatively scarce and under-investigated area. In this work we provide a comprehensive overview of the current statistical & machine learning methodologies on characterizing distribution shift from the feature level via leveraging seemingly familiar methodologies and techniques - ranging from shapley-value, meta learner which originally rooted from causal inference regime as well as the random forest. The benchmark methodologies are carefully overviewed and our experiments showed that the parsimonious methodologies sometimes may have very powerful and compelling performance in real-data applications. In terms of covariate shift the random forest is leveraged for the criterion of feature selection with highly competitive performance in comparison with the current state-of-art methodologies. The optional post-hoc subset level localization procedure is conducted to quantify the degree of distance between the two subsets via MMD(covariate shift) and PO-risk(concept drift). On the other hand, for the feature selection in concept drift, the recently proposed meta-learner based methodologies purely for distance estimation worked well in very complex, highly nonlinear concept drift. The variable importance metrics' efficiency for characterizing the feature level contribution in distribution shift is discussed carefully under various circumstances - including but not limited to the. Sensitivity analysis is conducted in terms of the overlap issues and the choice of the propensity score model and the outcome model in the distribution shift detection procedure. The production deployment considerations are carefully addressed - ranging from parallelization strategies and runtime constraints to auto-fallback mechanism under poor overlap among batches of data. A wide range of real-data applications are conducted including user preference shift(Appendix 10) and the text shift(Appendix 9) This procedure transforms the Statistical & Machine Learning methodology into an operational diagnostic tool and can be efficiently embedded in the production ML system. By selecting the ranked, interpretable feature outputs, it shifts the MLOps from the reactive trouble-shooting to active data shift management, enabling the high-frequency & timely alerts on the deteriorated feature quality & shift checking with concise feature selection procedures. **We emphasize that the supplementary materials present a few product-landing use cases, but these are limited to product cases only. The separation is clear: the main text focuses on the methodological contributions.**

1 Introduction

Understanding the change of data is of high importance in the real-world ML systems - which can be characterized into understanding the distribution shift and understanding the performance degradation. We focus on the first aspect - distribution shift where the distribution of data evolves as time passes. Despite its practical significance, the feature-level characterization of the distribution shift remains a largely under-explored area. Spanning the full spectrum from characterizing the data change at the feature level tracing their origins back to specific subset of the variables, it arouses highly computationally efficient and robust characterization strategies for production machine learning systems - both from the variable importance level and the feature selection scenario. Although most of the existing methods are primarily targeted for the feature selection, the hypothesis testing and the online adaptation procedures do give people active insights regarding the how to understand the distribution shift from the feature level. We focus on the situation onto two batches of data where the existing batch is $\mathcal{D}_{exist} = \{\mathbf{X}_{exist}, \mathbf{Y}_{exist}\}$ and $\mathcal{D}_{new} = \{\mathbf{X}_{new}, \mathbf{Y}_{new}\}$. We split our discussion into two parts, the first is primarily focusing on covariate shift where the difference of covariate distribution is imposed on $P(\mathbf{X}) - P(\mathbf{X}_{new}) \neq P(\mathbf{X}_{exist})$. The second aspect mainly focus on the concept drift where the difference of the conditional dependence are imposed, e.g. $P(Y_{new}|\mathbf{X}_{new}) \neq P(Y_{exist}|\mathbf{X}_{exist})$. The current methodologies related to understanding the model performance degradation & distribution shifts can be summarized into the following three streams of the researches

- Hypothesis Testing procedures to detect whether there's a significant difference between the distribution of the two batches of datasets $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ - either for covariate shift or concept drift, or both. There are many existing methodologies along this direction, include but not limited to C2ST([33]), conditional dependence test([31]) and the recently proposed **RFPerm**(Random Forest Permutation Test) and **CFPerm**(Meta-Learner Objective Function Permutation Test).
- Understanding the distribution shift the sample/subgroup level - especially which and how each of the subgroup is driving the distribution shift or the model performance degradation. The pivotal methods include ProFILE([75]), ShapShift([10]) and SHIFT([70]) which aims to localize the change of the data to the subgroup & subset level.

- Understanding the change of the data from the feature level, decomposing the model performance degradation into different feature levels will require strong prior knowledge about the causal graph/variable dependence graph among different variables - specifically related to the. On the other hand, the distribution shift characterization techniques include a few existing works [14], [44], [15] and [89] including but not limited to characterizing concept drift from the feature level and decomposing the covariate shift into the individual directions. However, accurately and concisely characterizing the distribution shift is a highly under-investigated area without any assumptions - as in practice, the assumptionless attribution is definitely preferred than the two-step procedure starting from a causal discovery procedure followed by the performance drop decomposition strategy. Our research addresses this gap by developing an assumption-less procedure which is agnostic to the data-generating process, compatible with arbitrary data modalities, and robust to varying model complexities.

We advance feature selection under distribution shift by systematically evaluating existing methods against our own, and we use per-feature variable importance as a proxy, with a careful justification of its validity. We begin with two use cases that demonstrate its practical relevance.

1.1 Use Cases

Here we want to talk about two industrial use cases related to the concept drift and covariate shift detections - especially under the scenarios of first **Fraud Detection**, second **Ecommerce-Transaction Modeling**. Firstly in the models built for tackling with the fraud behavior in the continuous and ever-evolving fraud types and logits, both of the covariate shifts(fraudsters mixture shifts) and the concept drift(fraud behavior & pattern shifts) exist. Therefore, accurately and concisely identifying the subset of features that contribute significantly to either fraudster mixture shift or fraud pattern shift is crucial to give interpretable & real-time insights to guide the downstream model update and iterations. In transaction modeling of the E-commerce Platform, both covariate shift (user mixture shift) and concept drift (purchasing pattern shift) are pervasive. Proactively identifying a robust and accurate subset of features—among both embeddings and handcrafted ones — is of high practical significance. From another aspect, the Beyond enhancing model robustness, it provides business teams with interpretable, near-real-time insights into how data changes relate to the business context from both the system level and the feature level.

In addition to benchmark existing methods, this work present the following two contributions. First, we provide a systematic review of feature selection methodologies for distribution shift under both covariate shift and concept drift, covering a broad spectrum from ensemble learners to meta learner based distance estimation. Second, we show that the random forest classifier with variable importance achieves competitive performance for covariate shift feature selection, and is more computationally efficient than the existing SOTA methods. Third, we demonstrate that meta learner based approaches, specifically PermuCATE(permutation variable importance) and grf(causal forest split VIMP) - consistently outperform SOTA benchmarks in feature selection efficiency. We also offer production-level recommendations to machine learning practitioners, covering deployment practices and operational considerations, to facilitate deployment in real-world ML systems for feature-level distribution shift characterization. The rest of the work is organized as follows, we present the methodologies we employ in Section 2. The introduction of the datasets we leveraged and the experimental settings are carefully described in Section 3. The performance summarization and comparisons are discussed in Section 4. The production landing level consideration is presented in Section 7. The elaboration of the difficulty of disentangling the covariate shift and concept drift is presented in Section 6 along with the empirical studies. Summary and the future directions are presented in Section 8. The supplementary materials discusses the implementation details in section 11 and the complementary procedure(Section 15) of the meta-learner as feature selection procedure proposed.

2 Methodology

To address this gap in the the research landscape in Statistics and Machine Learning, we provide comprehensive evaluation & comparison for the current strategies and methodologies for **Feature-Selection-for-Distribution-Shift** leveraging a wide variety of current Statistical & Machine Learning methodologies and technologies. The introduction of each of the method we benchmark and compare are listed and discussed below carefully.

2.1 Covariate Shift

Understanding or attributing the covariate shift from the feature level is a ubiquitous challenge that almost every data scientist will face in their daily works when they faced with two batches of data which have been investigated for many years [39]. A natural question would be to answer which variable drives the difference - thus the targeted downstream business strategy can be conducted. We benchmark the following methodologies for feature selection via leveraging a wide variety of techniques. Traditional methodologies can be categorized into the following three streams, first is to capture the marginal dependence of the association between each individual feature and the response like mutual-information methods(cMIM,[24]) and minimal redundancy maximum relevance(MRMR, [59]). Second, the sparse learning based approach like Lasso([32]) and group-lasso([86]) via imposing the regularization on the model structure to select the features. Third, the model based procedure to capture the the wrapper based methods for the recursive feature selection procedures including [17] and [27]. Despite their popularity, these methods predominantly focus on marginal associations

and often fail to account for complex, intractable correlation structures among features. Moreover, they won't be able to perform well in the low signal-to-noise ratio settings. Recent advances along this direction primarily focus on the aspect of leveraging highly flexible deep neural networks([7] and [8]) with claimed SOTA performance - making the results cleaner and more efficient.

Separately, several methods have been developed to be robust to distributional shifts([28], [68]). Our objective, however, is distinct: we aim to identify features that contribute most significantly to the shift itself, rather than merely being robust to it. In this work, we propose to leverage one of the most popular, robust off-the-shelf predictive modeling procedures for feature selection - through the OOB(out-of-bag) variable importance, on top of the traditional MMD/HSIC statistical measures equipped by the LOCO(leave-one-covariate-out) procedure. We introduce the methodology carefully in this subsection.

First and foremost we borrow idea from the traditional two sample testing procedure MMD(maximum mean discrepancy, [65]) and HSIC(Hilbert-Schmidt Independence Criterion ,[1]) in estimating the distance via statistical measures(MMD/HSIC) between the two sets of covariates $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, equipped by a LOCO(leave-one-covariate-out) procedure to estimate the contribution for each of the feature in the covariate shift. Under covariate shift, if a direction(feature) is contributing significantly to the shift, the statistical measure we leveraged would be smaller w.r.t. a large degree after removing that specific feature. Thus the ranking of the LOCO difference for this measure can serve as a proxy of its contribution of the covariate shift. We leverage the standard hyperparameter setting in their original works where the $\sigma = 1$ for the RBF kernel in the MMD. In the benchmark comparison we denote them as **MMD-LOCO** and **HSIC-LOCO**. Additionally a couple of other variable importance methodologies are leveraged for evaluating the feature selection efficiency - ShadowVIMP ([51], shadow variable importance 2026), Vita Permutation Variable Importance([34], variable importance testing) and Boruta([37], one of the most popular methods for feature selection). All three methods are based on the random forest as the component base model for binary classifier to distinguish $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ where the first two methods are evaluated through both AUROC(with variable importance) and F-1 score, on the other hand, Boruta only outputs the F-1 score with the binary feature selection decision.

Delta-Shapley(Δ shapley value, [10]): We compare the difference of the feature-wise shapley values on the two batches of data, although no existing work describe this in detail, the absolute difference of the feature-wise shapley value across the two batches can represent their relative difference of their individual contribution to that specific predictive performance(magnitude as the sign only guides whether a positive or a negative dependence exist). From the definition, the $\Delta(shap)_j = |shap_{j,exist} - shap_{j,new}|$, although this procedure is originally developed for characterizing the feature's contribution to the predictive performance, it can be leveraged to understand each feature's contribution to the distribution shift. In the benchmark experiments we denote this procedure as **Delta-Shapley**.

VIMP for the classifier. Borrowing the ideas from one of the streams of the hypothesis testing procedure for the distribution shift - building a classifier to distinguish the two batches of data $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ (like classifier two-sample test, [33] and **AutoML** two-sample test, [36]), we leverage the variable importance of the binary classifier built to distinguish the two batches of data as the proxy for their contribution in the distribution shift. The random forest([13]), xgboost([16]) and the MLP([74]) are leveraged and the performance is evaluated on a separate holdout set. No tuning is conducted for all of these procedure to ensure the robustness. In the domain classifier we designed by the MLP, the 2-3 layers procedure is adapted and we denote this methodology as **Domain-Classifier**(rf-domain).

Localization Network We leverage the two recently proposed procedures for the covariate shift detection([7] and [8]) for the benchmark evaluations. They proposed to compress the statistical properties to the embedding and train a discriminator to maximize the separability between two batches of data. Notably, these procedures won't need fine-tuning where per the discussion in their original paper, the pre-trained neural network can be potentially generalized to arbitrary distribution shift. It also significantly reduce the computational cost in comparison to the pre-training from scratch.

We want to emphasize that for all of these procedures that conducting operations directly on the covariate space, although they can be directly extended onto the joint distribution $P(\mathbf{X}, Y)$, it won't be able to give people any useful information regarding the feature-level contribution on the concept drift - thus these are strictly focused on characterizing the feature's contribution on $P(\mathbf{X})$. They propose to compress the statistical properties of each batch into embeddings and train a neural network to detect feature-level shift localization methodologies - we include these as the benchmark comparisons in our adversarial perturbation experiments described in Section 3.1.

2.2 Concept Drift

We present the recently proposed meta-learner based procedures to characterize the concept drift from the feature level. Firstly, the **LOCO with R-learner**: We compute the relative importance of each of the feature X_j in terms the difference of the R-risk([53]) before and after removing that feature. Primarily targeted for concept drift. Secondly **DRPerm**: Pseudo-Outcome Learner([35] with the LOCO procedure(the PO-risk), primarily targeted for concept drift while also to some degree, covariate shift. Thirdly, GRF: Causal Forest Variable Importance([6] with the split frequency VIMP, primarily targeted for concept drift while also to some degree, covariate shift. Fourthly, **PermuCATE** - Conditional

Permutation Variable Importance for the DR learner through the conditional fitting([54]), primarily targeted for concept drift while also to some degree, covariate shift. Enormous flexibilities are included in this model selection for both of the propensity score models and the outcome models. We leverage random forest for the outcome model and the conditional fitting model $X_j \sim X_{-j}$ (where X_j is continuous) and logistic classifier for the propensity score model and the binary outcome conditional fitting model $X_j \sim X_{-j}$. On the other hand, when the outcome is multi-levels, the conditional fitting model here would be $X_j \sim X_{-j}$ whereas the multinomial logistic regression is leveraged when cardinality of the feature is larger or equal to 3 for robustness considerations.

We want to mention that the causal interpretation is out of the scope of the current proposed Statistical Methodology - specifically, the meta learners we leveraged including R-learner, DR learner, causal forest and the Dragon-Net are strictly restricted under the scope of the distance/deviation estimation rather than drawing detailed causal interpretation. This is a very important reminder for every modeler that aim to leverage our proposed methodology.

The current literature for feature selection of the distribution shift is fairly scarce, we benchmark the following methods. Firstly **SGShift**([44]) incorporating a couple of methodologies for the concept drift explanation including but not limited to de-randomized knockoff, sparse linear regression with correctly specified and mis-specified component models. We want to further mention that in their papers, the experimental design was fairly simple whereas only the linear concept drift is injected where the fact that the penalized linear regression works well was expected. The methods in their papers include the **l1-spec**(penalized regression with l1 penalty, correctly specified), **l1-misspec**(penalized regression with l1 penalty, mis-specified model) and **knockoff**(de-randomized knockoff) procedure. Secondly **ARD**(automatic relevance determination, [81]), primarily designed for feature selection and dimensionality reduction, the current implementation incorporates a optimization framework to characterize the distribution shift(primarily concept drift, to some degree covariate shift). Additionally all the other methods for the feature selection in covariate shift are leveraged for characterizing the concept drift crossly.

3 Datasets & Experiments

In this section we present the datasets and experimental settings we leverage for evaluating the efficiency and robustness of our proposed procedure. First we leverage the linear and nonlinear data-generating processes for evaluating the efficiency of the procedures. Specifically the non-linear data-generating processes are leveraged in the existing batch of data

$$\mathcal{D}_{new} - Y = T * (3X_1^2 + 3X_2^2 + 3X_3^2 + 2X_1X_2 + 2X_2X_3 + 2X_3X_4 + 1.5X_1X_4 + 1.5X_2X_4) + \beta * X + \epsilon$$

with nonlinear shift imposed otherwise the linear shift is imposed on the first 4 covariates where the $\Delta(\beta) = (1, 0.8, 0.6, 0.4)$ for $X_1 - X_4$ respectively. The visualization scripts can be found in the notebook - <https://colab.research.google.com/drive/1w5fKyTqnWoKEfixGnpaZCMSzQg0I0r2-#scrollTo=MuxSjR6uuxcP>.

Secondly we perform the variable importance on the distribution shift through simulated datasets(both covariate shift and concept drift), semi-synthetic datasets([43], artificial injection of the linearity and nonlinearity) and the real-world datasets([26]) with the ground-truth indices for significant contribution to the distribution shift. Both linear and nonlinear concept drift & covariate shift(interaction between features, quadratic and the sin functions) are injected into the $\mathcal{D}_{new}$ on known indices of the features for evaluating the efficiency of the recovery for all those features. The Whyshift([43]) dataset is leveraged for performing the pairwise via imposing different types of shifts, including but not limited to adversarial perturbations, linear perturbations and nonlinear perturbations as well as the adversarial perturbations - across the 49 pairwise comparisons across different states.

3.1 Adversarial Perturbation

On top of the relative simple linear & nonlinear distribution shift imposed on the new batch of data $\mathcal{D}_{new}$, we leverage a couple of adversarial perturbation strategies to inject a variety of types of shift. Originally proposed for robustness training, the adversarial procedure can be readily leveraged to inject the complex distribution shift for evaluating the efficiency of the **Feature Selection for Distribution Shift procedure**. Additionally the procedure is highly adaptive to a variety of types of the distribution shift, including but not limited to the following 5 methods: NesterovFGSM, MIFGSM, VMIFGSM, rFGSM and Jitter. All are evolved variants of the fast gradient sign method (FGSM), designed to generate adversarial examples with improved transferability across black-box models. They address the fundamental limitations of the vanilla FGSM - namely, susceptibility to local optimal and lack of diversity - by introducing distinct optimization enhancements. The MI-FGSM([2]) incorporates the momentum accumulation to stabilize the gradient directions and escape the sharp local maxima, the NI-FGSM([41]) extends this with a Nesterove based momentum update procedure that anticipates the future gradients, enabling a more stable updates. The **VMI-FGSM**([80]) further adds variance tuning procedure to reduce the gradient variance, preventing the optimizer from being trapped in the suboptimal regions. These 3 methods form a progressive family of the momentum-based attacking procedure, each built upon the previous to achieve increasingly reliable & transferrable perturbations.

On the other hand, the **rFGSM**([3]) and **Jitter**([66]) take the randomization oriented procedure. The rFGSM procedure injects a small random perturbation before computing the gradient, diversifying the attack path and mitigating the catastrophic overfitting commonly seen in the adversarial training. The Jitter procedure incrementally add the stochastic noise directly to the loss function to enable it escape the local optimal. Together these techniques helps enhancing the robustness and transferability of the adversarial attack. In the implementation procedure, we incorporate a mask to isolate the subset of features to attack - during the gradient update step and the momentum recording step to enforce the adversarial attack on a certain subset of the features. The evaluation protocol is similar to that in the other datasets with F1 and AUC recorded with the ground-truth feature indices attacked.

In the implementation stage, the neural network is leveraged to predict (Y_{exist}, Y_{new}) using features $(\mathbf{X}_{exist}, \mathbf{X}_{new})$, most importantly, through the gradient based procedure and the regularization on the 30% of features that has the highest gradient $\nabla(X)$, under this circumstance, the covariate shift of X dominates the procedure. On the existing batch of data, $X_{perturbed} = X + \epsilon * \text{sign}(X) * \nabla_X l(X, Y)$ on the top 30% of the features here. We denote this procedure as **FGSM30**. Apart from imposing the perturbation on the top 50% of the features $\mathbf{X}_{exist}$, we impose the gradient attack for Y_{new} in the new batch of data. Here the gradient based attacks are imposed whereas the gradient is imposed on the gradient $\nabla_Y l(X, Y)$ as well as the responses Y where $Y_{perturbed} = Y + \epsilon * \text{sign}(Y) * \nabla_Y l(X, Y)$. We denote this procedure as **FGSM50**. Under this circumstance, both highly nonlinear concept drift and covariate shift exist. A variety of the **OpenML**([11]) datasets are leveraged for the model evaluation. The details and the scales of the datasets we leveraged here are displayed in Table 15.

We want to mention that to inject the synthetic shift, the FGSM procedure is indeed imposed on the regression problem where the neural network is fitted to predict (Y_{exist}, Y_{new}) using $(\mathbf{X}_{exist}, \mathbf{X}_{new})$ - then after imposing the FGSM attack on 30% of the features in $\mathcal{D}_{new}$, we indeed injected complex concept drift, efficiently mimicking the real-world scenarios with untractable perturbations on the conditional dependence of $Y|\mathbf{X}$. To summarize, although subtleties remains, most of the adversarial perturbation procedures are primarily injecting covariate shift more than concept drift, unless specifically attack the regressor on one of the batch of data extensively through a momentum based procedure, where they mainly serves as the benchmark comparison methods for the covariate shift feature selection procedures.

3.2 Overlap Sensitivity Analysis

Poor overlap between the treatment group and control group is known to severely degrade the estimation quality of both propensity score model and outcome model in causal inference regime, in our meta-causal-learner framework, it would be also very important to demonstrate the robustness of feature selection efficiency under poor overlaps. We assess the impact of this overlap issue on feature selection, for both causal and non-causal methods. This challenge has received ongoing attention across both the causal inference and machine learning communities with recent work addressing high-dimensional settings([22]), overlap quantification via o-values([38]) and the representation learning under weak overlap([20]).

Under our framework of causal-learner as distance estimation, poor overlap between the supports of $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ would, in principle, undermine the quality of the distance estimation - a well-known challenge in the causal inference literature. Thus we conduct numerical studies for the lack of overlap between the two batches of data through the repliclust package([88]) with flexible specification of the overlap ratio between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ (ranges from 0.01 to 0.5 in our experiments) - the AUC for the logistic regression built to distinguish the two batches of data $\mathcal{D}_{exist}$ and $\mathcal{D}_{new}$ is recorded for quantifying the feature selection efficiency across various settings. As demonstrated in tables 28 and 31, the non-causal methods are fairly robust with respect to the distribution shift which aligns perfectly of their designs. Moreover, as demonstrated in Figures 13(concept drift) and 14(covariate shift), the covariate shift feature selection methods, ranging from RF-classifier, FSL-net to DataFix are very robust across different overlap proportions; The penalized regression and the knockoff procedure are also fairly robust under poor overlaps. On the other hand, the tree-based model as the outcome model and the propensity score model are very concise and helpful for robust estimation. Thus it aligns with the theoretical support in meta-learner methods where to ensure its efficiency in practice, at least moderate degree of overlap between the two batches of data(groups) are required & highly desired.

Similarly, for the high-dimensional dataset with more than 300 covariates, the eigen-shift is implemented on top of the SVD procedure whereas the degree of overlap can be characterized by the AUROC for the logistic regression model fit to distinguish the two batches of datasets where we borrow the ideas from the Classifier Two-Sample Test([33]). The logistic regression is leveraged to characterize the degree of linear separation to avoid the extra sensitivity introduced by this procedure. Moreover, the model here is highly flexible in terms of the dataset itself - providing opportunities to characterize the overlap instead of the traditional statistical measure for distance estimation.

In our framework, we treat the objective function of meta-learner(pseudo-outcome risk as well as the R-risk) as a tool for distance estimation between two batches of data - empirically we observe that the tree-based outcome models, including random forest and the causal forests, exhibiting strong robustness even under poor overlap. This is largely due to their recursive partitioning structure, which adapts to local covariate distributions and mitigates the curse of weak common support. This robustness makes them a practical and defensible default choice in real-world deployments, where

overlap conditions are often unknown or changing over time. We emphasize that the robustness of distance estimation can be substantially enhanced by choosing a robust outcome model specification — specifically, random forests or causal forests instead of the neural network or the support vector machines — not for treatment effect estimation or interpretation, but rather, only for reliable distance measurement.

3.3 Outcome Model and Propensity Score Model

A critical practical consideration in both machine learning modeling and meta-learner frameworks by causal inference is the appropriate choice of the outcome model - selecting parsimonious yet efficient models for those components would be essential to ensure robust and efficient estimation in practice. In this subsection we justify that the random forest model is among the best in both propensity score model and outcome model for feature selection of distribution shift. To demonstrate this choice, we conducted comprehensive sensitivity analysis in the version of the propensity score model(logistic regression, random forest, xgboost and mlp) as well as the outcome model(kernel ridge regression, random forest, xgboost and mlp) in both the permuCATE and DRPerm procedures to evaluate the overall efficiency for the feature selection both in terms of F-1 and AUC through various perturbations. As demonstrated in Table 31, the random forest outcome model is consistently among one of the best methods in terms of the feature selection, better than MLP and kernel ridge by a large margin and more robust than XGBoost.

Similarly for the propensity score model, we compare the feature selection efficiency for different propensity score models - MLP, logistic regression and XGBoost with the random forest. Random forest is also capable of achieving enhanced efficiency(as demonstrated in Figures 29 and 30) in comparison with the existing methods such as logistic regression and the MLP - further demonstrating the practical recommendations that if we adapt the meta-learner to do the distance estimation, the random forest is highly preferred both in terms of the propensity score model and the outcome model. While we neither expect nor require the propensity score model to achieve strong predictive fit in practice, the random forest classifier will preserve improved robustness along with the flexibility to capture more complex decision boundaries than the logistic regression model which focus on the linear decision boundary for distinguishing $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$. Although it's contradict to the common practice that the logistic regression model is preferred for the propensity score model in biomedical applications([29], [71], [76]) and financial domains([23], [60], [57]), we want to emphasize this insights to every ML practitioners that in conducting the accurate feature via the causal learner, the random forest is definitely what you will need - align with that in the regime for feature selection in distribution shift.

3.4 Random Forest's role in Feature Selection for Distribution Shift

We adopt random forest as both the propensity score model and the outcome model within the meta-learner framework for feature selection under concept drift and covariate shift. Relative to logistic regression, MLP, and XGBoost, random forest offers a key practical advantage: it effectively captures highly nonlinear and interacting decision boundaries without manual specification. More importantly its robustness under a wide variety of challenging scenarios have long been demonstrated([90], [49] and [48]) - enabling the robust fitting of the binary classifier. As established in [67], the random forest is capable of adapting to sparsity efficiently, with convergence rates depending on the number of active features rather than the ambient dimension. This ensures reliable distance estimation between the existing and new data distributions, making the meta-learner procedure both efficient and robust for feature selection.

The adaptivity of random forests to the intrinsic low-dimensional structure, combined with their inherent robustness to noisy and high-dimensional covariates, naturally lends itself to improved out-of-distribution detection. By focusing on the active feature subspace and avoiding overfitting to irrelevant noise, the forest produces stable and reliable predictions that are sensitive to meaningful shifts in the covariate distribution. This adaptivity-robustness duality enables more reliable identification of distributional changes, which is precisely the foundation for effective feature selection under covariate shift and concept drift - in terms of the reliable OOD characterization across each of the dimension.

4 Performance Summarization

The proposed Feature Selection for distribution shift procedure has shown its high competency in detecting complex, nonlinear concept drift, achieving superior performance over all of the existing methods ranging from C2ST, ARD(automatic relevance determination), Delta-Shapley value, knockoff based procedure as well as the penalized regression - as vividly demonstrated in the nonlinear concept drift and adversarial perturbation injection setting. Moreover, the DR(doubly-robust) pseudo-outcome learner also shows its highly competitive performance in both covariate shift and concept drift(Figures 9, 10, 11 and 12). We want to emphasize that, although very powerful and efficient for linear concept drift(Figures 11), the penalized regression procedure proposed in SGShift([45]) is much less efficient under the circumstance of covariate shift and nonlinear concept drift which aligns perfectly with the linearity nature when they were proposed - illustrating their less favorable positions for understanding the complex shift of the conditional dependence which further enhance the capability of our meta-learner based hypothesis testing methodology. In the meta-learner

based methods(PermuCATE and DR-Learner), we recommend random forest as the outcome model as they are relatively less sensitive to poor overlap - and since our goal is distance estimation, not causal interpretation, this robustness makes them a good fit.

On the other hand, under the circumstances of covariate shift, the **MMD-LOCO** and **RF-domain** procedures dominates in terms of the efficiency - under very complex attack on the classifier, the random forest procedure works slightly better(Table 19) whereas for simpler attack or lower dimensional data, the **MMD-LOCO** procedure is slightly better in terms of tradeoff between computational efficiency and robustness. The degradation of performance of the **l1-misspec** and **l1-spec** under nonlinear, complex adversarial attacks have demonstrates the significant limitation for the benchmark methodology([45]) - further enhancing the necessity for the MMD/RF based procedures. We want to emphasize that the low F1/AUC for the R-risk procedure is perfectly expected for datasets across modalities, as this procedure is intentionally designed to be invariant under the covariate shift. Moreover the Delta-Shapley value(absolute difference of the shapley value across each feature) procedure performs well under covariate shift instead of concept drift, aligning with its deviation of the objective from characterizing the distribution shift to characterize the change of the contribution of predictive accuracy. Similarly like that in the meta learner, the random forest and MMD based procedure both possess exceptional robustness under poor overlap between the supports of $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$.

Apart from the feature selection procedure, we want to emphasize that the highly competitive performance for the C2ST(classifier two-sample test) via random forest is also very practical in production where as long as one random forest classifier is fit well, the feature selection performance would be superior than most of the benchmark methods, highlighting the potential for the random forest to shine in production environment. The MMD based procedure is highly recommended given enough computational resource because of its highly robust and accurate feature selection performance. More importantly, the density estimation nature of the random forest can further enhance the robustness & efficiency of the variable importance procedure. Thus the random forest procedure would be highly preferred for understanding the covariate shift from the feature level.

The 8 real world datasets(Table 15) with various perturbation strategies are presented before. The Tables 16(concept drift injected), 17(nonlinear concept drift), 18, 19(covariate shift), 20(concept drift dominated), 21(covariate shift dominated) and 22(concept drift dominated) illustrate the aggregated performance and the individual performance in terms of the performance. As can be observed, the testing procedure based on meta-learner is highly competitive among all of the benchmark methods, especially under the circumstance where the concept drift dominates the covariate shift. On the other hand, the **Random Forest** based procedure has demonstrated high efficiency for the characterizing the covariate shift, regardless of how the overlap looks like. The SGShift(l1misspec and l1spec, [45]) based procedure works well under linear shift but struggles under nonlinear distribution shift which aligns with its original objective for penalized regression. Among the testing procedures via the meta-learners, the permucate procedure is among one of the most efficient(especially in circumstance where concept drift dominates covariate shift like Table 16) while the computationally efficient **grf**(causal forest with split VIMP) possess highly competitive performance in comparison to the existing methods, making both of the procedures highly practical in real-world application scenarios.

To further evaluate the efficiency of the random forest classifier, we benchmark a variety of other datasets primarily for understanding the effect of the covariate shift. Across other modalities of the datasets(FMoW Image dataset - [19], ChronoBerg - [30] semantic shift datasets where all of them are transformed via the representation extraction procedure - Gemma3([64]) and Mongo-Leaf([78]) for the text data and Video Transformer([62]) for image data. We want to mention that for the ChronoBerg dataset, FMoW dataset([19]) as well as the Wilds Medical dataset, the covariate shift is strictly dominated as the labels is constructed via the order of the observations - thus the between group covariate change dominate the change of the logic of labeling across different batches.

Specifically, the 5 types/forms of the adversarial perturbations are added on the whereas this procedure is highly efficient in terms of characterizing the change of the data in the continuously evolving data-generating processes. By construction, the Chronoberg and Wilds datasets are constructed for covariate shift with the pseudo-labels imposed with respect to the time order(from the indices 1 to the number of observations in this dataset). As demonstrated in Tables 23 and 25, the **LOCO-MMD** and **therf-domain** are among the most efficient one, much more efficient than the knockoff one and the penalized regression versions - although in some of the circumstances, the Delta-Shapley value based procedure also works efficient, the objective mismatch is a significant drawback for interpretation.

From the time-profiling aspect, as demonstrated in Table 26 and 27, the random forest classifier procedure followed by the variable importance is among one of the most accurate and most computationally efficient ones for feature selection in context of the covariate shift, thanks to the very robust performance for the random forest in the high-dimensional covariates and the efficient prediction in a wide variety of very challenging data-generating scenarios - while in low-to-moderate number of covariates, the LOCO-MMD do have its advantage, in the high-dimensional setting, the random forest procedure will play its role, which is highly competitive with respect to the NN based methods including FSL-Net([8]) and the DataFix([7]). Thus the very low training cost and deployment requirement with near-real-time inference nature equipped by the robustness of random forest make it preferred in the production environment.

4.1 Additional Benchmark Comparison

We conduct additional benchmark comparisons for the most recent benchmark for the covariate shift - DataFix([7]) and the FSL-Net([8]) both in terms of the time complexity and the detection efficiency. The random forest based VIMP procedure is highly competitive in comparison to these two neural network based procedures. We want to mention that these are two parallel lines of research where their methods also employ good flexibilities but in practice, the random forest procedure does play its role in feature selection for covariate shift. As demonstrated in the experimental results, the **FSL-Net** procedure works slightly better than the proposed methods but more computationally expensive and the **DataFix** procedure is less competitive than either the Random Forest based procedure or the MMD based procedure.

We also benchmark other feature selection methods for covariate shift built upon the tree-based models including Boruta ([37]), ShadowVIMP ([51]) and VITA-permutation variable importance([34]). It's fairly obvious that these methods won't provide additional benefits on top of the vanilla OOB-variable importance procedure based on the random forest. Unlike the three alternatives — which involve non-trivial resampling procedures and multiple-testing corrections — RF-VIMP produces a direct importance ranking from the trained forest at virtually no extra cost. This simplicity not only speeds up daily re-evaluation but also makes the results easier to interpret and communicate across non-technical audiences, a practical advantage that often outweighs marginal gains in selection accuracy.

4.2 Summary

To summarize, the random forest classifier(RF-domain, high dimensional covariates) and MMD-LOCO(lower dimensional covariates) are best for feature selection for covariate shift. On the other hand, for feature selection of concept drift, the permuCATE(concise feature selection) & causal forest split VIMP procedure(most parsimonious, relatively efficient) are recommended in practice, **consistently outperform the existing benchmarks in any types of the change of data more complex than the linear concept drift**. These are very valuable explorations providing data science people a very concise version of helping people understanding the change of data from the feature level. On the other hand, the RF-VIMP/MMD based procedure is highly competitive w.r.t. the existing SOTA(FSL-Net) in terms of performance and more computationally efficient than the existing methods which demonstrate their practical applicability. Additionally the careful sensitivity analysis has revealed the enhanced robustness of the classifier based procedure.

The practical implications of these findings is compelling: several of our recommended methodologies can offer practical advantages - from relative light tuning, straightforward parallelization and exceptional robustness under poor overlap. We discuss their deployment in production in Section 7.

5 Variable Importance for Distribution Shift and its proxy to feature selection

In this section, we establish the duality between variable importance (VIMP) and feature selection within the proposed framework. We consider two complementary settings: a random forest based domain classifier for covariate shift, and a DR pseudo-outcome learner with PermuCATE and LOCO importance for concept drift. In both cases, the VIMP ranking serves as a coherent and interpretable proxy for feature selection, justifying our proposed Feature Selection for Distribution Shift (FSDS) procedure. Crucially, this ranking also acts as a reliable proxy for prediction accuracy, offering practical guidance for model diagnosis and feature pruning under distribution shift.

For characterizing covariate shift via random forest, the variable importance derived from a binary classifier trained to distinguish between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ offers an efficient and interpretable measure of each feature's marginal contribution to the distributional change. This approach is rooted in the classic random forest framework of Breiman (2001), with extensive subsequent developments on the interpretation and calibration of variable importance measures (e.g., [72] and [40]). In particular, [79] established the superior efficiency of out-of-bag (OOB) variable importance in both regression and classification settings, supporting its use as a reliable diagnostic tool for feature importance. More importantly, this procedure is highly efficient and agnostic to the underlying data generating process. In our specific context—where the binary classifier distinguishes between the existing and new batches—the resulting variable importance can be directly interpreted as each feature's individual contribution to the observed covariate shift. This aligns with the classifier-based testing framework proposed and leveraged in [33], where the separability between two batches serves as the basis for detecting and characterizing distributional changes in the covariates X . We want to further mention that this notion of this variable importance is indeed generalizable to other types of models beyond the random forest, such as MLP and XGBoost where each feature's contribution collectively characterize the covariate shifts.

On the other hand, we leverage the pseudo-outcome risk equipped with the permutation variable importance and the LOCO(leave-one-covariate-out) procedure for estimating each individual feature's contribution to distribution shift with a specific focus on concept drift. The pseudo-outcome learner is originally proposed to provide causal effect estimation where as long as one of the propensity score model(binary classifier to distinguish the two covariate spaces) and the outcome model(regress the response on the covariates) are correctly specified, the causal effect would be unbiased. In

this work we leverage this pseudo-outcome risk as the objective function for estimating the distance between the two batches of data $(\mathbf{X}_{exist}, Y_{exist})$ and $(\mathbf{X}_{new}, Y_{new})$ within the meta-learner-as-distance-estimation framework we proposed.

The pseudo outcome is calculated as

$$\psi_i = \frac{T_i - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)(1 - \hat{e}(\mathbf{X}_i))} (Y_i - \hat{\mu}_{T_i}(\mathbf{X}_i)) + (\hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)) = (1) + (2)$$

Statistically, this object admits a direct interpretation: part (1) is the propensity-score weighted residual, which measures how well the current model fits the data under the existing covariate distribution.(introducing the shift in term 1). Consequently, a covariate shift in the new batch will alter the propensity score $e(X)$ (as reflected in its classification performance, the binary classifier to distinguish two covariate spaces), thereby directly influencing the downstream PO-risk. The second term captures the change in the conditional difference between the two batches, which is precisely the estimand for the conditional mean difference $\tau(X) = E[Y|X, T = 1] - E[Y|X, T = 0]$. This term was originally designed to quantify the conditional mean structure—the first-order effect on the conditional expectation of φ .(introducing the shift in term 2, the outcome model). Thus, the two components together provide a concise interpretation of variable importance for both concept drift and covariate shift, as each part contributes to a distinct aspect of the shift. Then estimate the doubly robust estimator distance $\tau_{DR}(\mathbf{X}) = E[\psi|\mathbf{X}]$ with the PO-risk calculated as $R_{PO} = \sum_{i=1}^n \tau_{PO}(X)^2$ - can efficiently deal with both of the concept drift and covariate shift - which serves as the distance estimator between the two batches of data $\{\mathbf{X}_{exist}, Y_{exist}\}, \{\mathbf{X}_{new}, Y_{new}\}$. From this perspective, the first component serves primarily as a covariate shift detector, while the second component is more directly aligned with concept drift. Together, these two components offer a concise and coherent interpretation of variable importance under both types of distributional change, with each part addressing a distinct source of shift. Under this framework, both permutation-based variable importance—defined as the difference in PO-risk before and after permuting a given feature—and the LOCO procedure—defined as the difference in PO-risk with and without the feature—effectively capture each feature’s marginal contribution to the objective function (PO-risk) as a distance estimator. This naturally induces a notion of variable importance as the feature-specific contribution to the overall distribution shift between the two batches. We emphasize that the doubly-robust property of this pseudo-outcome learner is established for treatment-effect estimation under counterfactual reasoning, not for our distance-estimation regime. The efficiency of our procedure is demonstrated directly from the construction of the pseudo-outcome(PO) risk, without causal identification.

Similarly we leverage the R-risk([53]) to characterize the distance between the conditional distribution of response given predictors $D(Y_{new}|\mathbf{X}_{new}, Y_{exist}|\mathbf{X}_{exist})$ where it is invariant under covariate shift justifiably from both geometric aspect and double/orthogonal machine learning([18]) aspect. Thus it provides solid theoretical guarantee and statistical insights innate to the distribution shift, primarily focusing on the different aspects of the R-risk and the PO-risk - focusing primarily on the concept drift. The detailed justifications is deferred in the supplementary materials 12.

Thus combining all of the previous justifications, we would be able to demonstrate that the variable importance for the are capable of serving a proxy of the feature selection in both covariate shift and concept drift.

6 Feature Selection & Driver for Distribution Shift

A very important problem in practice for feature selection in distribution shift mainly lies in the fact that, given the selected top-k features, $\mathbf{X}_1, \dots, \mathbf{X}_k$, how to give people concrete ideas of whether it’s driving covariate shift or concept drift or both, can we attribute the distribution shift into individual features? Although empirical evidences and recommendations can be proposed given the experimental results from the benchmark methods, a concrete and concise procedure is highly desired. Our answer regarding this question is "impossible" - under no other assumptions on the correlation structures among features or the complexity of the data-generating processes, it would be impossible to attribute the global distribution shift into individual feature level, not to say disentangling the covariate shift and concept drift. This is corresponding to the identifiability issue in terms of the impossibility of decomposing the covariate shift and concept drift into separate sources and individual features. We carefully justify this via empirical studies.

From a statistical standpoint, the decomposition of covariate shift and concept drift into individual feature level or even disentangle their effects are inherently ill-posed. This challenge parallels the difficulty of attributing the degradation in predictive performance—such as an increase in mean squared error (MSE)—to a single feature, since any marginal contribution is invariably conditional on the presence of all other covariates. Without imposing explicit or very strict structural assumptions on the underlying covariance or correlation matrices, achieving a unique and identifiable decomposition of the global distribution shift is fundamentally unattainable. We conduct a series of empirical data studies to illustrate this issue. Two batches of data are simulated through highly nonlinear data-generating processes with both concept drift and covariate shift with the distribution shift measure: MMD, HSIC(for covariate shift) and the R-risk(for concept drift) are measured for the two batches of data. Then the random feature inclusion paths are simulated as the order of each feature to be included are simulated with 20 repeats, for each of the feature included into the data-generating model, the increase of the MMD/HSIC and the decrease of the R-risk are recorded. In this way the path of the difference of the measures are recorded and constructed for each of the variable, the path as well as the uncertainty

of the risk measure value are visualized. We want to notice that under this process we have never made any assumptions regarding the structure of the dataset itself.

As demonstrated in Figures 15 and 16(a couple of different repeats are drawn on this plot), the feature insertion order is repeatedly permuted, and the corresponding incremental MMD and HSIC—both serving as proxies for covariate shift—are recorded at each step (with purple markers denoting the inclusion juncture of the genuine covariate-shift features). Crucially, the incremental paths exhibit considerable variability across permutations. Complementarily, the concept drift decomposition is evaluated via an R-risk-based procedure, with results for linear and nonlinear settings presented in Figures 17 and 18, respectively. In both cases, no consistent or uniform attribution path emerges across different inclusion orders - which gives people very vivid illustration that the decomposition of the global distribution shift(measured by MMD or R-risk) into different features is impossible as it depends heavily on the feature inclusion path and the increment of the distribution of the feature X_k is conditional on the existence of features $X_1, \dots, X_{k-1}$.

These empirical observations align precisely with our theoretical intuition: the change in R-risk is strictly conditional on the current set of **existing variables**, just as the incremental **MMD** is contingent upon the features already incorporated into the model. Consequently, any attempt to allocate distributional shifts to individual features remains inherently ambiguous. In summary, disentangling distribution shifts at the feature level is fundamentally impossible without imposing remarkably strong—and often empirically unverifiable—assumptions on the data-generating process itself. Given the theoretical non-identifiability, we propose a pragmatic production-level heuristic that integrates MMD/RF-based detectors and meta-learners for feature attribution: (i) unique to MMD/RF implies a covariate-shift driver; (ii) unique to meta learner based procedure implies a concept-drift driver; and (iii) overlap implies joint driving. While this rule does not resolve the fundamental identifiability issue, it offers actionable statistical guidance for feature monitoring and prioritization in practice. Based on our analyses—including MSE error decomposition, MMD-based covariates shift measurement, and R-risk concept drift decomposition—we reveal a fundamental obstacle: the task of separating distribution shift from ordinary data changes is infeasible without extra assumptions. Despite extensive prior research in causal representation learning and online drift detection, achieving effective disentanglement in practice remains challenging. The core difficulty arises from the fact that the observed drop in predictive performance cannot be uniquely decomposed into independent contributions from concept drift and covariate shifts without any assumptions which is impossible in practice.

Crucially, disentanglement remains intractable without imposing strong, often untestable assumptions on the correlation structures and the data-generating processes. Therefore, the guidance we offer should be viewed as a statistically grounded, actionable prioritization tool, rather than a formal causal or attribution claim - the only aspect we can make concrete argument is that, the feature selected are primary source of driver of concept drift(DRPerm or PermuCATE) or covariate shift(RF-VIMP) but we truly won't really know each feature's individual contribution to the distribution shift in the whole feature space and won't know that whether a feature is driving the other type of shift given they are contributing significantly to driving this type of shift. This is a fundamental limitation: without imposing a causal graph or running a randomized controlled trial, the "unique" attribution of the distribution shift into individual feature is infeasible, unless very strong and concise causal graph assumption and the dependence among features are imposed.

On the other hand, while the proposed feature selection for distribution shift procedure can provide concise guidance for identifying the root-causes of the distribution shift from the feature level, it cannot, by itself, indicate which upstream data sources should be implicitly investigated. It should therefore be complemented by marginal shift monitoring and missing rate diagnostics to enable through upstream data-source-level root-cause analysis. Figures 20 and 19 show that RF-domain and MMD-LOCO are among the most efficient and robust approaches compared to the alternatives. In addition, the DRPerm procedure demonstrates strong capability in capturing distribution shifts, corroborating the insights we previously obtained. Collectively, these findings validate the comprehensiveness and practical utility of our proposed feature-selection framework, which integrates MMD-LOCO with a random forest classifier. We emphasize that, despite the wide range of existing methodologies for quantifying overlap, a unified proxy remains notably absent, which inevitably complicates systematic evaluation. Nevertheless, meaningful empirical guidance can still be extracted from these approaches. More importantly, the inherent robustness of random forests—and tree-based models in general—under poor overlap between batch feature spaces positions them as a concrete and dependable choice in practice.

7 Product Consideration

Traditional monitoring methods across feature levels include Population Stability Index([87]), Kullback-Leibler Divergence([21]) for the predictive score distribution monitoring and the missing ratio statistics monitoring. Although all of those methods can only deal with individual dimension & marginal information, they do provide the complementary information on top of the proposed methods. Since only one random forest with default hyperparameters is needed, apart from serving it directly in the production system with near-real-time inference, this essentially provides every data scientist with a portable tool for diagnosing the covariate shift from the feature level.

7.1 Statistical Aspects for the production system

The computational cost would not be a bottleneck for this procedure to be embedded in the production ML system. Like the permutation test, the LOCO variable importance procedure can be straightforwardly parallelized, yielding substantial computational savings—**JobLib** in Python and **multiprocess** in R serve as the standard backends. In practice, **PermuCATE**, particularly when implemented with a doubly-robust pseudo-outcome learner (DR-Learner) under conditional outcome mechanisms, consistently ranks among the top-performing concept drift feature selection methods. Importantly, both feature-wise permutation and LOCO-based importance can be efficiently distributed with minimal engineering overhead, making the regular(every 2-3 minutes) testing procedure reasonable in the deployed ML systems.

Thus in practice, when people are trying to characterize the covariate shift, the **Random Forest Classifier(for high-dimensional covariates)** is what they will prefer after looking at the variable importance to serve as one of the most parsimonious yet efficient ways to characterize the covariate shift from the feature level - thanks to its capability for modeling the nonlinearity as well as the exceptional robustness under many challenging scenarios, users have the flexibilities to choose from the different notions of the variable importances. The MMD-LOCO procedure, although slightly more computationally expensive, are still among one of the most efficient and trustworthy way to characterize the covariate shift from the feature level. On the other hand, to understand the concept drift from the feature level, we want to mention that both the conditional permutation variable importance in the doubly-robust pseudo-outcome learner(DR-learner) or the split frequency variable importance in causal forest are what we would recommend - beating the current state-of-art methodologies(**ARD**, automatic relevance determination and **SGShift**, penalized regression procedure for feature selection) by a relatively large margin. The conditional permutation version of the R-risk is slightly less favorable yet still outperforms the existing methods. We want to further emphasize that although there are many existing methodologies trying to estimate the distance between the two batches of datasets, the random forest keeps performing exceptionally well both in hypothesis testing, feature selection as well as the variable importance - highlighting its practical applicability in many of the real-world circumstances, expanding out of the traditional tabular dataset, but also many other modalities of the datasets include but not limited to image, text, audio, video data and so on. It aligns perfectly with what we have observed in the **CFPerm** procedure, the **PermuCATE** and **Split-Freq** variable importance outperform the **LOCO** version to detect concept drift. However, to purely understanding concept drift to the feature level, the **R-learner** is what we recommend regardless of the sensitivity under higher degree of model mis-specifications and ultra-complexity of the data-generating processes. Our work makes solid contribution towards understanding distribution shift(both covariate shift and concept drift) from the feature level - especially under concept drift with tractable practical guidance.

We want to further justify that, upon the procedure for the domain classifier, it can be viewed as a class of models - starting from the logistic regression, random forest, XGBoost, MLP and etc, we compare the feature selection efficiency for all those models - the random forest is highly preferred over the other variants requiring a lot of hyperparameter tuning which will introduce a lot of bias and the lack of robustness for the feature selection procedure. However, it doesn't mean that the proposed R-risk/PO-risk based procedure is perfect - first is the much higher computational cost than the RF/MMD method(as demonstrated in Table 26 where the meta-learner based procedures are in the high range of computational cost even after parallelization) under the circumstances where the overlap between support of $\mathbf{D}_{exist}$ and $\mathbf{D}_{new}$ are poor, the meta-learner will not only struggle to fit the propensity classifier & outcome model but also introduce significant bias in estimating the ITE accurately - thus undermining the efficiency for the proposed meta-learner based methods for concisely characterizing the distribution shift. Under poor overlaps, both of the MMD-LOCO procedure and random forest one Under that circumstance, both of the MMD-LOCO and random forest based procedure work well especially under the circumstances with large deviation of support between two batches of data $\mathbf{D}_{exist}$ and $\mathbf{D}_{new}$. The minimum sample size for effectively characterizing distribution shift via the hypothesis testing procedure based on the meta-learner is around 200 per batch to enable the robust & efficient estimation via the meta-learner testing procedure. On the other hand, the random forest variant would only require 100 observations per batch for efficient feature selection. Here from the monitoring & alerting aspect, we want to mention about the following aspects:

- Suppose H_0 in the **CFPerm** is not rejected, it provides strong statistical evidence that the data distribution has not meaningfully changed. Therefore, no full-retraining is required. This itself delivers tangible economic values via avoiding unnecessary computation & engineering overhead, where at the same time providing the deployment engineers the confidence to proceed with the current version of the models.
- Suppose H_0 (both CFPerm or the Random-Forest Classifier procedures) is rejected and the top-K features are known to be low-risk(e.g. business-as-usual features) - then people should log and notify the data scientists to review them carefully throughout the ranked feature list based on the capacity of manual reviews. On the other hand, suppose H_0 is rejected and the top features belong to critical business metrics or the upstream data pipelines, then the immediate actions are required for the root-cause identification workflow. Under this circumstance, a proactive pre-warning threshold should be setup efficiently even before the significant distribution shift is observed.

From another aspect, we want to extend this process to a general-purposed class of procedures of building a binary classification model to distinguish the two batches of datasets $\mathbf{X}_{new}$ and $\mathbf{X}_{exist}$ - starting from the logistic regression, to random forest to gradient boosting to the MLP, we compare the overall efficiency and the robustness of the different

versions of the models - especially comparing the MLP and XGBoost versions.

7.2 Potential Risks

The potential risk for the feature selection procedure primarily focus on the lack of overlap between the covariates in $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ under which circumstance the meta-learner based procedure will struggle, under this circumstance where the support shifts significantly between different batches, the random forest classifier would be highly efficient, maybe among the best performing procedure in comparison to the other versions like the l_1 penalized regression procedures. Thus in practice, we propose the following general strategy for the product best practice:

- Obvious covariate shift is observed, where there's almost no overlap between the distribution of the covariates in both $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, the random forest classifier is what we may need, however, the random forest VIMP can only tell concisely which feature shifts significantly conditional on the existence of all the other features, even including Y , which is fundamentally different from the concept drift detection use cases. For the relatively minor or gradual concept drift with covariate shift, the meta-learner method would work slightly better than the SGShift, Knockoff as well as the MMD for characterizing concept drift.
- In production or real-data analysis, the support shift can be quantified by the overlap score proposed by Matsouakav et.al. in 1989([58]), the threshold suggested are around 0.3 where if the $OVL = \int \min\{f_1(e), f_0(e)\}de \leq 0.3$ the fallback procedure will be triggered. Alternatively, the o-values([38]) or the Bhattacharyya distances can also be leveraged for quantifying the degree of the overlap between $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ - the fallback procedure is also implemented in the submitted code.

On the other hand, continuously monitoring variable importance in production enables interpretable, feature-level characterizations of model behavior—specifically, the detection of both gradual and abrupt shifts in feature importance over time. This capability is particularly valuable in dynamic online environments where data streams will exhibit concept drift and models are updated incrementally. Several existing methods have been developed along this direction. iPFI([25]), iSAGE([52]) and Streaming SHAP([50], maintaining the running average for the shapley values) are among a few of them. We emphasize that while full configurations of these online interpretation methods falls outside the scope of our proposed Statistical Methodology, they would serve as a powerful complement in the production monitoring regime. Specifically, it can enable the dashboard visualizations that provide operators with interpretable, real-time insights into which feature is driving the significant distribution shift.

7.3 Deployment Checklist

In this subsection we discuss the topics directly related to its deployment in production ML system. Firstly the random forest classifier procedure with high degree of robustness & efficiency and computational efficiency, can serve as both a real-time feature selection procedure for the distribution shift and a portable and efficient tool for people to check the distribution shift among batches of datasets Secondly, this pipeline operate within the data ingestion and model inference cycles - whether that is micro-batch(every a couple of minutes, enabling the alerting mechanism in a very short period of time), hourly or daily. Rather than specifying the one-size-fits-all configuration setting, the system exposes a couple of factors including the number of permutations B , degree of parallelization and the sample size(granularity) for each of the batch testing. These would allow the deployment engineers to carefully tune the runtime based on their specific cluster capacity and the monitoring window. Thirdly, the guiding principle is that the pipeline would be required to complete its run before the next batch arrives in the inference engine - if this constraint can't be met during the peak load, the system would automatically downgrades to the computationally cheaper random forest classifier VIMP fallback. This procedure would ensure the monitoring loop never become a bottleneck with ensuring the high-quality feature characterization achieved regardless of resource constraints.

From the cost-efficiency considerations, the primary economic value of feature selection for distribution shift lies in drastically shortening the latency between the drift onset and the actionable feature-level detection. Rather than waiting for the requests from the downstream business metrics to degrade before initiating a costly investigation, the pipeline proactively flags the exact variables driving the shift - enabling teams to diagnose the root causes before it propagates into the revenue - impacting the outcomes directly. Upon output, the ranked feature list empowers deployment scientists to prioritize debugging efforts with surgical precision, narrowing a sprawling, multi-hour or even multi-days investigation into the focused selection for the top-K candidates. The rapid, almost real-time feature selection(especially for covariate shift) translates directly into improved **unit economics** across the following aspects. First, it minimizes compute waste by avoiding unnecessary full-model retraining. Second, it substantially reduces operational expenditure (OpEx) by shrinking engineering triage time. Thirdly, it mitigates the loss of revenue by intercepting performance-degrading shifts before they cascade into the production - intervening at the feature level rather than reacting to the degraded business metrics after multiple days.

Validation of the this pipeline is conducted through rigorous **offline back-testing** procedure rather than online A/B testing, leveraging the historical data with known distribution-shift events(recorded by the logs). This procedure

quantifies the key performance metrics - including recall@K for the truly shifted features, precision of the flagged set and the calibration of the feature-specific p-value or the features with the highest variable importance in the Random Forest Classifier procedure, under controlled and reproducible conditions. By establishing a robust baseline accuracy offline, the engineers and ML scientists can confidently calibrate the decision threshold and validate the ML pipeline’s diagnostic power before exposing to the live traffic, ensuring the production level deployment both safe and empirically grounded.

Overall, the feature selection as distribution procedure transforming the ambiguous distributional changes into concise, prioritized and **Comprehensive** action list, it effectively shifts the operational cost curve from costly reactive fire-fighting to lean, proactive health management. The economic impact is not merely coming from saving the retraining efforts, but also embedding continuous & interpretable diagnostics into arbitrary ML systems - delivering measurable ROI at model & batch and the incident level - simultaneously reducing the cognitive burden on the engineering team.

7.4 ML System Level Relevance

We focus on three key aspects that connect distribution shift procedures to ML system-level relevance. First, when should a given feature be removed from the deployed ML models? Second, when is an in-depth root-cause analysis required for downstream error diagnostics? To efficiently address these three problems, the statistical methodology introduced in this work should be complemented by traditional statistical measures and system-level statistics. We assume that the deployed ML models are already equipped with standard monitoring dashboards, including but not limited to missing-value statistics (e.g., the proportion of missing values per batch) and marginal shift monitoring procedures based on KL divergence [21] and the population stability index (PSI, [87]). Additionally, leveraging the existing methods for online monitoring of the variable importance in the streaming settings are also required including incremental permutation feature importance([25]) and the incremental SAGE([52]). We then elaborate on each of these issues in details.

Regarding the removal of the corrupted features, we want to elaborate that the current statistical methodology should be complemented with the variable importance of the deployed ML models, especially during the back-testing stages. We would recommend knocking out this feature has the following 3 conditions/prerequisites. Firstly, the distribution of the feature itself oscillates a lot across different time periods where the PSI and the KL-Divergence consistently exceed the critical threshold set by the business relevances. Secondly, the feature has been contributing actively either to concept drift or covariate shift(stays among the top-k features for a good period of time via our **Feature Selection for Distribution Shift Procedure**) - contributing to the distribution shift significantly. Thirdly and foremost, the feature has no significant predictive power in the deployed model, either in the back-testing stage(in the past several days) or the testing stages. Then after satisfying these 3 conditions, we would claim that the feature should be removed immediately.

Second, to systematically determine whether and when a feature should be escalated to the downstream root-cause analysis(with respect to a certain time window, we denote it as the degradation window) and the deep-dive system, we propose a three-evidence criterion that must be jointly satisfied. This criterion is grounded in the observation that a single monitoring signal—such as a high PSI or a drop in variable importance—is often insufficient to justify a deep investigation, as it may arise from benign data fluctuations or model adaptations. Instead, we require a convergence of independent statistical signals that collectively point to a feature’s anomalous behavior during a performance-degradation window, relative to a stable reference period. The three evidence types are designed to capture different facets of the feature’s relationship with the model and the data: its changing role within the model, its net contribution to predictive performance, and the structural nature of its distribution shift across subpopulations.

To talk about this in detail, the first evidence requires a substantial rank shift in the variable importance across different time windows(the reference window and the degradation window) for these features - for example, a rank change of at least five positions or a relative VIMP change exceeding 50%. This indicates that the model’s reliance on the feature has materially redistributed, signaling a potential instability. The second evidence is obtained by a leave-one-covariate-out (LOCO) evaluation on the degradation window: removing the feature should yield a statistically significant improvement in performance, such as an AUC gain above 2% with a 90% or 95% confidence interval that would not include zero(these values are just for illustration, would subject to discussion with the business teams). It indicates that during the degradation period, the feature has become an informative predictor. This suggests that, during the problematic period, the feature has become a net source of noise or misleading signal rather than a useful predictor. The third evidence examines the feature’s marginal distribution shift across subpopulations(although the meta-learner testing as feature selection procedure can also induce subgroup level insights, it would be out of our scope); we require that the shift be concentrated in a specific subgroup, as measured by a subgroup-level PSI(Population stability index) that exceeds the global PSI by more than 0.15, with that subgroup accounting for at least 5% of the total sample. Only when all three conditions are concurrently met does the system generate a ‘recommend for the downstream deep-dive’ signal. This joint criterion provides a statistically grounded and evidence-based trigger for further investigation, while explicitly avoiding any causal claim—the evidence is intended to guide, not replace, human diagnostic efforts, and serves as a prioritization tool for allocating limited analytical resources. We want to emphasize that among all of these procedures, are only for reference, no deterministic output, the results should be blended with the business context & relevance for the decision-making process. **We focus on the evidence — not the decision — and frame all outputs as recommendations to guide, rather than replace, human diagnostic efforts.**

We also acknowledge that ML monitoring mechanisms are far more comprehensive than the aspects covered here. They include, but are not limited to, deciding whether a new feature should be incorporated into currently deployed models and how frequently the model should be retrained. The former can leverage existing methods such as the adversarial random forest ([12]), while the latter is often heuristic-based rather than purely data-driven. Streaming VIMP and incremental feature acquisition and selection ([85]) represent another important class of statistical and ML problems. Although outside the scope of this work, we note that these techniques would be highly desirable to ensure that scientists and engineers can extract interpretable, concise, real-time insights for smoother and more tractable decision-making.

8 Summary

We present a comprehensive evaluation of feature selection strategies for distribution shifts, encompassing both concept drift and covariate shift—a critical yet often overlooked procedure in machine learning system design. For concept drift detection, the meta-learner approach outperforms current SOTA methods. For covariate shift, we adopt a random forest-based variable importance procedure on the out-of-fold samples, which achieves highly competitive performance relative to existing SOTA(FSL-NET, [8]) while exhibiting slightly better computational efficiency. We demonstrate these by both simulations and extensive real-data applications equipped by a variety of types of the adversarial perturbations. The variable importance based procedure has been justified for efficiently characterize the feature-level contribution to the distribution shift. Most importantly, the optimality of random forest—both as the outcome and propensity models within the meta-learner, and as the binary classifier trained for covariate shift with VIMP-based feature selection — is empirically justified, demonstrating its strong practical value. Integrated with standard ML monitoring best practices, the proposed framework can be readily deployed in production systems, delivering interpretable top-k interpretable feature rankings, actionable recommendations for root-cause analysis, and near-real-time business insights into variable changes—as validated across diverse real-world use cases. Empirically, our results reaffirm the fundamental impossibility of unambiguously disentangling overall distribution shifts into feature-level contributions of covariate shift versus concept drift. Looking ahead, extending this feature selection framework to the online mini-batch or streaming settings presents a promising direction, where adaptive updating strategies - potentially drawing from the online learning or bandit algorithms - could offer more flexible solutions targeted to industrial demands. We leave this as a future research avenue.

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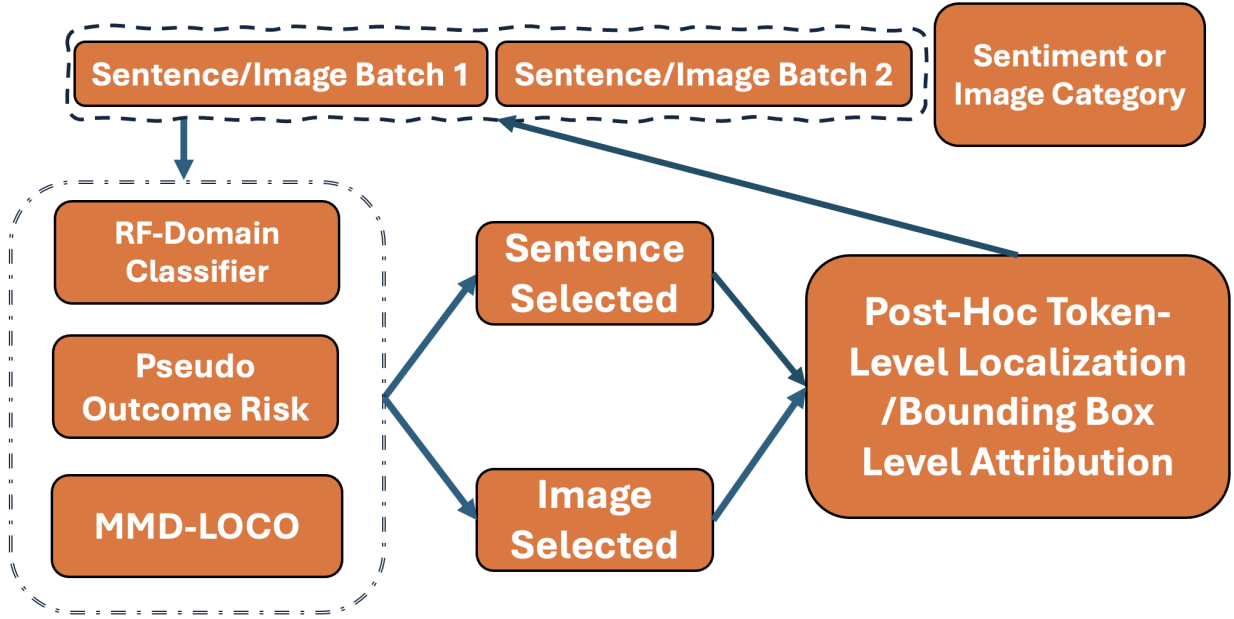
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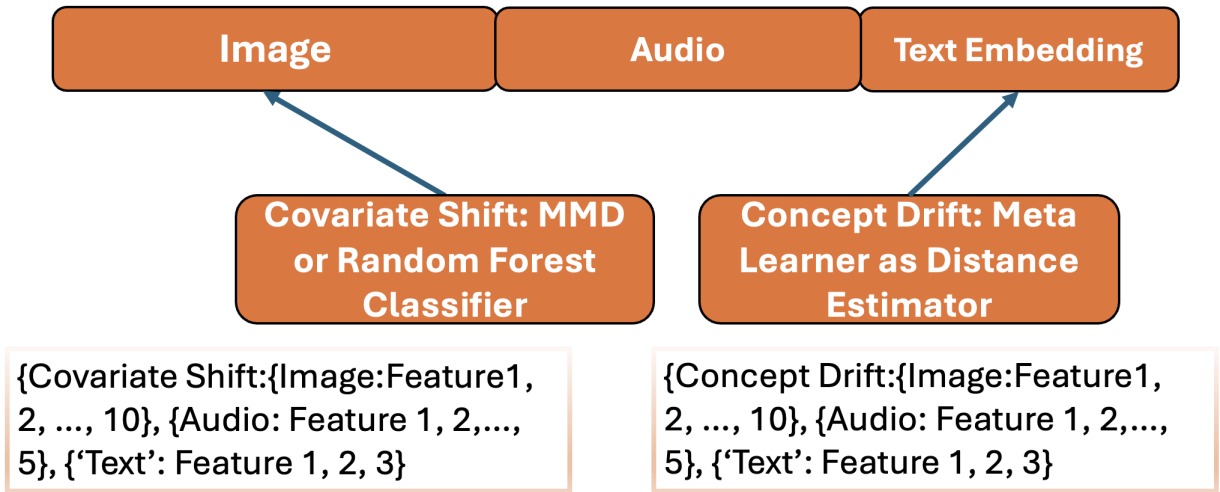
Post-Hoc Localization Procedure for Attribution



(a)

Figure 1: Diagram for the post-hoc localization procedure through image& text modalities

How the feature selection for distribution shift(FSDS) be adapted to the user preference/user mixture shift modality attribution?

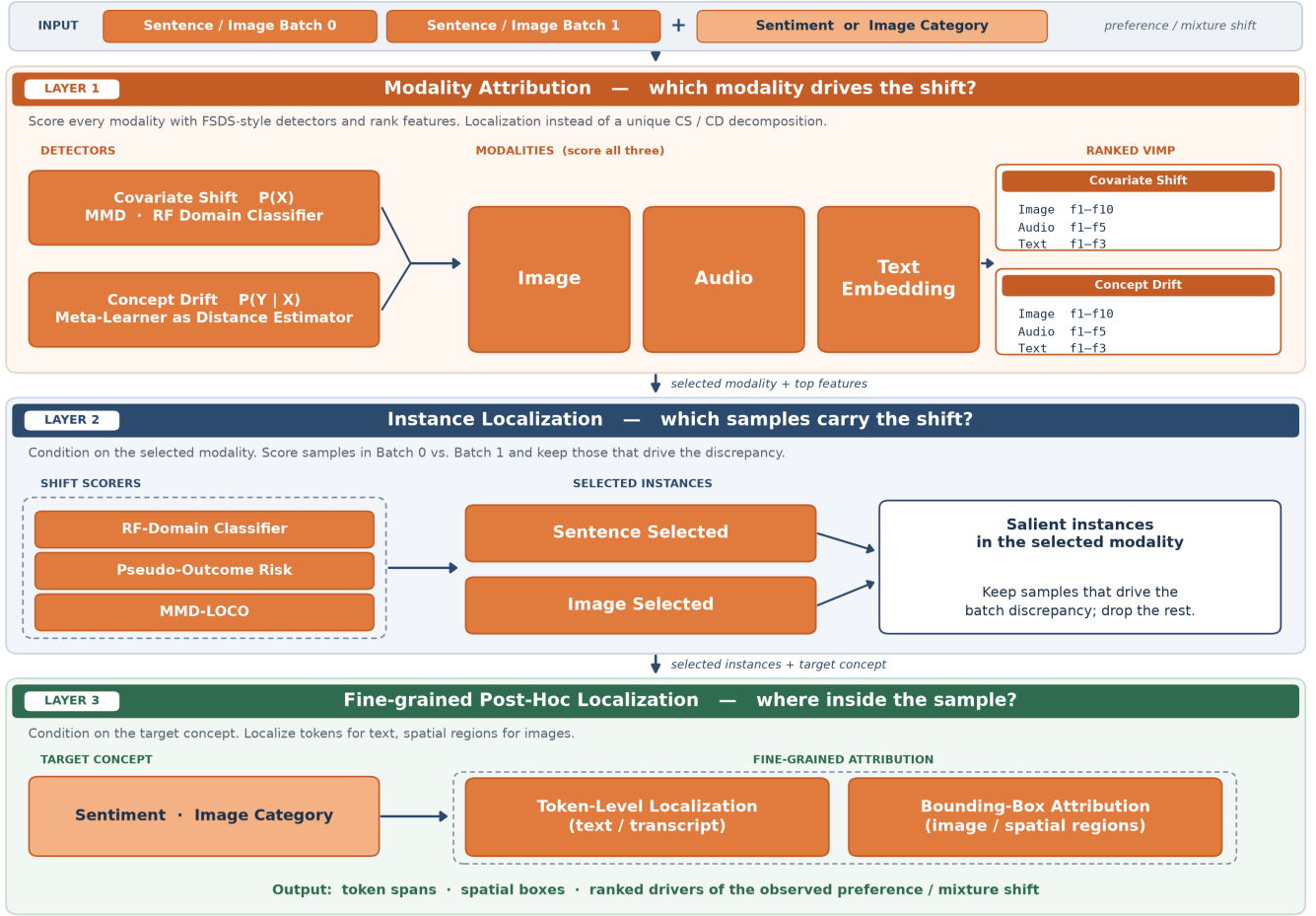


(a)

Figure 2: Diagram for the feature selection application to the multi-modal dataset distribution shift

Hierarchical Multimodal Attribution Procedure

Adapting Feature Selection for Distribution Shift (FSDS) to user-preference / mixture-shift attribution



(a)

Figure 3: Illustration Diagram for the Multi-Modal Attribution through the multi-modality from feature selection to post-hoc Localization

MSR-VTT · Batch 0 vs Batch 1 region board

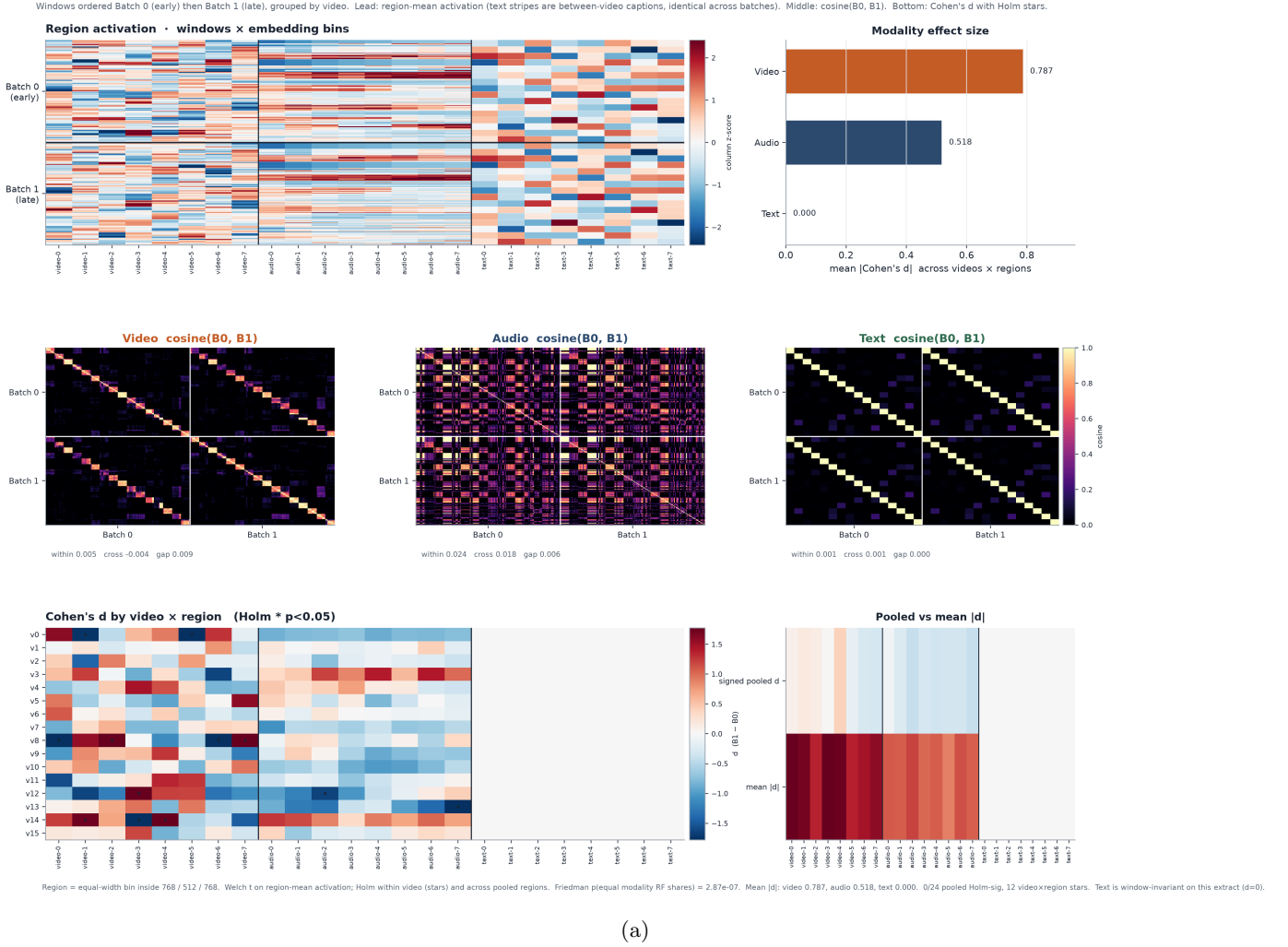


Figure 4: Illustration for the modality specific attribution procedure for the MSR VTT video & audio & caption dataset

9 Supplementary Materials: Spanish & Text Data Applications

Data. Reviews: X is the review text (word n -grams, plus an `is_spanish` flag); Y is sentiment on $\{-1, 0, 1\}$ [46], [4]. NER: X is the token sequence (plus WikiAnn-style entity rates `rate_0`, `rate_PER`, ...); Y is entity density, not token accuracy [4] [55]. Split is chronological, not i.i.d.: freeze the *train tail* as $T=0$, compare it to the *production head* (English-only, before Spanish) or to the *Spanish-window head* (the mid-stream pulse in production). Pooling train and production and reshuffling would mix the pulse into both sides and invent a shift that operations never saw. The three cuts are Table 55; the same locked train-tail embedding geometry is Table 56. **Two-step Feature selection procedure.** Step 1 screens X only. A domain classifier ranks features by how well they separate the two time blocks; that list is covariate shift, a statement about $P(X)$, and is Table 57. Step 2 confirms on the screened set only. CFPerm treats the block label as W and asks whether those same features also move Y ; the last column of Table 57 is that confirm statistic, not the CS rank. Table 3 is the hashing-embedding analogue of the same screen: which 64-d bins separate the blocks. Do not rank Step 1 by the Step 2 p -value: language and domain markers that do not change the scorer would disappear, and the ticket would be wrong.

Now we talk about the aspects in context of domain of the NLP. On the English production head, Table 57 keeps fulfilment diction (`product`, `this`, `never received`): same language and same sentiment schema, different review type. The business action is to refresh the English baseline or sample that source—not to wait for accuracy to drop, and not to retrain Y while RFPerm p in Table 55 stays large. On the Spanish window the same table keeps `is_spanish` and function words (`de`, `que`, `muy`): route that traffic to a Spanish model or to review; do not score it with the English head. On NER it keeps entity-mix rates (`rate_0` up, `rate_PER` down), not a fake location spike from an i.i.d. draw: the tagging contract is still English NER, but the mention mix changed, so the baseline for X must move even if $Y \mid X$ does not. A single embedding distance (Table 56) cannot issue any of these tickets; the FSDS list can. Hashing dimensions in Table 3 locate the shift in embedding space but are not the operations ticket—the readable n -grams in Table 57 are.

Table 1: Arize train-tail vs production-head monitors. RFPerm tests $Y \mid X$; RF-domain accuracy and eigen-shift (SAS misalignment, RIR, PET) read embedding covariate shift. PET state from locked train-tail covariance.

Dataset	cut	RFPerm p	RF-dom. acc	SAS mis	RIR	PET _u	PET
xtreme	prod.head	0.463	0.777	0.722	1.36	0.166	CAUTION
ecomm early EN	prod.head	0.195	0.944	0.662	1.43	0.195	CAUTION
ecomm Spanish	Spanish-window head	0.463	0.913	0.643	1.42	0.189	CAUTION

Table 2: Eigen-shift geometry panel, scaler locked on the train tail. Spectrum TV is total variation between eigenvalue masses; cov rel-F is $\|\Sigma_1 - \Sigma_0\|_F / \|\Sigma_0\|_F$.

Dataset	SAS mis	RIR	eig TV	cov rel-F	$\ \Delta\mu\ _2$
xtreme	0.722	1.36	0.043	0.536	0.551
ecomm early EN	0.662	1.43	0.016	0.504	0.520
ecomm Spanish	0.643	1.42	0.028	0.563	0.928

10 Supplementary Materials: Application on User Preference Shift

Frozen $T=0$ monitor on Amazon Reviews 2023 Gift Cards and Magazine Subscriptions [hou2024bridging]. Text = title+body; one review per user on the time and catalog cuts. $Y = (r - 3)/2$. X = frozen hashing (1,2)-grams [weinberger2009feature]; not a sentence-transformer. Cuts: **time_magazine** (same catalog, calendar mix); **catalog_swap** (gift cards vs magazines); **user_pref** (generous vs harsh magazine users; the held review does not label the user). RFPerm freezes μ_0 [breiman2001random]; the decision is the adaptive p -path plus online FDR [ramdas2018saffron], not a single permutation p . Online AR-bootstrap [palm2024online] tracks frozen PO-risk with batch and running intervals; it does not refit and is not a second RFPerm. FSDS-CS (no Y) owns $P(X)$ via RF-domain, MMD-LOCO [gretton2012kernel], and raw-text n-grams. CFPerm ranks embedding coordinates of the joint (X, Y, T) ; DRPerm / R-risk own $Y \mid X$. Post-hoc $\psi = (Y - \hat{\mu})(T - \hat{e})$ names the user/item cohort. Not NDCG, not generator quality. Table 5 is the only numeric record.

Catalog swap is a feature-store event, not a model-rollback: language separates the two catalogs, and Table 5 leaves $Y \mid X$ quiet, so the frozen star map stays in service. Time is a watch rather than a reject: the embedding mix and the star level both move, CFPerm marks a few hashing coordinates, and post-hoc ψ sits on helpful votes, length, time, and verified purchase, but the path gate in Table 5 does not call exchangeability broken. A running PO-risk interval that leaves the reference level only keeps that watch open. User preference is the strongest cohort shift and the largest downstream residual lift, yet CS and CD do not share coordinates, and Table 5 still passes the path gate, so harsh versus generous users do not authorize replacing μ_0 . Across cuts, CS ranks agree with each other and disagree with CFPerm: the coordinates that move $P(X)$ are not the coordinates that carry $Y \mid X$. Action follows the owner, not the size of the star drop: covariate signal refreshes retrieval features; a significant ψ subset is routed, not globally rolled back; only a gate other than pass in Table 5 authorizes preference-model fallback.

11 Supplementary Materials: Implementation Details

We provide all the experimental results across diverse types and modalities of the datasets, ranging from linear & nonlinear synthetic data to tabular synthetic datasets and the chronoberg text corpus dataset as well as the FMoW dataset. Additionally sensitivity analysis in terms of both component models in meta-learners and the overlap are also provided via justifiable empirical studies. The shadowVIMP and VitaPIMP procedures are adapted from **R** through the rpy2 package with python benchmark for evaluation. The rf-domain, delta-shap, MMD-LOCO, DRPerm, PermuCATE, ARD methods are implemented in Python, the adversarial perturbations procedures on the domain regressor trained is adapted from that on the image dataset. The FSL-Net, DataFix, knockoff, l1-spec and l1-misspec are leveraged from the existing implementations.

11.1 Detailed Settings for the FSDS

We present some of the detailed settings for the **CFPerm** procedure in production.

1. The default hyper-paramter setting for the Random Forest on the dataset with n observations and p features are: for continuous response, the $mtry = \frac{p}{3}$, for discrete response, set the $mtry = \sqrt{p}$. Then the minimal node size is set as $node_size = \frac{\sqrt{n}}{2}$ and the number of trees in the outcome model is set as $n_{trees} = 150$. For the logistic regression procedure, the **lbfgs** procedure is adapted for optimization and the number of iterations are set as 2000 by default.
2. The cross-feature quantile (95%) and the within feature quantile (99%) are pre-set by Statistical Researchers, won't recommend any modifications. On the other hand, regarding the notion of variable importance, we would suggest the **permuCATE**(conditional permutation variable importance, [54], more accurate with lower uncertainty than

Table 3: CFPerm top-5 hashing embedding dimensions (CS feature rank on 64-d hashing). Minimum $p = 1/31$ at $n_{\text{perm}} = 30$.

Dataset	1	2	3	4	5
xtreme	dim 35	dim 9	dim 34	dim 16	dim 6
ecomm early EN	dim 58	dim 26	dim 35	dim 34	dim 21
ecomm Spanish	dim 54	dim 30	dim 14	dim 58	dim 17

Table 4: FSDS covariate-shift feature list, ranked by RF-domain importance (which n -grams / extras separate the chronological batches). CFPerm p is a confirm statistic on the screened set, not the CS rank. $\Delta = \text{mean}^{\text{new}} - \text{mean}^{\text{train}}$.

Dataset	rank	feature	RF-dom. imp	Δ	CFPerm p
xtreme	1	rate_0	0.049	+0.034	0.871
	2	the	0.039	+0.013	0.065
	3	state	0.035	+0.009	0.839
	4	in	0.034	-0.003	0.645
	5	rate_PER	0.033	-0.019	0.226
	6	album	0.025	+0.006	0.484
	7	to	0.024	+0.011	0.516
	8	national	0.023	+0.006	0.387
	9	george	0.023	-0.009	0.968
	10	of	0.022	-0.011	0.677
ecomm early EN	1	of	0.042	+0.012	0.452
	2	it	0.020	+0.008	0.419
	3	this	0.017	+0.011	0.065
	4	one	0.016	+0.007	0.935
	5	product	0.015	+0.005	0.774
	6	he	0.013	-0.006	0.323
	7	never received	0.012	-0.007	0.065
	8	this one	0.012	+0.005	0.581
	9	were	0.011	-0.005	0.355
	10	could	0.011	+0.005	0.548
ecomm Spanish	1	is_spanish	0.102	+0.258	0.806
	2	de	0.053	+0.046	0.710
	3	un	0.051	+0.023	0.677
	4	que	0.036	+0.028	0.710
	5	en	0.033	+0.026	0.484
	6	muy	0.030	+0.028	0.516
	7	no	0.029	+0.024	0.129
	8	lo	0.024	+0.018	0.387
	9	es	0.023	+0.028	0.484
	10	con	0.021	+0.020	0.548

the LOCO) for more accurate calculation and the split frequency version by Causal Forest for parsimonious output generation(with significantly faster computational speed).

- From the Statistical aspect, we want to emphasize again that the assumption for the causal inference procedure is no longer satisfied in terms of the strong ignorability condition especially under the circumstance where the H_0 is rejected.
- In production, there is fair degree of flexibility in choosing among the outcome model and the propensity score model. However, for the tabular data or the embedding representation data, the Random Forest outcome model and the Logistic Regression propensity score model are recommended for sake of robustness and system reliability.
- One additional aspect worth emphasizing concerns the sample size requirements for each batch in the testing procedure in **CT-as-FS**. In particular, the new batch size n_{new} should not fall below a certain minimum threshold(at least 300 to 500); otherwise, the lack of adequate fit in both the outcome model and the propensity score model may introduce non-negligible bias into the estimated variable importance. We want to mention that in production environment, this constraint is reasonable because the existing batch is typically much larger than the new batch by default (e.g., accumulated historical data).
- Although additional computation is introduced, the number of permutations should not fall below 100 to enable a valid data-driven threshold for the testing procedure. Given ample computational resources, the number of permutations in **CFPerm** should be set to 200 or 500 by default to ensure the false alarm control as well as the robustness of the feature-selection stage.

Table 5: Amazon review embedding shift. Gate is the path plus online FDR. RF-domain and MMD² own $P(X)$. DRPerm owns $Y | X$. A single RFPerm p is not the decision.

Cut	RFPerm p	gate	RF-domain	MMD ²	#rej	DRPerm p	stars
time	0.034	pass	0.77	0.021	2	0.12	4.14→3.59
catalog	0.276	pass	0.72	0.026	0	0.88	4.52→4.07
user pref.	0.034	pass	0.75	0.016	0	0.059	4.40→3.23

12 Supplementary Materials: Justification of R-learner for isolating Concept Drift

We now formalize the rationale of the capability of the proposed objective function through the R-risk for isolating the concept drift from covariate shift from both the double ML aspect and geometric aspect. The R-risk is calculated as

$$R_risk = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{m}(X_i) - \hat{\tau}(T_i - \hat{e}(X_i)))^2$$

where the $\hat{m}$ is the fitted outcome model and the $\hat{e}(X)$ denotes the fitted propensity score model. The key mechanism relies on the **Neyman’s Orthogonality** of the R-learner objective function. First we consider the nuisance parameter vectors $\eta_0 = (m_0, e_0)$ where $m_0(x) = E[Y|X = x]$ and $e_0(x) = P(T = 1|X = x)$ with observed data $Z = (X, T, Y)$. The R-learner, fitted via weighted least squares, associated with the following score function(robinson’s transformation [63]) $\varphi(Z; \theta, \eta) = (Y - m(X) - \theta(T - e(X)))(T - e(X))$ satisfies the moment condition $E[\varphi(Z; \theta_0, \eta_0)] = 0$, note that the parameter of interest θ is the treatment effect τ . **We justify this phenomenon from both the Theoretical aspect and the Geometric Aspect.**

Under the regularity conditions established in [53], this score function satisfies the **Neyman Orthogonality Condition** $\partial_\eta E[\varphi(Z; \theta_0, \eta)]|_{\eta=\eta_0} = 0$. In studying the impact of the estimation of the nuisance parameters, we establish the Taylor’s expansion through **Gauteux** derivative([18]) around η_0 where

$$E[\varphi(Z; \theta_0, \hat{\eta})] = E[\varphi(Z; \theta_0, \eta_0)] + \partial_\eta E[\varphi(Z; \theta_0, \eta_0)](\hat{\eta} - \eta_0) + |Res| = (1) + (2) + |Res|$$

. Then the first-order term(term 2) vanishes via Neyman’s Orthogonality Conditions. Based on the derivation in [18], the Res term is calculated as $Res = E[(m(X) - m_0(X))(e(X) - e_0(X))]$. Applying the cauchy-schwarz inequality induces the fact that the residual term satisfies the rate $|Res| \leq \|\hat{m} - m_0\|_2 \cdot \|\hat{e} - e_0\|_2$. Thus if the nuisance estimators satisfy $\|\hat{m} - m_0\|_2 = o_P(n^{-r_m})$ and $\|\hat{e} - e_0\|_2 = o_P(n^{-r_e})$ with $r_m + r_e > \frac{1}{2}$, then the residual term satisfies the rate $|Res| = o_P(n^{-1/2})$. Consequently, the estimation admits an asymptotically linear representation, $\sqrt{n}(\theta - \theta_0) \rightarrow_d N(0, \sigma_\theta^2)$. The most important takeaway from this justification mainly focus on that the errors in estimating the outcome model m_0 and propensity score model e_0 - which are primarily driven by the marginal distribution of X - only depends on the second-order. Thus as a result, this estimator would be asymptotically insensitive to **covariate shift** where still highly responsive to the changes of the conditional dependence $Y|X$ (concept drift) by definition of the treatment effect estimator.

Indeed, disentangling the covariate shift and concept drift completely is intrinsically impossible for any kinds of the meta-learners where the equation of

$$D(P(\mathbf{X}_{exist}, Y_{exist}), P(\mathbf{X}_{new}, Y_{new})) = D(P(\mathbf{X}_{exist}), P(\mathbf{X}_{new})) + D(P(Y_{new}|\mathbf{X}_{new}), P(Y_{exist}|\mathbf{X}_{exist}))$$

only satisfies under the KL-divergence where under other meta-learners, such decomposition won’t be satisfied anymore(MMD or HSIC) where in general decomposing the shift in the joint distribution into the covariate shift and concept drift is impossible. That’s exactly why we need the double/debiased machine learning to isolate the concept drift from the covariate shift(remove the effects of the covariates and looking at the residuals - the residualized outcome model) - we simply ensure that our estimate of the conditional discrepancy is not contaminated by shifts in the covariate distribution. Alternatively the invariance of the R-risk function can be justified from the geometric point-of-view. It borrows insights from the Hajek projection with conditional expectation in the Hilbert Space. We consider the outcome model and the propensity score model $m(X) = E[Y|X]$, $e(X) = E[T|X]$ which can be interpreted as the projection of responses Y and treatment(batch label) T onto the hilbert space $L^2(P)$ spanned by the column of covariates $\mathbf{X}$. We denote the projection as $\Pi_X(Y)$ and $\Pi_X(T)$ respectively. We define the residuals of the propensity score model and the outcome model as

$$\tilde{Y} = Y - E[Y|X] = Y - \Pi_X(Y), \tilde{T} = T - E[T|X] = T - \Pi_X(T)$$

respectively. By construction both $\tilde{Y}$ and $\tilde{T}$ falls into the orthogonal complement of $\mathcal{H} = \overline{span(\mathbf{X})} - \mathcal{H}^\perp$. Then, the invariance under covariate shift follows naturally. Any objective function built from the form of $(\tilde{Y} - \theta\tilde{T})$ satisfies

$$\langle \tilde{Y}, f(X) \rangle = 0, \quad \forall f \in \mathcal{H},$$

and therefore resides entirely in $\mathcal{H}^\perp$. As a result, regardless of how the marginal distribution of the covariates $P(\mathbf{X})$ shifts, as long as the conditional dependence $P(Y | \mathbf{X})$ remains unchanged, the R-risk objective remains invariant. Therefore, the negligible power under covariate shift is readily justified. In particular, under the absence of concept drift, the permutation test will not exhibit power as also demonstrated in the empirical results, since the objective function remains unchanged regardless of the covariate shift.

Thus the R-risk here can be interpreted as the distance estimator for the conditional dependence between the two batches of data, where the feature ranking for the R-risk under the leave-one-covariate-out procedure is highly efficient in terms of characterizing their individual contribution to the concept drift.

13 Supplementary Materials: (Optional)Post-Hoc Shift Localization Procedure

After identifying the features that actively drive the observed distribution shift (encompassing both covariate shift and concept drift), we refine the post-hoc localization procedure for the next rollout phase. Specifically, we leverage MMD([65]) and HSIC([1]) to assess covariate shift, and the pseudo-outcome risk (PO-risk, [35]) to evaluate concept drift, quantifying the extent of change in the respective subsets. This quantification yields a principled measure of the shift’s significance between the two subsets, which in turn informs the subsequent root-cause analysis and the subset selection procedure - in parallel to the **CFPerm** procedure with respect to the feature selection via the Doubly-Robust Pseudo-Outcome Learner. The post-hoc localization results for both covariate shift and concept drift are summarized in Tables 8, 10, and 9. Specifically, Table 8 reports the selected feature subsets identified by the RF-Domain classifier under various shift scenarios. Table 10 presents the (optional)bootstrapped confidence intervals for each feature’s variable importance, providing a measure of uncertainty. Table 9 extends the analysis to concept drift, where the subset level concept drift is quantified by the PO(pseudo-outcome) risk, enabling subset-level shift characterization. Together, these tables constitute an optional and helpful post-hoc procedure following the primary feature selection pipeline, offering ML practitioners with deep insights into the structural nature of the detected shifts. We want to elaborate again that this procedure won’t yield any of the causal interpretations which can help people identifying the subset level distribution shift quantified by explicit quantitative measures.

14 Supplementary Material: Extension to multi-layer feature selection

In modern machine learning systems, a feature store is essential for ensuring timely, consistent feature updates and for keeping both models and business metrics aligned. In an e-commerce platform, features are generated at multiple levels—merchant, order, item/product, and user—and these levels interact: merchants sell products, which are converted into orders by users. A feature store is therefore critical not only for ML model monitoring but also for user behavior modeling in recommendation systems. The key challenge is that a metric change in one dimension—say, merchant—can propagate to other dimensions, such as orders. Understanding these effects at the subset level is therefore crucial. More importantly, subset localization enables two-step or multi-step localization and feature selection, allowing us to pinpoint precisely which subsets of business units contribute most significantly to a given data change. From a business-decision perspective, this transforms monitoring into action: it helps determine which merchants to support, which subset of users to give promos and how to allocate the incentives to the individual merchant/user level. In this section, we demonstrate how this procedure can be extended beyond two batches or two layers of data.

We first briefly describe our procedure, we rank and select groups by the leave-one-group-out ablation $\phi_g = T(\text{all}) - T(\text{all} \setminus g)$, the drop in the shift metric when a group’s features are removed, because it directly answers the operational question—“if we ignore or repair this group, does the detected shift disappear?”—and is a **conditional** importance (given all other features present), crediting only non-redundant drivers rather than everything correlated with the shift; it is cheap ($K+1$ metric evaluations), its sign is meaningful, and at the raw-attribute group level it is immune to within-group correlation that would dilute single aggregations. By the non-uniqueness result, the total is order-invariant but the per-feature split is not, so the invariant and estimable object is the contribution of a whole subset (include or exclude the group as a unit); empirically the group-level answer is stable across attribution conventions (standalone, LOGO, Shapley) and resamples, while the per-feature answer is not, and since a real shift typically enters through a raw signal—a collection/definition change in GMV, a pricing change, a new buyer cohort—that moves all of that signal’s aggregations together, the raw-attribute group is the natural causal and operational unit, making selection both statistically well-posed (invariant, stable) and operationally meaningful (mapping to an owner, a rule, and an action). For the modalities, a characteristic-kernel MMD is zero iff the two batches share the same $P(X)$, and its LOGO drop is a genuine reduction in a distributional distance sensitive to shifts in any moment (mean, scale, tails), so it does not miss non-linear covariate shifts; the PO-/R-risk targets the orthogonal, outcome-relevant question—has $P(Y | X)$ changed—and is insensitive to pure covariate shift, so the two modalities cover the two failure modes without double-counting. Both are permutation-exact and assumption-light, which makes a LOGO drop built on them a trustworthy, convention-free number, in contrast to Shapley, which becomes unique only by averaging over an arbitrary set of orders.

Given the impossibility issue discussed in Section 6, while we cannot uniquely attribute contributions to individual features, subset localization is highly preferred for multi-layer feature selection, enabling accurate positioning of the drivers of distribution shift (i.e., business changes). We therefore leverage the Tencent Recommendation System dataset [56] to build item-level and user-level interaction data by subsetting to 10,000 users’ past behaviors and engineering 150 features from user, item, and user–item interaction levels, then generating synthetic orders and merchants for modeling purposes. To make feature selection and estimation more concise, we incorporate a bootstrap module for feature-level MMD (for covariate shift) and an R-risk module (for concept drift) to establish confidence intervals (with the stability estimated via proportion of times this feature is among the top), yielding a concise procedure ready for next-stage rollout. Additionally, we adapt a leave-one-group-out procedure to estimate each block’s contribution to the distribution shift at the feature level, analogous to multi-modal attribution procedures (e.g., for audio, video, and image data).

The two-layer localization procedure is demonstrated here. We take the week of 2024-09-01 - 2024-09-07 and the 2024-10-08 - 2024-10-14 for the comparison. In the second week’s batch, the **CFPerm** procedure yields the significance. The first-level localization on the *region* axis selects the activity subset — concretely the order-count feature **n_orders** (and order-value aggregations in the past 1, 14 and 30 days to avoid the data leakage) shifted *up* for a subset of users in region *R*: those buyers are placing more (and larger) orders than the first batch. We want to emphasize that we do *not* claim “region *R*’s users account for 43% of the shift” — by the proposition that number is order/convention dependent. We claim the shift is localized to region *R*’s user subset, with a stability of, say, 0.98 and a split rule **n_orders per user** $> t$. Then we conduct the next-level subset localization. We first restrict the order table to exactly the level-1 subset (the flagged region-*R* users, **n_orders per user** $> t$), and only then aggregate those users’ orders on the item / product-vertical axis and re-run the two-stage selection. This conditional drill localizes the extra orders to a specific **product vertical** (item category *C*) within that user subset: among the flagged region-*R* users the surge concentrates in category *C* (e.g. “fresh grocery”), not uniformly across the catalogue. The localization is thus **product-vertical** $\times$ **user-subset** (category *C* conditioned on the region-*R* user segment), output as subset = {region *R* users, category *C*} with its own split rule and stability. Evaluating level 2 on the whole population for 1 subset would confound the two levels and is explicitly avoided. The table 7 vividly demonstrated this aspect.

Action / closed loop. The subset routes to the category-*C* / region-*R* owner; the candidate rule becomes a monitor; the decision (e.g. reweight if the mapping is unchanged, collect labels or retrain if the outcome relationship moved) is attached; the trend field marks it resolved once the subset stops flagging. The business learns where and who and what to do — never a spurious percentage. We hereby justify the question: “Why this is the right granularity” “43% from category *C*” would be unfalsifiable (non-identifiable) and un-actionable. “The new-batch order surge is localized to category *C* among region-*R* users; here is the threshold and the owner” is checkable, routable, and monitorable — it closes the loop.

We emphasize that multiple testing procedures are not preferred for accurately attributing data change to different subsets, as they fail to secure business stakeholder buy-in, let alone address the reduced statistical accuracy and confidence level. Instead, we stress that the total distribution shift is identifiable; its per-feature decomposition is not. This is not a limitation of any particular estimator but a property of dependent features under any additive path attribution. The productive response is to **localize responsible subsets** recursively down the entity hierarchy and attach owner, rule, decision, and trend to each—turning an unanswerable “how much” into an operational “which subset, who, and what to do.” This approach can be extended to arbitrary levels of data systems, particularly feature stores or metric systems, where it can help mine the relationships between business metrics.

From another angle, this procedure can be extended to concurrent business metric systems to automatically root-cause changes in a given metric—say, from merchant-level metrics down to order-level metrics. The **FSDS** procedure can play its role here: we can regard the feature known to have changed as the response *Y*, and the potential candidate features as *X*, aggregated to the same level, then conduct next-stage attribution and dig deeper. **It would probably need some graph-based algorithms which need slightly more investigation here. The cross-layer causal chain is desired - it may needs slightly more investigation - will serve as an important exploration direction.** We hereby provide a smallest prototype for this multi-layer feature selection procedure via the Hilbert-Schmidt Independence Criterion (HSIC, [1], for the covariate shift) and PO-risk (primarily for concept drift) - **as in practice, not all of the relationship between business metrics can be revealed by deployed models.**

14.1 Understanding Vague Relationship between business metrics- HSIC

Business metrics live at different levels (order, user, merchant) and have **no explicit formula** linking them. We only want to know **which metrics move together**—the association **linkage**—not any causal direction. We leverage HSIC and PO-risk to provide illustrative use cases for ML practitioners to better understand.

The Hilbert–Schmidt Independence Criterion (HSIC) is a kernel measure of statistical dependence that is zero iff two variables are independent and, unlike correlation, detects **non-linear** association without assuming a functional form. Running HSIC on every pair of metrics reveals the linked chain and isolates the unrelated ones. For paired samples $\{(a_i, b_i)\}_{i=1}^n$, with RBF kernels $k(a, a') = \exp(-\|a - a'\|^2 / \sigma_a^2)$ and $l(b, b') = \exp(-\|b - b'\|^2 / \sigma_b^2)$ (bandwidths by the

median heuristic), form Gram matrices $K, L \in R^{n \times n}$ and the centering matrix $H = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$. The empirical HSIC is

$$\widehat{HSIC}(a, b) = \frac{1}{(n-1)^2} \text{tr}(KHLH) \geq 0, \quad (1)$$

large when a, b are dependent and ≈ 0 when independent. It is symmetric and makes **no** directional / causal claim—it is purely an association.

The steps are presented below:

1. Compute each metric per entity (a common index), giving columns $m_1, \dots, m_p$.
2. For every pair compute $\widehat{HSIC}(m_i, m_j)$.
3. Keep pairs with $\widehat{HSIC}$ above a small threshold as **linkage edges**; the connected component is the metric **chain**.

On a synthetic marketplace with a linked chain **delivery_mins** – **user_rating** – **n_orders** – **gmV** (metrics across the order, user, and merchant levels) plus two unrelated metrics (**quantity**, **discount**), the pairwise HSIC is:

metric a	metric b	$\widehat{HSIC}$
delivery_mins	user_rating	0.055
delivery_mins	n_orders	0.056
delivery_mins	gmV	0.046
user_rating	n_orders	0.062
user_rating	gmV	0.054
n_orders	gmV	0.070
any pair involving quantity or discount		≤ 0.002

Table 6: Pairwise HSIC. The four chain metrics are mutually dependent ($\widehat{HSIC} \in [0.046, 0.070]$); every pair touching **quantity** or **discount** is ≤ 0.002 (independent). HSIC recovers the linkage chain **delivery_mins**–**user_rating**–**n_orders**–**gmV** and isolates the unrelated metrics—with no explicit relationship supplied and no causal direction claimed.

Thus HSIC alone—one symmetric, assumption-light, kernel dependence number per pair—is enough to **discover the metric linkage across levels**. Direction and root cause are deliberately **not** claimed here; this prototype only recovers which metrics are associated.

14.2 Understanding Vague Relationship between business metrics - PO-Risk

Business metrics live at different levels (order, user, merchant) with no explicit relationships. Two questions, two tools: **(A) Which metrics move together? — HSIC (association linkage)**. For metric pairs (a, b) with RBF Gram matrices K, L and centering $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$,

$$\widehat{HSIC}(a, b) = \frac{1}{(n-1)^2} \text{tr}(KHLH) \geq 0, \quad = 0 \iff a \perp\!\!\!\perp b.$$

Symmetric, non-linear, assumption-light; large $\widehat{HSIC}$ = a linkage edge. It recovers the chain and isolates unrelated metrics (*no direction claimed*).

(B) Did a target’s $Y|X$ change, and on which X ? — PO-risk (one model per target). With batch W as “treatment”, cross-fit $\mu(x) = [Y|x]$ (with random forest as the outcome model) $e(x) = \Pr(W=1|x)$; pseudo-outcome $\psi = (Y - \mu)(W - e)$; $\tau = \text{argmin}_f[(\psi - f(X))^2]$;

$$\text{PO-risk} = [\tau(X)^2] \ (> 0 \Rightarrow P(Y|X) \text{ shifted}), \quad \phi_j = \text{PO-risk}(X) - \text{PO-risk}(X_{-j}).$$

One meta-learner takes all X jointly; LOCO ϕ_j localizes the change – cost $O(\#\text{targets})$, not $O(p^2)$ pairwise models.

(A) HSIC linkage			(B) PO-risk $Y=\text{GMV}$ (test $p = 0.010$)	
a	b	$\widehat{HSIC}$	metric x_j	LOCO ϕ_j
delivery	rating	0.055	rating (driver)	+0.106
delivery	gmV	0.046	orders	+0.046
rating	n_orders	0.062	delivery	+0.039
n_orders	gmV	0.070	promo	+0.032
* w/ quantity/discount		≤ 0.002	tenure	+0.028
Chain delivery--rating--n_orders--gmV recovered; $P(Y X)$ shifted; change loads on rating . noise metrics isolated.				

Level (unit)	Candidate	LOGO / gap	Stability	Perm. p
Level 1 — merchant axis (global $MMD^2 = 0.281$, $p = 0.003$)				
	gmV (selected)	+0.283	1.00	—
	quantity	−0.277	0.85	—
Level 2 — product item category (conditional on gmV, new vs. existing)				
	Fresh Grocery (c^* , selected)	55.88	—	0.002
	Home & Kitchen	0.77	—	0.158
	Apparel	0.58	—	0.240
	Electronics	0.29	—	0.621
	Beauty	0.26	—	0.583
	Toys	0.03	—	0.932

Table 7: Two-level conditional subset localization (**fsds_cross_level.py**). Level 1 selects the **gmV** attribute subset (LOGO ablation + bootstrap stability); the conditional Level 2 drills into the **product item category** and selects exactly the shifted one, **Fresh Grocery** (gap = new−existing mean GMV within the category; merchant-block permutation p). The order surge is localized to a single product item category, not spread as proportions; ground truth is recovered at both levels.

Putting these together, the maps the metric linkage across levels; for any target metric, PO-risk says whether its $Y|X$ relationship shifted and which linked metric carries it — so a change in a merchant-level metric can be localized to an order-level metric, without modelling every pair and without a causal-proportion claim.

To summarize, this **feature selection for distribution shift (FSDS)** procedure can be extended beyond two batches of data, unlocking the potential for multi-layer feature selection and localization in both feature stores and concurrent metric systems. In many circumstances, the vague associations between metrics are not yielded by business resource-intensive models(which requires a lot of man/days from the business requirement to the deployment) and are not deployed in production systems—this is where another value of our FSDS procedure lies.

Table 8: Feature selection (RF-domain) and channel routing. CS = covariate shift (MMD); CD = concept drift (PO-risk). Counts = subset cells with bootstrap CI significance ($B=20$, $\alpha=0.10$).

Data	Dom. acc	Selected features	#CS	#CD
taxi	0.724	trip_distance, PULocationID, DOLocationID, passenger_count	18	16
waymo	0.987	next_y, y, x, next_x, z	30	7
diabetes	0.654	num_lab_procedures, num_medications, number_inpatient, time_in_hospital, number_diagnoses	4	3
M5	0.595	month, past_7_days_snap_CA_mean, past_10_days_snap_WI_mean, past_28_days_snap_CA_mean, past_21_days_snap_CA_mean	7	2

15 Supplementary Materials: Meta-Learner Testing as Feature Selection

This section briefly discusses the meta-learner testing procedure that employs the permute-then-refit framework to assess variable importance. This procedure serves as a complement to the main contribution of our work, which focuses on feature selection for distribution shift. From an overarching perspective, this supplementary procedure is more stringent and concise, yielding higher confidence in the selected features, while with lower sensitivity compared to our proposed Feature Selection for Distribution Shift approach.

15.1 Efficiency Justification

We justify the efficiency for the proposed **Meta-Learner Testing as Feature Selection Procedure** in both concept drift and covariate shift. As demonstrated in Figures 6 and ??(the 'X' and dot are those point with the rejection), the features with concept drift $\mathbf{X}_4 - \mathbf{X}_7$ are ranked among the top identified features with the smallest feature-wise p-value as the degree of concept drift increase. Moreover, as demonstrated in Figures 8 and 7 both **PermuCATE** and **LOCO** have also demonstrated their efficiency in characterizing mean shift and variance shift with conditional permutation VIMP slightly more efficient.

The crucial metrics for evaluating the efficiency for the **Meta Learner-Testing-as-Feature-Selection** including the following given ground-truth shifted feature set $\mathcal{S}_{gt}$

Table 9: Selected feature $\rightarrow$ channel. Channel = CS if MMD CI>0, CD if PO-risk CI excludes 0, both if both, else none.

Data	Feature	Channel	RF-imp	#cells	Primary measure
taxi	trip_distance	both	0.400	6	MMD + PO-risk
	PULocationID	both	0.302	6	MMD + PO-risk
	DOLocationID	both	0.257	6	MMD + PO-risk
	passenger_count	both	0.041	4	MMD + PO-risk
waymo	next_y	both	0.146	6	MMD + PO-risk
	y	both	0.136	6	MMD + PO-risk
	x	both	0.089	6	MMD + PO-risk
	next_x	both	0.074	6	MMD + PO-risk
	z	both	0.071	6	MMD + PO-risk
diabetes	num_lab_procedures	CS	0.156	6	MMD
	num_medications	none	0.138	6	—
	number_inpatient	none	0.116	3	—
	time_in_hospital	CS	0.093	6	MMD
	number_diagnoses	both	0.070	5	MMD + PO-risk
M5	month	CD	0.930	6	PO-risk
	past_7_days_snap_CA_me	CS	0.005	2	MMD
	past_10_days_snap_WI_m	CS	0.004	2	MMD
	past_28_days_snap_CA_m	CS	0.004	2	MMD
	past_21_days_snap_CA_m	CS	0.004	2	MMD

- The kendall’s correlation of the ranking of the identified feature set $\mathcal{S}$ and ground-truth shifted feature set $\mathcal{S}_{gt}$.
- The recall for the features identified : $\frac{|\mathcal{S} \cap \mathcal{S}_{gt}|}{|\mathcal{S}|}$ and the precision@k: $\frac{|\mathcal{S} \cap \mathcal{S}_{gt}|}{|\mathcal{S}_{gt,k}|}$

We want to mention that although in real-world applications, no ground-truth indices of features with significant contribution to the distribution shift is labeled, the Statistical and numerical results in our **CFPerm** methodology has showcased the efficiency for the procedure in different types of distribution shift justifiably.

Thus, the primary utility of **Meta-Learner Testing-as-Feature-Selection** procedure lies in its feature-selection efficiency — and that’s precisely the dimension that truly matters for production. Unlike the R-RiskPerm (via leveraging the R-risk objective function, only tells you whether a shift has occurred from the batched dataset level), CFPerm enables a ranked and **interpretable** set of features that are primarily driving the shifts. This feature-level output directly translates into actionable diagnostics: engineers can trace the identified features back to upstream data pipelines, schema changes, or operational anomalies, dramatically reducing the time spent on root-cause investigation in the production environment.

We want to further mention that the **Meta-Learner Testing-as-Feature-Selection** procedure further outputs the ranking of variable importance for each considered notion, including but not limited to PermuCATE, LOCO, and GRF. These rankings can be viewed as an alternative notion of out-of-distribution (OOD) feature importance. Specifically, the ranking of feature-specific p-values characterizes each feature’s relative contribution to the distribution shift between the two batches of data. The dashboard can be built in production environment for ML engineers to monitor each metric/feature’s contribution to the change of the Topline Business Metrics.

15.2 System Actions & Operations

The primary value of the CFPerm detection framework lies not in providing a binary ”retrain or not” signal, but in delivering a structured, feature-level diagnostic that directly informs downstream engineering and analytical workflows. The **Meta-Learner Testing-as-Feature-Selection** procedure’s primary value is not limited to providing a binary ’retrain or not’ signal, but in delivering a structured, feature-level diagnostics that directly informs downstream engineering and analytical workflows.

1. **Guiding downstream root-cause analysis and subgroup investigation.** The features flagged by the significance test serve not only as proxies for distribution-shift contributors, but also as natural stratification axes for deeper diagnostic exploration. Distribution shift localization can be achieved by partitioning the population along the ranges of these selected features. For example, if a categorical feature is flagged, comparing the distribution shift across its different levels provides a principled and practical basis for differential analysis. This transforms the testing result into a directly actionable input for case-study investigation and root-cause identification.
2. The features selected given the significance testing results will not only serve as the proxy for the ones contributing to distribution shift but also guide the downstream subgroup analysis or the in-depth case-study for root-cause analysis of the change of data. The distribution shift localization can be achieved via stratifying the population with respect to the grids expanded by the ranges of these selected features. Specifically suppose a categorical feature is selected, the subgroup characterized by the different levels of that feature will enhance the differential analysis for

Table 10: Subset measures with bootstrap confidence intervals. CS measure = MMD (sig if CI entirely > 0); CD measure = Δ PO-risk (sig if CI excludes 0). Resampling: stratified bootstrap within OLD/NEW inside each G ($B=20$, 90% CI).

Data	Feat.	Channel	Measure	Subset	Est.	Lo	Hi	n	Sig
taxi	DOLocationID	both	MMD	High	0.0435	0.0207	0.0675	4148	•
	DOLocationID	both	PO-risk	High	-0.0516	-0.0701	-0.0333	4148	•
	passenger_count	CS	MMD	Low	0.0323	0.0062	0.0988	115	•
	DOLocationID	both	MMD	(100.0, 161.0]	0.0290	0.0100	0.0569	2361	•
	DOLocationID	both	PO-risk	(100.0, 161.0]	-0.0653	-0.0771	-0.0510	2361	•
	DOLocationID	both	MMD	(230.0, 265.0]	0.0289	0.0169	0.0600	1734	•
	DOLocationID	both	PO-risk	(230.0, 265.0]	-0.0643	-0.0983	-0.0496	1734	•
waymo	next_x	CS	MMD	(13177.556, 13..	0.5126	0.4203	0.5257	1407	•
	x	CS	MMD	(13177.556, 13..	0.4874	0.4195	0.5070	1407	•
	next_x	both	MMD	(13256.952, 13..	0.0845	0.0622	0.1000	1438	•
	next_x	both	PO-risk	(13256.952, 13..	0.0091	0.0016	0.0171	1438	•
	next_y	both	MMD	(8161.73, 8193..	0.1043	0.0891	0.1447	1363	•
	next_y	both	PO-risk	(8161.73, 8193..	0.0080	0.0029	0.0161	1363	•
diabetes	number_diagnoses	CS	MMD	(0.999, 6.0]	0.0124	0.0019	0.0215	2868	•
	num_lab_procedur	CS	MMD	(0.999, 42.0]	0.0050	0.0004	0.0107	2057	•
	number_diagnoses	CD	PO-risk	(8.0, 9.0]	-0.0003	-0.0006	-0.0001	2979	•
	number_diagnoses	CD	PO-risk	High	-0.0002	-0.0006	-0.0000	4074	•
M5	past_28_days_sna	CS	MMD	Category 0	0.0098	0.0009	0.0210	1241	•
	past_21_days_sna	CS	MMD	Category 1	0.0079	0.0027	0.0201	602	•
	month	CD	PO-risk	High	0.0039	0.0019	0.0059	1070	•
	month	CD	PO-risk	Low	-0.0011	-0.0023	-0.0004	773	•

the distribution shift among the subgroups. It will provide a practical and statistically principled diagnostic tools for the downstream case-study and root-cause analysis.

3. The variable importance/feature-specific-p-value identified via the causal ML models can serve as a notion of **Online** or **OOD** variable importance which characterizes their contribution to the distribution shift, specifically, given proper interpretation, would implicitly give deployment engineers concise information about the relative contribution of each metric/feature in the change of data.
4. Most importantly, from the system level, the top-k features selected can serve as an early alert for the model performance/feature quality degradation with no additional computational cost on top of the hypothesis testing procedure. From a system engineering perspective, this enables a proactive reliability paradigm: the monitoring stack can be configured to trigger a 'yellow' alert(i.e., a pre-warning) simply when the VIMP for a certain feature exceeds a pre-defined threshold, without waiting for business metrics degrades actually in real-world scenarios.

The entire diagnostic can be run offline, decoupled from real-time inference, so it adds zero latency to serving; within the monitoring window (e.g., hourly or daily), the parallelized pipeline comfortably completes well before the next batch arrives. Scalability is further ensured by the highly scalable $O(p)$ post-fit operations (threshold comparison and ranking) without requiring additional model fitting, which remain negligible regardless of p . The data-driven threshold efficiently avoid the external data dependencies which allow the system to handle fluctuating batch-sizes without re-engineering.

15.3 Multi-Modal Attribution and Post-Hoc Localization

A particularly important and interesting use case focuses on the modern multi-modal learning scenario: how to attribute distribution shift to a specific modality, especially at the feature level. We leverage our Feature Selection for Distribution Shift (FSDS) procedure to perform feature selection on data changes across multiple modalities, providing concise insights into which subsample of features actively contributes to the shift. The diagram is presented in Figure 2. We use the following three datasets to demonstrate the efficiency of the multi-modal attribution procedure: the Affec dataset [69], the Fashion-IQ dataset [82], and the Tiny-MM-IMDb dataset [77].

Specifically, for the Affec dataset, six modalities are present—including eye-tracking, pupil, cursor, GSR-EDA, and EEG. We leverage the same context length to align the context window for the next-stage rollout, thereby constructing a concatenated embedding for adapting the FSDDS procedure. This makes the attribution traceable to the individual modality level—specifically, how many features contribute to the distribution shift within each modality and how their relative importance appears under either covariate shift or concept drift. For example, in the Affec dataset, a 10k subsample is obtained via time-indexed subsample slicing. The results are presented in Tables 35 (concept drift) and 34 (covariate shift), with modality-specific ranks presented.

Similarly, for the Tiny-MM-IMDB dataset, the results are presented in Tables 44 for concept drift and 43 for covariate shift. For the Fashion-IQ dataset, the corresponding results are presented in Tables 40 and 41 for covariate shift, and Table 42 for concept drift. These results provide the ranking of each feature/metric's contribution within each modality.

Causal Testing as Feature Selection: X4-X7 Selected
Concept Drift Only on X4-X7

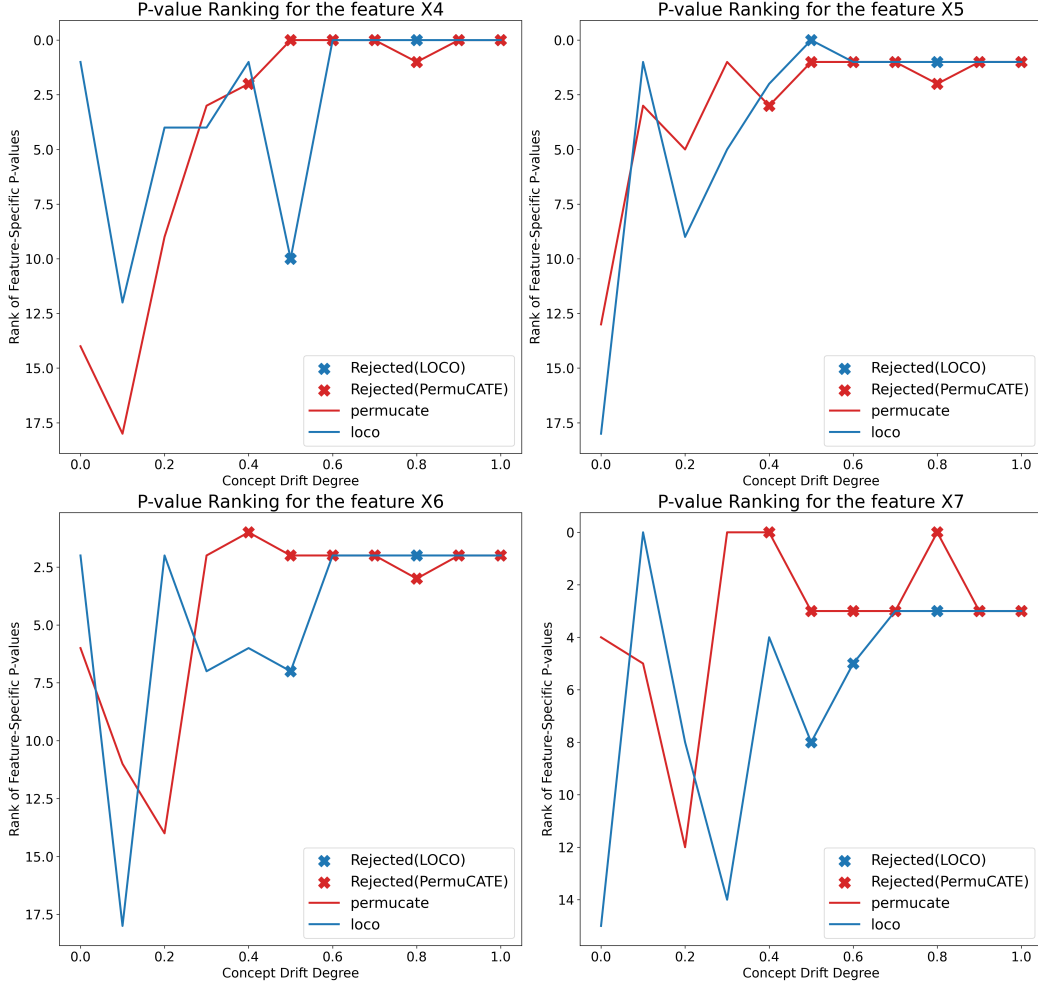


Figure 5: Causal Testing as Feature Selection: Concept Drift 1

The procedure is capable of yielding concise business interpretation and ranking across each modality. We want to elaborate that we would like to provide a certain degree of flexibility to practitioners, thus making it a concise procedure for multi-modal attribution.

More importantly, post-hoc localization can be conducted after incorporating bounding-box-related features for each image. If the localized subset of images is important, this post-hoc localization procedure can provide a better understanding of how each bounding box contributes, and it operates in a parallel manner. Therefore, we leverage the multi-modal **COCO** dataset [42] to conduct this level of **Post-Hoc Bounding Box Localization**, closely analogous to the post-hoc distribution-shift localization procedure in **CFPerm**. This makes it a parallel procedure to token attribution (leave-one-token-out PO-risk difference), further enhanced by manually engineered bounding-box features for the images, such as crowdedness, the Gini index of the covered areas, and the average IoU across bounding boxes. The entire diagram procedures is illustrated in Figure 1 which can be extend to various levels. As demonstrated in Tables 38 and 37, the important and informative bounding boxes can be readily extracted through the following procedures, which can be viewed as a two-layer attribution procedure that proceeds from the image/text level to the bounding-box level. More importantly, because bounding boxes serve as a bridge between the text modality and the image modality, this **LOBO (leave-one-bounding-box)** procedure effectively demonstrates how the text modality interacts with the image modality from both technical and feature perspectives.

The product relevance visualization is presented in Figure 3. It not only helps users identify which feature subset (tokens or patches) is important and supports instance/subset localization and fine-grained post-hoc localization, but also provides granular attribution for out-of-distribution (OOD) behavior through a leave-one-out procedure based on pseudo-outcome (PO) risk.

15.4 Code Availability

The code for implementing the procedure is available in <https://github.com/ruanhq-StatsML/CFPerm> in both **Python** and **R** with both parallelized and serial versions available. The google colab for reproducing the visualization can be

Causal Testing as Feature Selection: X4-X7 Selected Concept Drift Only on X4-X7

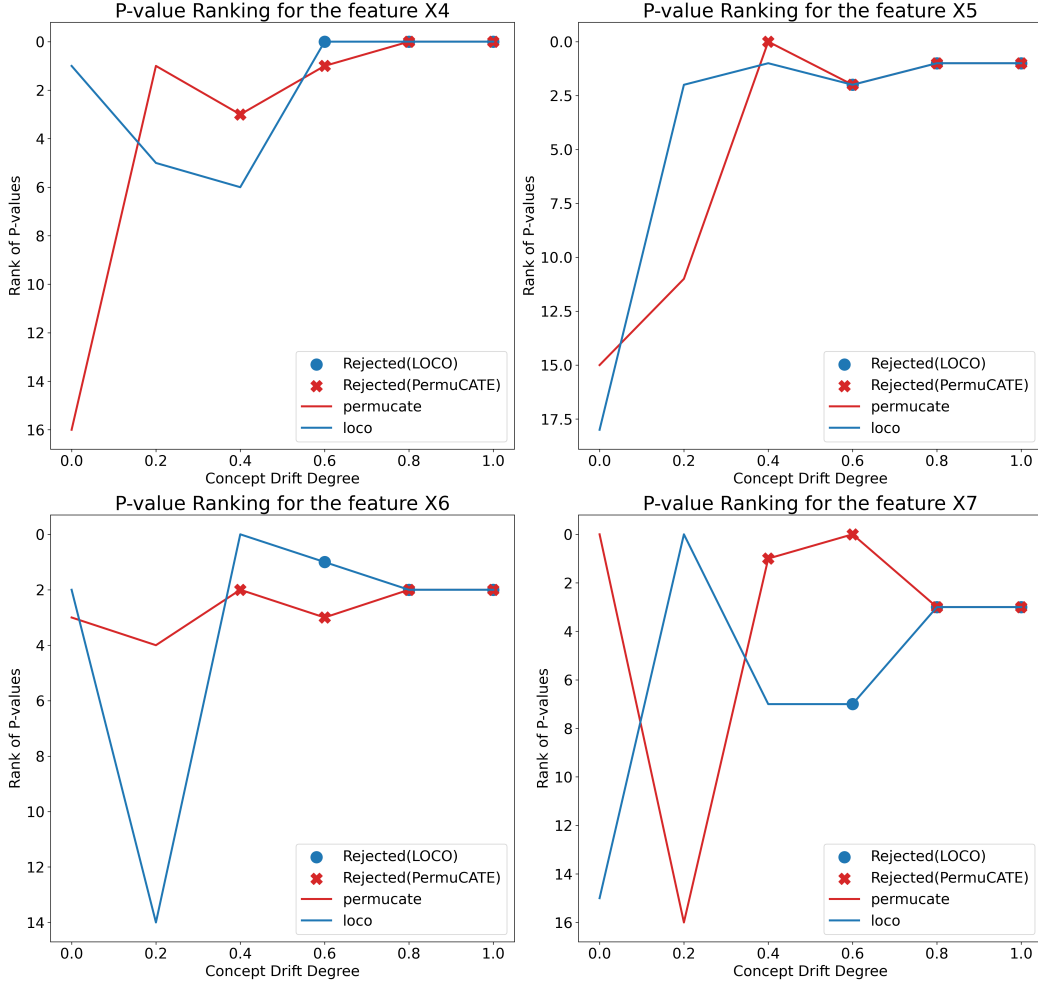


Figure 6: Causal Testing as Feature Selection: Concept Drift 1

found in <https://colab.research.google.com/drive/1EZ2CcPcETwegopk30hpPrDyRUF6a51NM>.

15.5 Computational Aspect and Efficiency

From the computational efficiency aspect, we want to emphasize that the feature selection procedure itself would enable near-real-time inference with almost no latency especially for the feature ranking indices as well as the rank correlations. Thus in production, the latency itself won't be an issue. The CFPerm procedure is straightforward to parallelize in both R and Python, as each refitting step in the permutation test can be distributed across independent computational nodes. In Python, this is implemented via the `joblib` library; in R, via the `foreach` package. As shown in Figure ??, parallelization reduces computational cost by approximately 60%–70% compared with the serial implementation on the 2-core CPU, achieving computation save comparable to the DRPerm procedure (based on the pseudo-outcome learner). Like the RRPerm procedure, the CFPerm test can be readily deployed in production environments by leveraging multiple computational nodes in a cluster for configuration of the batch testing procedure every a few minutes justifiably.

Another potential risk for this attribution procedure primarily focus on the lack of overlap between the covariates in $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$ under which circumstance the causal inference based procedure will struggle, under this circumstance where the support shifts significantly between different batches, the random forest classifier would be highly efficient, maybe among the best performing procedure in comparison to the other versions like the l_1 penalized regression procedures. Thus in practice, we propose the following general strategy for the product best practice:

- Obvious covariate shift is observed, where there's almost no overlap between the distribution of the covariates in both $\mathbf{X}_{exist}$ and $\mathbf{X}_{new}$, the random forest classifier is what we may need, however, the random forest VIMP can only tell concisely which feature shifts significantly conditional on the existence of all the other features, even including Y , which is fundamentally different from the concept drift detection use cases.
- For the latent or gradual concept drift with covariate shift, the causal inference method would work slightly better than the SGShift, Knockoff as well as the MMD for attributing concept drift.

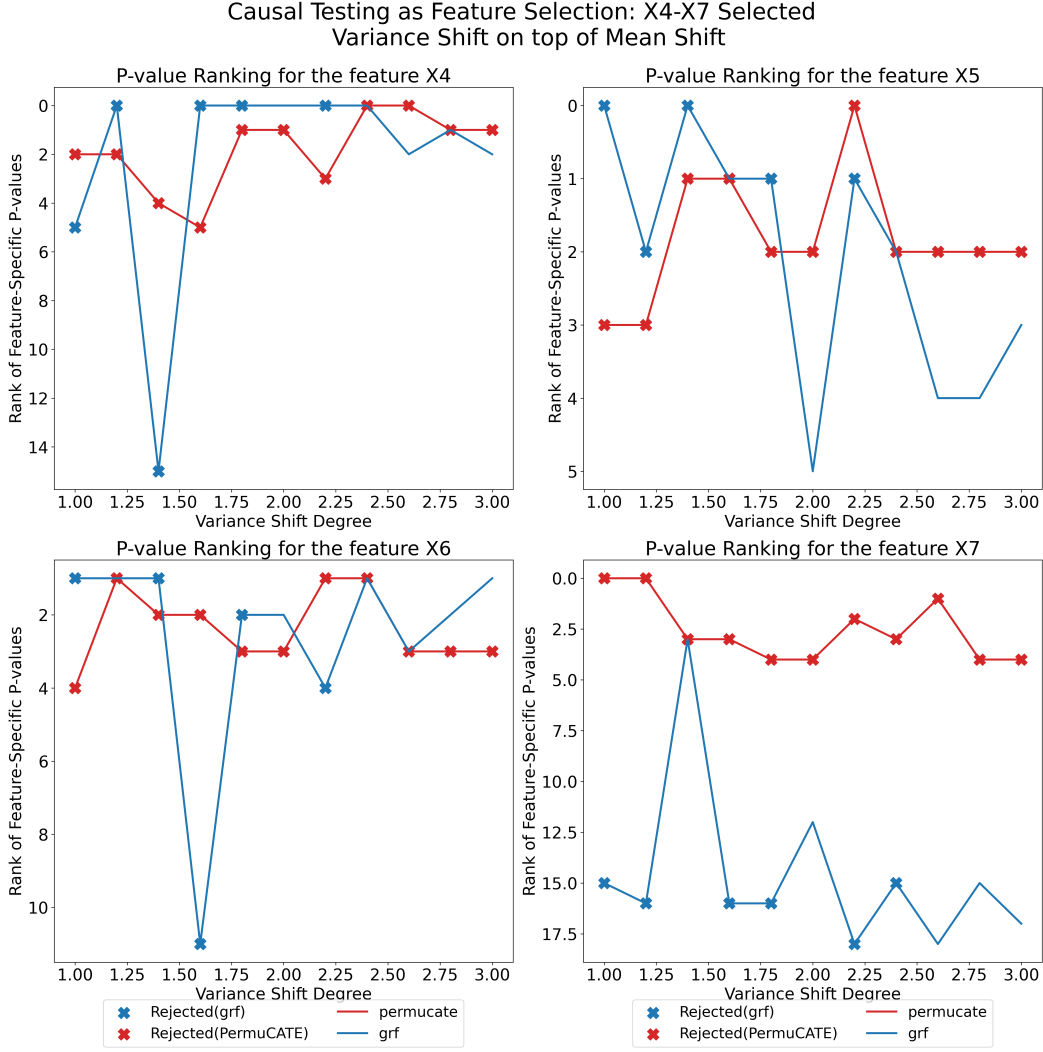


Figure 7: Causal Testing as Feature Selection: Variance Shift on top of Mean Shift

- Under the circumstances where both covariate shift and concept drift are significant, the fallback procedure is probably preferred where the combination of the **CFPerm** and the **RF-Domain** classifier is highly preferred.
- In production or real-data analysis, the support shift can be quantified by the overlap score proposed by Matsouakav et.al. in 1989

16 PO-risk as the reweighted estimator for adapting the distribution shift

The observation-level pseudo-outcome risk can serve as an efficient proxy for uncertainty quantification. Inspired by density ratio estimation [73], where observations with higher density ratios (i.e., more out-of-distribution) receive larger weights, we replace density ratio estimation with the observation-level pseudo-outcome risk. Specifically, we set $w_i = \sqrt{PO - risk_i}$ which acts as a proxy for out-of-distributionness. This procedure therefore provides an efficient proxy for adapting to distribution shift once a significant shift is detected. In particular, we use the p-value from the **OnlineRFPerm** procedure as a gate to control the degree of distribution-shift adaptation: only when a significant distribution shift is detected is the PO-risk reweighting procedure applied.

We adapt an online predictor on a consecutive-batch stream $B_0, B_1, \dots$ by *reweighting* the last two batches with an instance PO-risk score. $\sqrt{PO}$ weights fire only when a consecutive out-of-sample probe jump says $P(Y | X)$ moved, then anneal back. At hop t the learner has seen $B_0, \dots, B_t$ and is scored on the **next** batch B_{t+1} . The train set is always the last two batches $B_{t-1} \cup B_t$ (same rows whether the gate is quiet or on fire). Write $T_i = 0$ for rows in B_{t-1} (previous batch) and $T_i = 1$ for rows in B_t (this batch). The next-step model is RF / XGBoost (not Ridge). **Probe.** The PO probe is a shallow random forest ($n_{est} = 20$, depth = 4) fit on past batch as $T = 0$: $\mu_0 = \text{RF}(B_{t-1})$. This probe is *not* the next-step predictor. Consecutive OOS errors (no K -fold; an in-sample $e_1 > e_0$ on trees fires every hop)

$$e_{\text{now}} = \text{err}(\mu_0 \text{ fit } B_{t-1} \rightarrow B_t), \quad e_{\text{prev}} = \text{err}(\mu_0 \text{ fit } B_{t-2} \rightarrow B_{t-1}). \quad (2)$$

The hop fires iff $e_{\text{now}}/e_{\text{prev}} \geq \gamma$ (default $\gamma = 1.5$). The first hop is always quiet: there is no previous OOS.

Causal Testing as Feature Selection: X4-X7 Selected
Mean Shift on top of Variance Shift

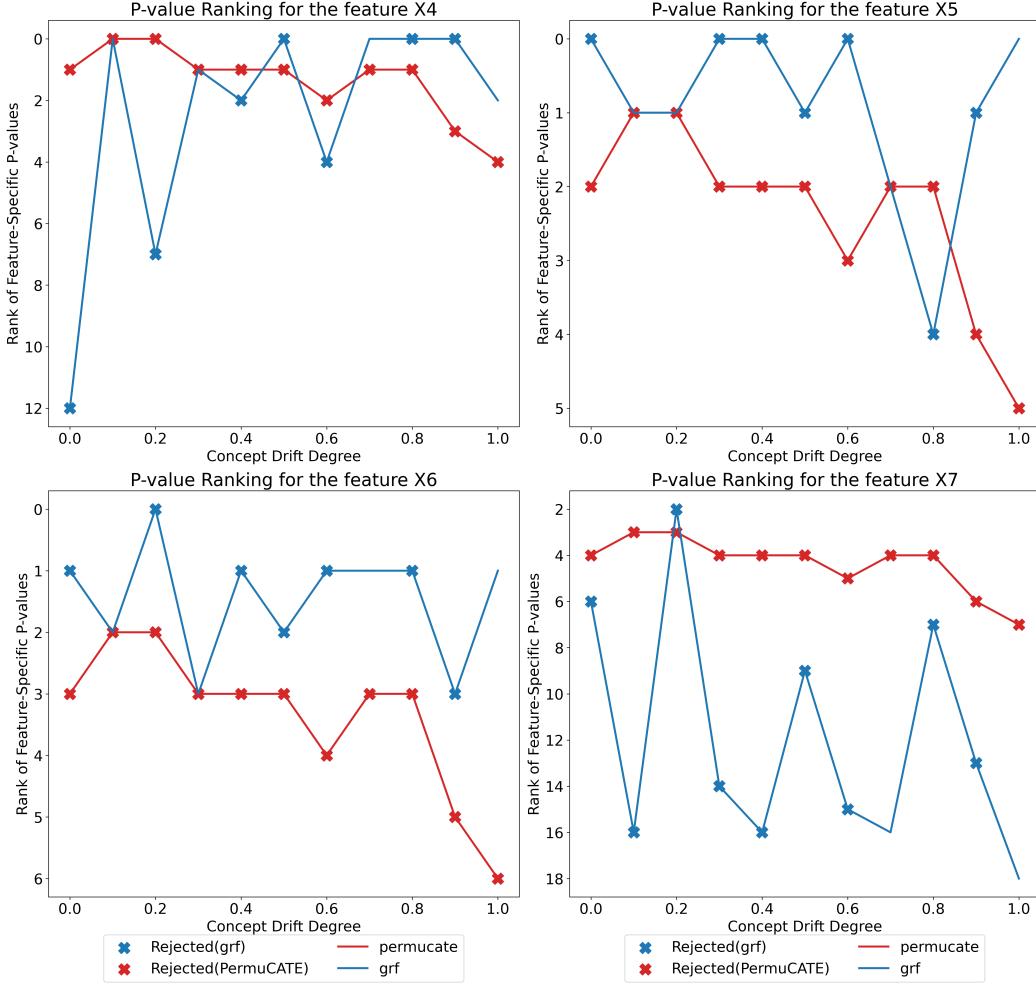


Figure 8: Causal Testing as Feature Selection: Mean Shift on top of Variance Shift

Observation-level PO-risk. Every observation on the last two batches gets its own score against the $T = 0$ probe. For regression, $r_i = |Y_i - \mu_0(X_i)|$. For classification, $r_i = 1 - p_{\mu_0}(Y_i | X_i)$. Mix with the batch gap $\delta = \max(e_{\text{now}} - e_{\text{prev}}, 0)$ at mix 0.5: $\text{po_risk0}_i = r_i (1 + 0.5 \delta)$. This is a *per-row* map, not a batch scalar.

Quiet hops stay uniform. On fire, keep the same last-two rows; $T = 0$ stays at 1 and $T = 1$ is upweighted by $\sqrt{\text{PO}}$:

$$\tilde{w}_i = \begin{cases} 1, & \text{quiet, or } T_i = 0, \\ \sqrt{\text{po_risk0}_i}, & \text{fire and } T_i = 1, \end{cases} \quad w_i = \text{clip}\left(\frac{\tilde{w}_i}{\bar{w}}, 0.05, 20\right). \quad (3)$$

The next-step model trains on $(X_i, Y_i, w_i)_{i \in B_{t-1} \cup B_t}$ and is scored on B_{t+1} . The ratio is a one-shot heat. After a fire, the next hop's e_{prev} is that large OOS error, so $e_{\text{now}}/e_{\text{prev}} \ll \gamma$ and the stream cools back to last-two uniform. A one-shot concept cut heats once. A periodic clock (day/night occupancy, PM2.5 episodes) reheats on a schedule; reweighting is built for the hop after a jump, not to ride a cycle. **Two levers, not one.** Reweighting does not drop the old batch. Dropping B_{t-1} and training on B_t only is a separate hard lever (residual hop-gate). PO-tail subset localization is off: at batch size 200 a $q = 0.3$ tail is too thin. Always-on $\sqrt{\text{PO}}$ and always-on DRE underperform the last-two uniform strategy during quiet periods and on real consecutive clocks, motivating the uniform default. As shown in Tables ??, 13, 11, and 14, the procedure is effective under gradual distribution shift, especially with the optional next-stage rollout procedure (most importantly, much much more efficient and robust than the density ratio estimation [73]). This demonstrates its potential beyond **distribution shift detection** to **distribution shift adaptation**, enabling the model to continuously deliver business impact in practice.

Table 11: PO-risk vs density-ratio (**rf**). Same last-two rows. $\Delta = \text{DRE} - \text{rfperm}$ (MSE $\downarrow$; positive means $\sqrt{\text{PO}}$ wins). DRE cannot see a $P(Y | X)$ hop with stable $P(X)$.

Scene	uniform	DRE	rfperm	Δ vs DRE
similar	0.540	0.541	0.540	0.001
covariate	0.728	0.874	0.720	0.154
concept	1.565	1.584	1.466	0.118
mixed	1.889	2.020	1.818	0.202

Table 12: Next-batch MSE (**rf**). Last-two uniform is the default. **rfperm**: gated $\sqrt{\text{PO}}$ on $T = 1$. **dre**: always-on density ratio $p(x)$ on the same rows. **resid**: drop the old batch. Gate $\gamma = 1.50$.

Scene	uniform	DRE	rfperm	drop-old	oracle	fire
similar	0.540	0.541	0.540	0.540	0.540	0.00
covariate	0.728	0.874	0.720	0.852	0.728	0.04
concept	1.565	1.584	1.466	1.332	1.379	0.17
mixed	1.889	2.020	1.818	1.657	1.765	0.17

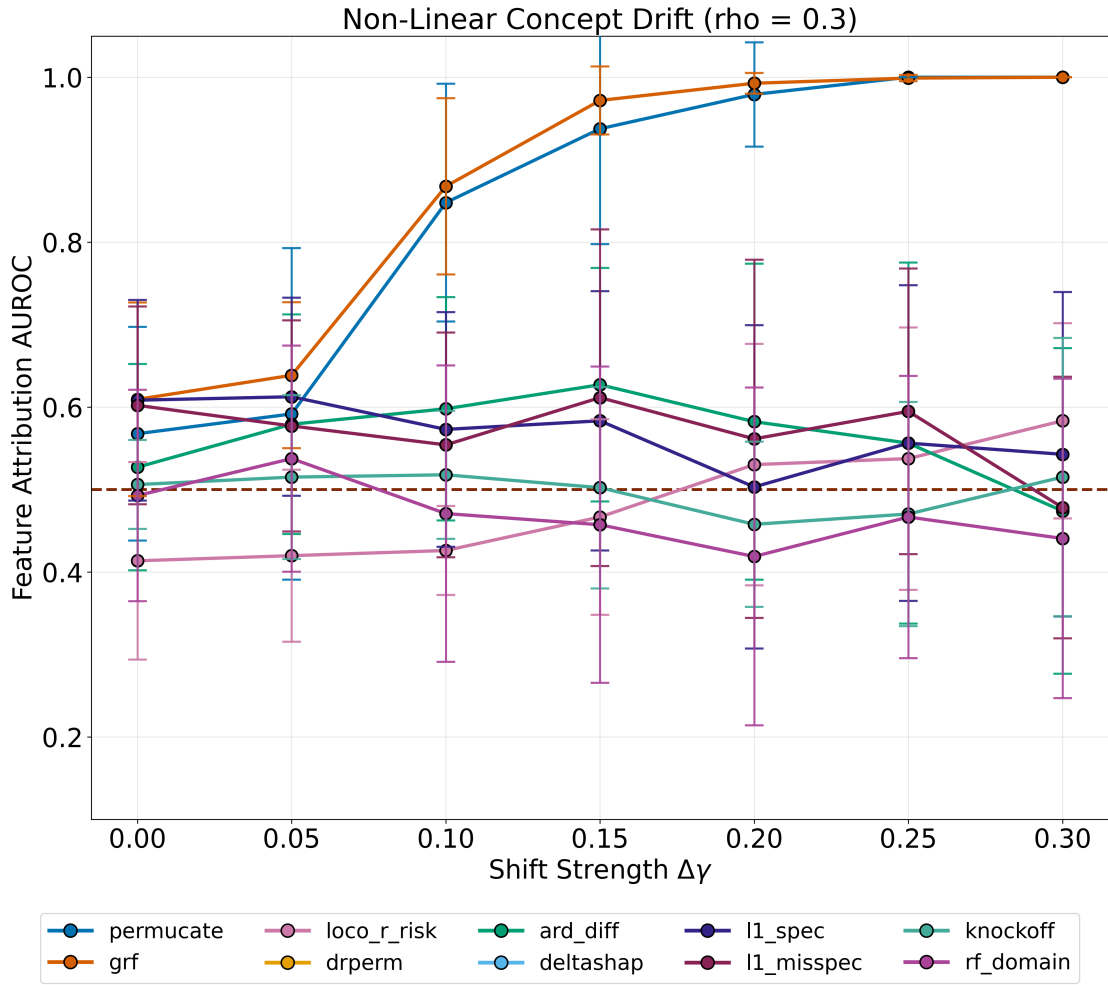


Figure 9: Feature Selection for the nonlinear Concept Drift

Table 13: Concept hop path (rf), cut at B_4 . Next-batch MSE. The cut hop is unforecastable; annealing pays on the hop after.

t	test	uniform	DRE	rfperm	drop-old	oracle
3	$B_4 \leftarrow \text{cut}$	5.054	5.051	5.054	5.054	5.054
4	B_5	2.149	2.254	1.556	0.755	0.755
5	B_6	0.503	0.506	0.503	0.503	0.619

Table 14: PO-risk vs density-ratio (xgb). Same last-two rows. $\Delta = \text{DRE} - \text{rfperm}$ (MSE $\downarrow$; positive means $\sqrt{\text{PO}}$ wins).

Scene	uniform	DRE	rfperm	Δ vs DRE
similar	0.396	0.393	0.396	-0.003
covariate	0.511	0.586	0.509	0.078
concept	1.639	1.651	1.592	0.059
mixed	1.957	2.132	1.920	0.211

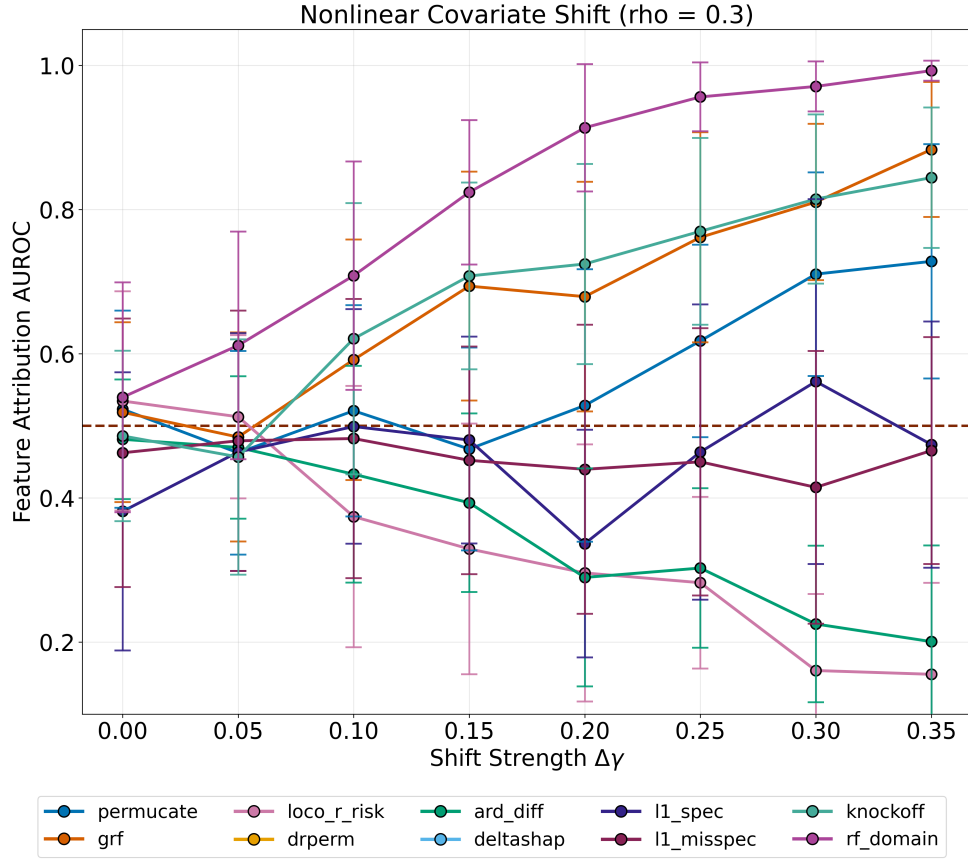


Figure 10: Feature Selection for the nonlinear Covariate Shift

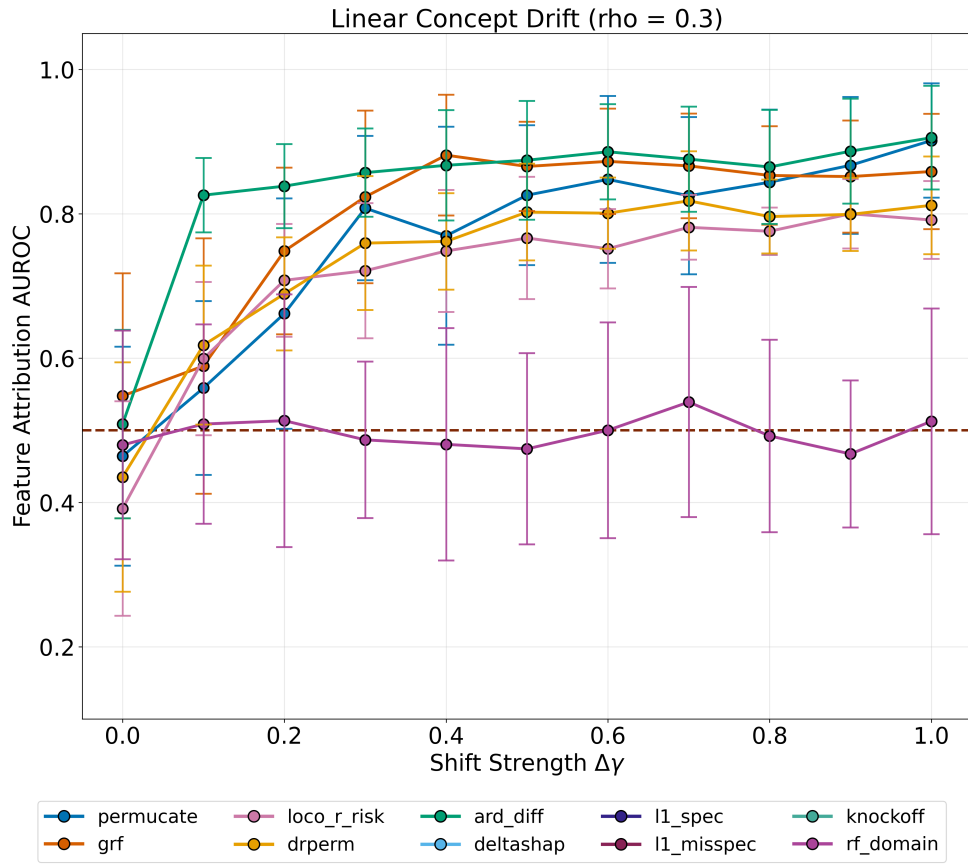


Figure 11: Feature Selection for the Linear Concept Drift

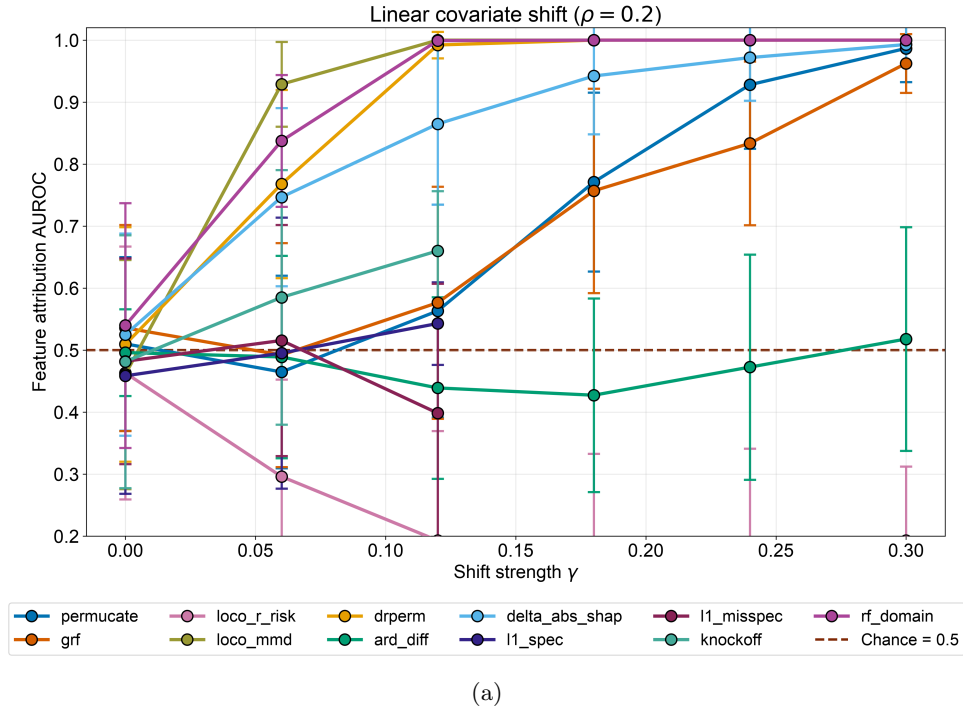
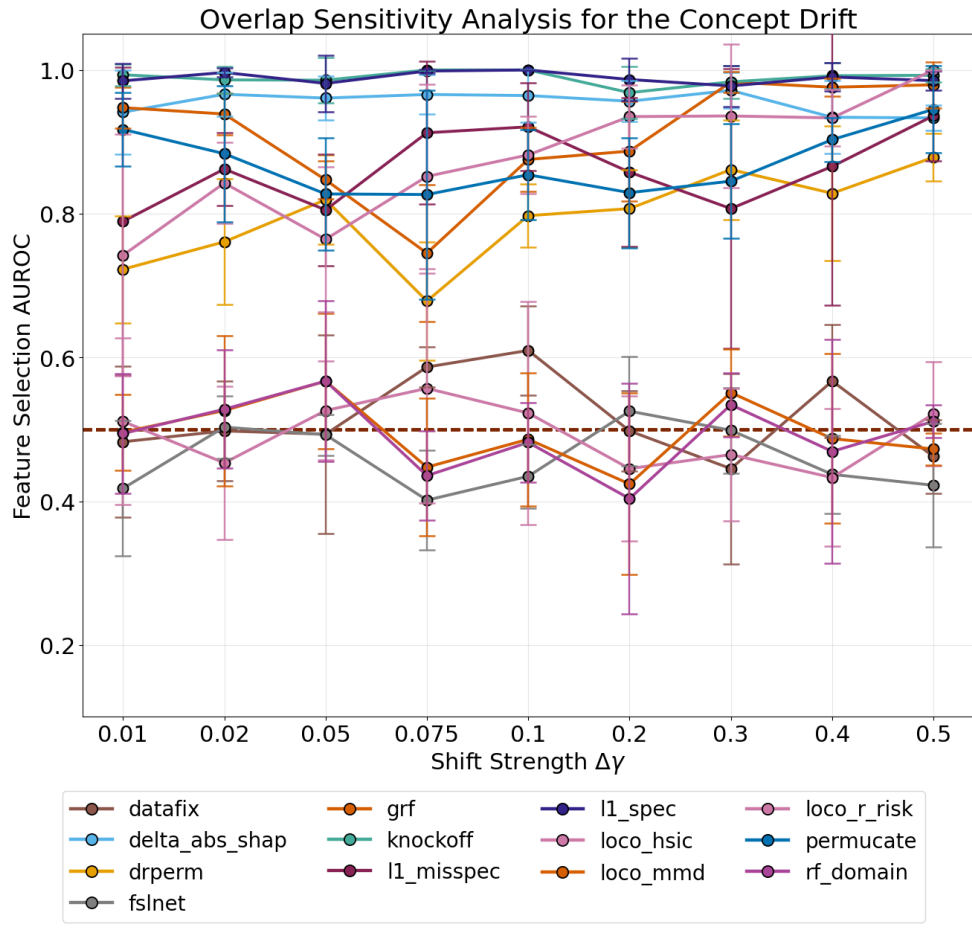
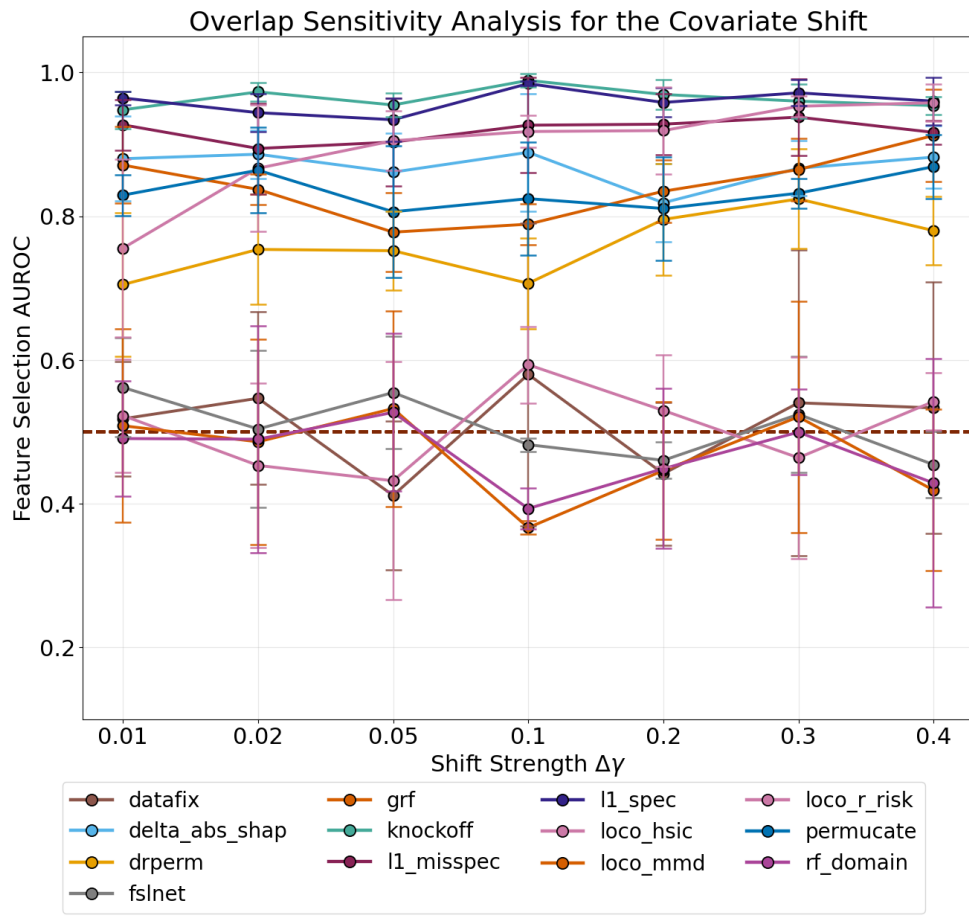


Figure 12: Feature Selection for the linear covariate shift



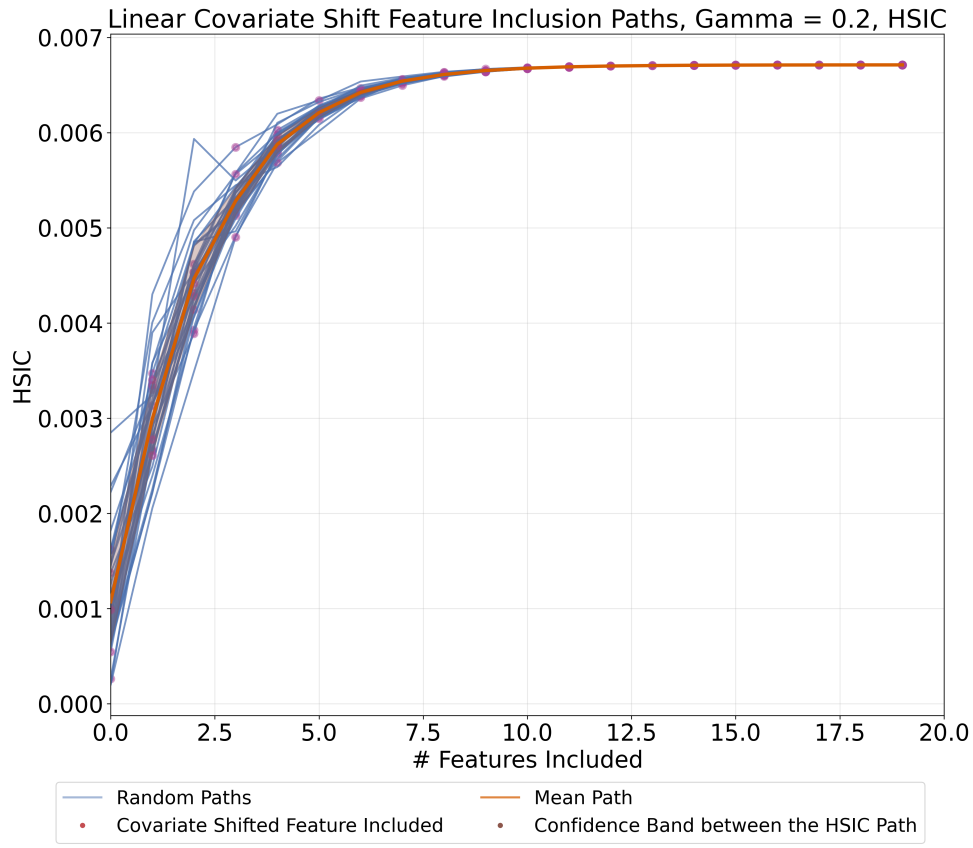
(a)

Figure 13: Overlap Sensitivity Analysis for Concept Drift



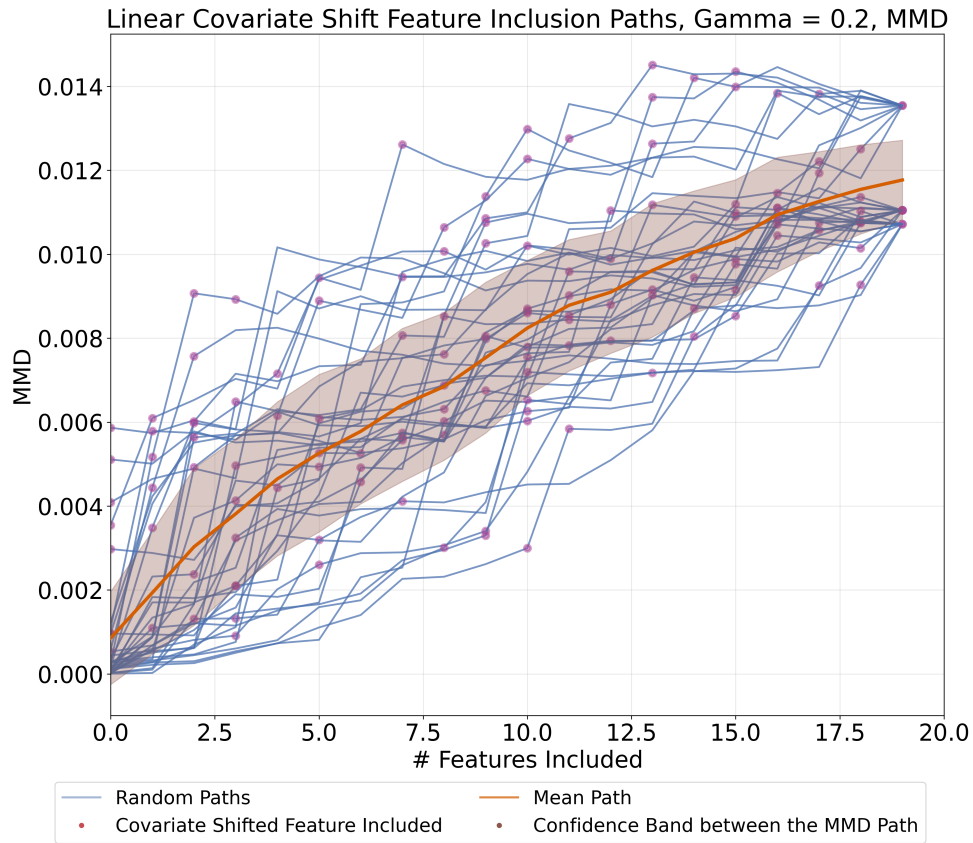
(a)

Figure 14: Overlap Sensitivity Analysis for Covariate Shift



(a)

Figure 15: Feature Path for the HSIC measure for Covariate Shift, the feature inclusion paths are permuted and the the incremental HSIC is recorded(the first 4 features are those with covariate shift)



(a)

Figure 16: Feature Path for the MMD measure for Covariate Shift, the feature inclusion paths are permuted and the the incremental MMD is recorded(the first 4 features are those with covariate shift)

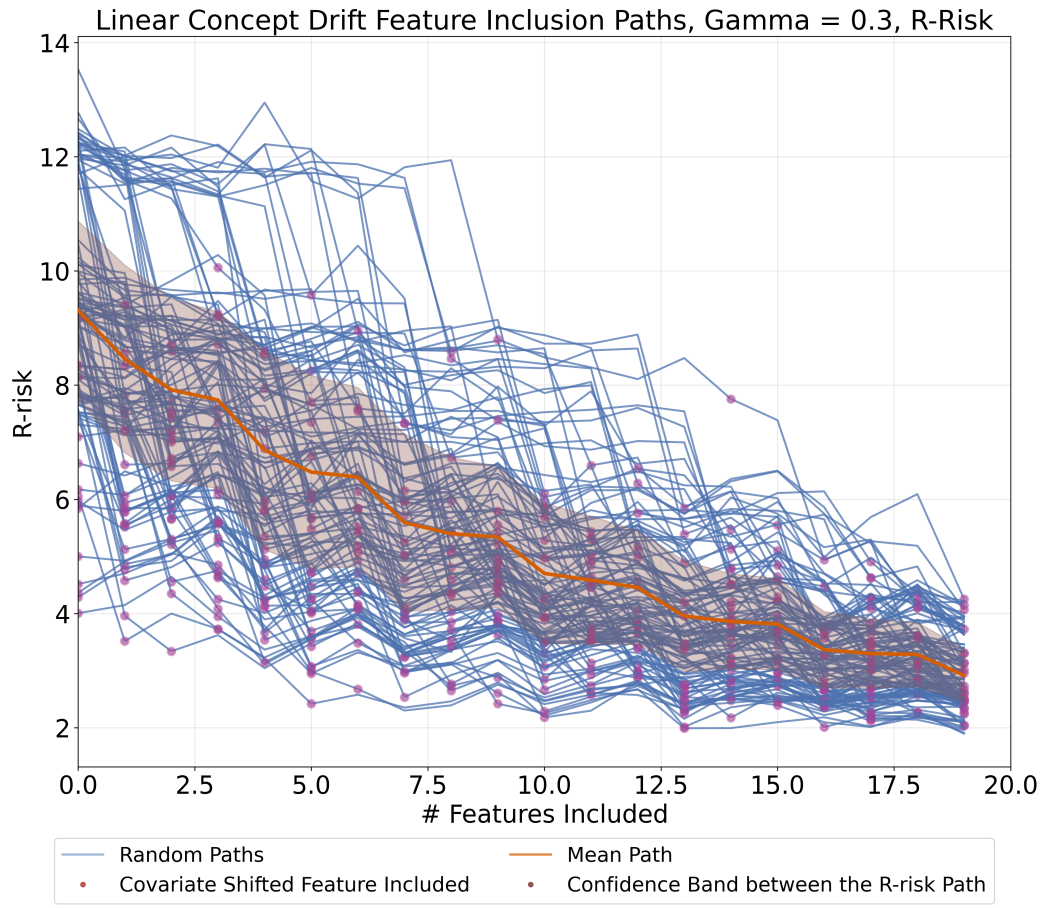
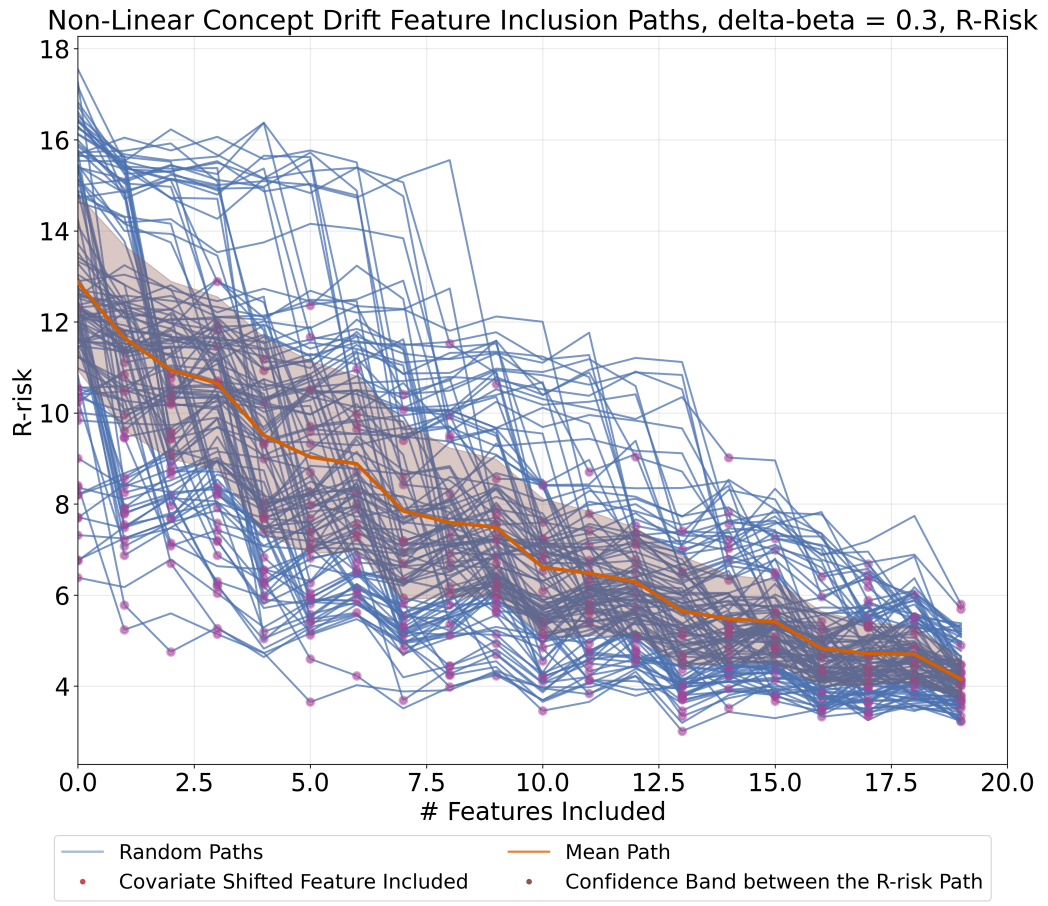


Figure 17: Feature Path for the R-risk measure for Linear Concept Drift, the feature inclusion paths are permuted and the R-risk is recorded(the first 4 features are those with concept drift)



(a)

Figure 18: Feature Path for the R-risk measure for Nonlinear Concept Drift, the feature inclusion paths are permuted and the R-risk is recorded (the first 4 features are those with concept drift)

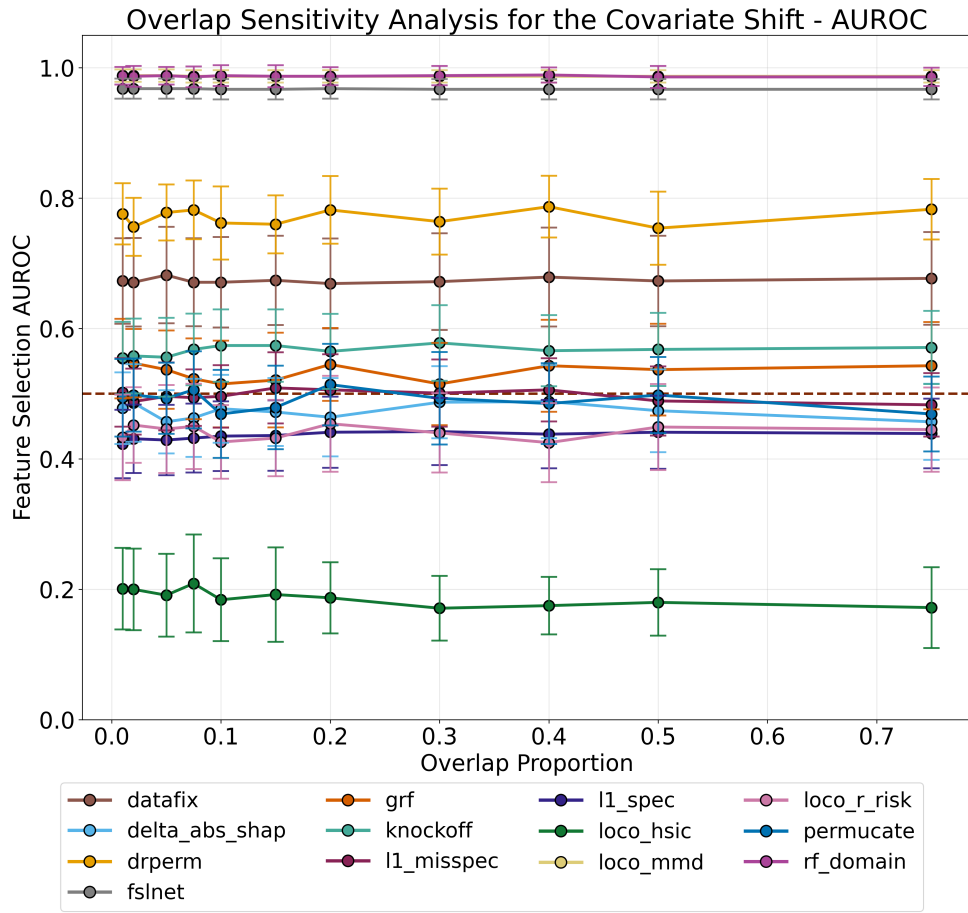


Figure 19: AUC vs. overlap sensitivity for the ChronoBerg Dataset with the Synthetic Covariate Shift

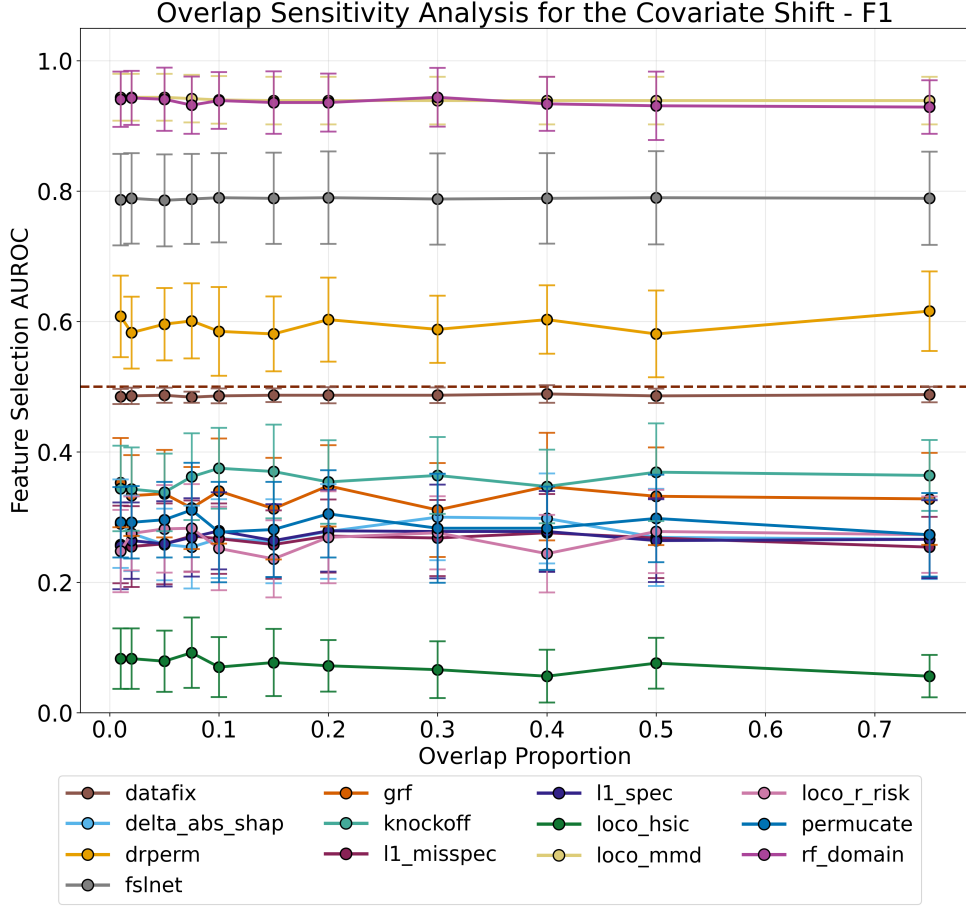


Figure 20: F1 vs. overlap sensitivity for the ChronoBerg Dataset with the Synthetic Covariate Shift

Table 15: Summary of the datasets we used - scales, feature types and the detailed descriptions

	0	1	2	3	4	5	6	7
dataset	dfadult	dfhouse16H	dfelevators	dfpol	dfcpu	dfkin	dfAil	dfbank32nh
nsamples	48842	22784	16599	15000	8192	8192	13750	8192
nfeatures	13	16	18	48	21	8	40	32
ncontinuousfeatures	0	16	18	26	21	8	40	32
ndiscretenumericfeatures	5	0	0	22	0	0	0	0
ncategoricalfeatures	8	0	0	0	0	0	0	0
featuremix	mixed	all continuous	all continuous	mixed	all continuous	all continuous	all continuous	all continuous
response	Y	Y	Goal	Y	usr	y	goal	rej
responsetype	continuous	continuous	continuous	discrete	continuous	continuous	continuous	continuous
responselevels	28523	2045	61	11	56	8191	35	6284

Algorithm. At online step t , AGOD treats the modality-specific gap as a clever covariate rather than a post-hoc diagnostic: given reference batch $\mathcal{D}_0$ and stream batch $\mathcal{D}_t$, it estimates $g_m^{(t)} = \text{Normalize}(\text{AUC}_m^{(t)} \cdot \text{VIMP}_m^{(t)} + \gamma \cdot \text{PO-risk}_m^{(t)})$ for $m \in \{\text{audio, image, text}\}$ via an RF domain classifier and a CFPerm meta-learner (with RF-Domain-only fallback when overlap < 0.3), maps $\alpha^{(t)} = \text{Softmax}(\mathbf{g}^{(t)}/\tau)$ through a gated EMA smoother, and updates the student by $\mathcal{L}^{(t)} = \mathcal{L}_{\text{global}}^{(t)} + \sum_m \alpha_m^{(t)} \mathcal{L}_m^{(t)} + \lambda \Omega(\alpha^{(t)})$. **Opportunities.** The control signal already routes capacity toward concept drift, but task gains remain small; the natural next steps are (i) replacing the linear teacher/student with frozen CLIP/CLAP/EmbeddingGemma on true ChronoBerg windows, (ii) allocating *update budget* (steps or tokens) by α with a global-loss floor so stable modalities are not starved, (iii) calibrating γ, τ, λ and using residualized CATE heterogeneity so strong covariate shift does not leak PO-risk across views, and (iv) extending alignment from modality-level MSE to slice/patch-token and sample-wise weights, evaluated with multi-seed retrieval, forgetting curves, and FLOPs when only one modality drifts.

16.1 From FSDS attribution to the next learning round.

We leverage the MSRVT (Microsoft Research Video Dataset, [84]) to illustrate how our modality attribution methodology via the **FSDS** procedure can guide the next stage of training. We leverage the clap [83] model for audio embedding,

Table 16: Concept Drift Injection Comparison Procedures for feature selection, aggregation across 8 real-world datasets - F1 & AUC recorded, FGSM50 denotes that 50% of the features are shifted,

method	FGSM30AUC	FGSM30F1	covariateshiftAUC	covariateshiftF1	FGSM50AUC	FGSM50F1
grf	0.883	0.756	0.82	0.682	0.753	0.712
l1misspec	0.835	0.705	0.817	0.675	0.682	0.673
l1spec	0.815	0.668	0.822	0.693	0.682	0.6245
locorrisk	0.827	0.685	0.764	0.592	0.726	0.7
permucate	0.892	0.727	0.785	0.651	0.772	0.734
rfdomain	0.572	0.378	0.896	0.87	0.762	0.734
locommd	0.497	0.336	0.933	0.877	0.713	0.679
deltashap	0.658	0.475	0.733	0.695	0.688	0.621
knockoff	0.796	0.645	0.762	0.715	0.713	0.658
datafix	0.881	0.757	0.866	0.708	0.854	0.768
fslnet	0.866	0.665	0.846	0.732	0.633	0.497
ard	0.703	0.754	0.728	0.726	0.756	0.638

Table 17: Nonlinear Concept Drift for the real-data applications - average across different datasets

	avgauc	avgF1	sdauc	sdF1
deltashap	0.77	0.6151	0.1656	0.1571
grf	0.8085	0.6731	0.1503	0.2155
knockoff	0.7379	0.6781	0.1898	0.258
l1misspec	0.7942	0.7457	0.1803	0.2087
l1spec	0.8019	0.7493	0.1929	0.2146
locommd	0.7612	0.5876	0.134	0.1725
locorrisk	0.6429	0.5683	0.2443	0.2577
permucate	0.7944	0.7096	0.1461	0.2056
rfdomain	0.7568	0.588	0.1027	0.1361
datafix	0.721	0.546	0.194	0.259
fslnet	0.783	0.597	0.185	0.296
ard	0.7675	0.7265	0.176	0.1672

ViViT([5]) for video embedding and the GPT2([61]) for the text(caption) embedding in the continuously batches in the video - there will definitely be some missing regions(not necessarily each window has tokens) - which we regard as the missing tokens - a special one via the pooling in the sliding windows. Then we adapt the FSDS procedure on this dataset via the temporal split - with the 40 resampling procedures adapted. The results are shown in Figure 4. Modality-specific FSDS on the within-video early-versus-late split W recovers the ranking implied by the application: *vision carries the temporal shift, audio is a genuine second channel, and clip-level text is a structural null* (Table 48). RF-Domain, coordinate-MMD, and PO-risk VIMP agree on the L1 shares ordering $\pi_v \gg \pi_a \gg \pi_t \approx 0$. A Friedman test of equal per-video RF shares rejects, and Holm-adjusted Wilcoxon tests reject all three pairwise equalities. This is the expected logic of a video clip: scenes and objects move across windows, and speech and soundtrack evolve with them, whereas a single caption is constant in t and therefore cannot explain $P(X | W)$. The text zero is structural—window-invariant embeddings on this extract—not a claim that language is useless for retrieval; audio is not noise, and some videos are even audio-dominant. Signed pooled d cancels across videos, indicating that the shift is heterogeneous rather than a

Table 18: Average Performance for the Adversarial Perturbation - alternative methods, mainly focus on covariate shift.

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.58	0.247	0.601	0.079
1	grf	0.819	0.174	0.859	0.174
2	knockoff	0.564	0.174	0.566	0.106
3	l1misspec	0.613	0.183	0.649	0.149
4	l1spec	0.597	0.152	0.626	0.126
5	locommd	0.868	0.187	0.883	0.183
6	locorrisk	0.366	0.162	0.346	0.146
7	permucate	0.817	0.174	0.839	0.068
8	rfdomain	0.888	0.148	0.936	0.129
9	fslnet	0.879	0.173	0.935	0.186
10	datafix	0.862	0.279	0.918	0.262
11	rfoob	0.901	0.178	0.958	0.095
12	xgboob	0.802	0.258	0.899	0.141
13	mlpoob	0.641	0.36	0.739	0.32
14	ard	0.844	0.075	0.789	0.141

Table 19: Adversarial for the covariate shift - 50% features attacked

	method	shifttype	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	cs	0.631	0.212	0.7	0.235
1	drperm	cs	0.656	0.201	0.689	0.251
2	grf	cs	0.74	0.13	0.797	0.171
3	knockoff	cs	0.644	0.228	0.651	0.233
4	l1misspec	cs	0.555	0.115	0.531	0.147
5	l1spec	cs	0.58	0.266	0.606	0.258
6	locommd	cs	0.704	0.243	0.713	0.258
7	locorrisk	cs	0.526	0.146	0.551	0.172
8	permucate	cs	0.745	0.229	0.802	0.186
9	rfdomain	cs	0.85	0.168	0.914	0.152
10	datafix	cs	0.757	0.368	0.854	0.194
11	fslnet	cs	0.711	0.212	0.866	0.259
12	ard	cs	0.656	0.199	0.66	0.253

single global direction—precisely the condition required for the next round of streaming updates to be video-clustered rather than pooled. To summarize the results for the modality specific contribution matches our expectations perfectly.

The modality-specific FSDS analysis is not only a retrospective decomposition of covariate shift $P(X | W)$: it supplies an operational prior for the subsequent online-learning (OL) round. Across RF-Domain, coordinate-MMD, and PO-risk VIMP, the within-video early-versus-late window shift is carried primarily by the visual block ($\pi_v \approx 0.77\text{--}0.90$), with a nontrivial audio contribution ($\pi_a \approx 0.10\text{--}0.19$) and a near-null text share (Table ??). A Friedman test on per-video RF

Table 20: Adversarial for the Case where Concept Drift Dominated - Whyshift Dataset

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.702	0.095	0.763	0.763
1	drperm	0.519	0.056	0.525	0.525
2	grf	0.548	0.122	0.597	0.597
3	knockoff	0.518	0.064	0.52	0.52
4	l1misspec	0.457	0.088	0.47	0.47
5	l1spec	0.373	0.103	0.383	0.383
6	locommd	0.21	0.067	0.188	0.188
7	locorrisk	0.343	0.058	0.293	0.293
8	permucate	0.634	0.091	0.666	0.173
9	rfdomain	0.698	0.101	0.732	0.732
10	fslnet	0.416	0.078	0.541	0.095
11	datafix	0.436	0.059	0.539	0.111
12	ard	0.574	0.061	0.593	0.126

Table 21: Adversarial perturbation - 50% features attacked, mainly covariate shift, minor concept drift

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.639	0.096	0.708	0.107
1	grf	0.703	0.186	0.735	0.188
2	knockoff	0.613	0.115	0.66	0.138
3	l1misspec	0.625	0.174	0.629	0.174
4	l1spec	0.619	0.186	0.655	0.158
5	locommd	0.741	0.228	0.796	0.273
6	locorrisk	0.496	0.124	0.516	0.097
7	permucate	0.72	0.231	0.752	0.271
8	rfdomain	0.732	0.208	0.754	0.189
9	datafix	0.709	0.353	0.824	0.187
10	fslnet	0.658	0.21	0.722	0.251
11	ard	0.56	0.253	0.65	0.199

shares rejects equal modality contributions ($p = 2.87 \times 10^{-7}$), and Holm-adjusted Wilcoxon tests reject all three pairwise equalities at $p < 10^{-4}$ (Table ??). Region-wise Cohen’s d reproduces the same ranking (mean $|d|$ of 0.787/0.518/0.000 for video/audio/text). Signed pooled d cancels across videos (0/24 Holm hits at the pooled-region level), so the shift is *video-heterogeneous* rather than a single global direction—which is the quantity a pooled OL gradient would otherwise average away.

These facts constrain the next OL protocol in three concrete ways. (i) *Where to adapt*. Update capacity and step sizes should be concentrated on the visual encoder and, secondarily, the audio encoder. Clip-level caption embeddings are

Table 22: Adversarial Perturbation: FGSM + Concept Drift with Concept Drift Dominated

	method	avgF1	stdF1	avgAUC	stdAUC
0	deltashap	0.483	0.15	0.696	0.172
1	drperm	0.525	0.248	0.693	0.063
2	grf	0.455	0.198	0.708	0.118
3	knockoff	0.511	0.265	0.703	0.086
4	l1misspec	0.565	0.309	0.725	0.093
5	l1spec	0.582	0.303	0.714	0.084
6	locommd	0.417	0.279	0.647	0.087
7	locorrisk	0.435	0.13	0.556	0.112
8	permucate	0.534	0.164	0.732	0.178
9	rfdomain	0.405	0.387	0.685	0.107
10	fslnet	0.392	0.138	0.578	0.092
11	datafix	0.317	0.104	0.529	0.106
12	ard	0.512	0.186	0.728	0.088

Table 23: FMoW dataset benchmark, covariate shift dominated

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.3115	0.0865	0.4992	0.0847
1	drperm	0.3881	0.171	0.5857	0.1543
2	grf	0.2462	0.1156	0.4442	0.1119
3	knockoff	0.4035	0.141	0.6247	0.1033
4	l1misspec	0.4244	0.137	0.6256	0.1071
5	l1spec	0.4027	0.1431	0.5921	0.1225
6	locommd	0.7029	0.2823	0.8222	0.2158
7	locorrisk	0.2762	0.0967	0.4619	0.0925
8	permucate	0.3029	0.0981	0.5039	0.1035
9	rfdomain	0.5426	0.1854	0.8125	0.1625
10	fslnet	0.5625	0.2469	0.8096	0.139
11	datafix	0.4273	0.1736	0.6822	0.1195

window-invariant on this extract, hence a *structural* zero for W -shift rather than evidence that language is uninformative for retrieval; a time-aligned ASR stream would be required before attributing residual W -shift to text. (ii) *How to monitor*. Drift triggers should be video-clustered—the same sampling unit as the bootstrap and Friedman tests. A global MMD or pooled d will miss the heterogeneous $|d|$ that FSDS detects. (iii) *How to fuse*. The RF shares provide a

Table 24: ChronoBerg Dataset, average over the many comparisons with all types of shifts - covariate shifts dominates

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.2666	0.1076	0.488	0.1012
1	drperm	0.4963	0.184	0.6924	0.1527
2	grf	0.3667	0.168	0.5485	0.1445
3	knockoff	0.3407	0.1303	0.5792	0.1294
4	l1misspec	0.2296	0.1127	0.503	0.1163
5	l1spec	0.2444	0.1182	0.4586	0.1011
6	locommd	0.6963	0.3561	0.7577	0.321
7	locorrisk	0.3	0.1373	0.4762	0.1373
8	permucate	0.3222	0.125	0.5443	0.1151
9	rfdomain	0.8056	0.319	0.7073	0.2665
10	fslnet	0.8169	0.1783	0.6772	0.2317
11	datafix	0.7229	0.1791	0.6289	0.2734
12	ard	0.4382	0.1527	0.5367	0.0534

hot-start for modality gates, with a documented per-video exception (video 13 is audio-dominant; Table ??) that argues against a frozen global fusion weight. Until a window-aligned text encoder is in the stream, a rising text VIMP in OL is a diagnostic for leakage, not a reason to retune the language tower against W . In short, the offline FSDS readout is the design matrix for the next round of streaming results: adapt vision and audio, cluster by video, and treat clip-level text as a negative control.

16.2 Application on the Continuous Inference for the LLM

What FSDS justifies in continuous multimodal inference. Continuous inference for a video-audio-language model is a streaming decision problem: at each window t the system must choose which tokens to re-encode, which KV entries to invalidate, which towers to adapt, and which evidence to write into the belief state. FSDS does not attribute next-token gradients of a frozen LLM; it attributes nonstationarity in the *input measure* $P(X | W)$ on the same 768/512/768 blocks that become those tokens (Table 48). That is the right object for scheduling continuous inference. If a modality is stationary across W , re-encoding it every window is wasted compute and a shortcut the language head can latch onto. If it is nonstationary, the KV cache and the reasoner’s working memory are stale unless that stream is refreshed. Concretely, the estimated shares $\pi_v \approx 0.77\text{--}0.90$, $\pi_a \approx 0.10\text{--}0.19$, $\pi_t \approx 0$ imply four inference-time controls. (i) *Encode-once versus encode-every-window*. Clip-level text is window-invariant on this extract (mean $|d| = 0$), so caption tokens should be encoded once per clip and reused. Visual tokens, and secondarily audio tokens, must be re-encoded on each window: that is where $P(X | W)$ actually moves. (ii) *KV invalidation and memory writes*. A streaming VLM should drop or compress visual (and audio) KV when region-wise $|d|$ or RF-Domain AUC on W crosses a video-clustered threshold, and should *not* refresh caption KV on the same trigger. A global pooled d is the wrong trigger: signed region effects cancel across videos, so a session-level (video-clustered) monitor matches the sampling unit of the Friedman/Wilcoxon tests. (iii) *Test-time adaptation mask*. If continuous inference includes a test-time LoRA or gated residual, the update support should be the vision tower and fusion, then audio, not the language tower against W . A rising text VIMP mid-stream is a leakage diagnostic (the model is fitting the static caption), not a reason to spend test-time compute on text. (iv) *Belief-state and chain-of-thought evidence*. For streaming VQA or long-horizon reasoning (“what is happening *now*”), FSDS ranks the evidence that is allowed to revise the belief: visual first, audio second, clip-level subtitle never—unless a window-aligned ASR stream is substituted and re-tested. Video 13 being audio-dominant further implies that fusion gates must be session-adaptive rather than a frozen (π_v, π_a, π_t) . The justified claim is therefore operational, not metaphysical: FSDS supplies the *nonstationarity budget* of the multimodal stream. Continuous inference should spend that budget on vision and audio, keep clip-level text as a control variate, and trigger refreshes at video granularity. What FSDS does *not* yet justify is a claim about which LLM layers implement that

Table 25: Whyshift Dataset, average over the many comparisons with all types of shifts - covariate shift & Concept Drift

	method	avgF1	sdF1	avgAUC	sdAUC
0	deltashap	0.7081	0.1273	0.7779	0.0883
1	drperm	0.5513	0.0678	0.555	0.0424
2	grf	0.4919	0.1169	0.5382	0.1309
3	knockoff	0.5189	0.0483	0.5636	0.0604
4	l1misspec	0.3784	0.0427	0.4221	0.0393
5	l1spec	0.3568	0.082	0.3544	0.1024
6	locommd	0.2486	0.0353	0.2245	0.0389
7	locorrisk	0.3622	0.0491	0.3215	0.0213
8	permucate	0.6703	0.0673	0.7091	0.076
9	rfdomain	0.7297	0.0689	0.7561	0.0483
10	fslnet	0.4159	0.1863	0.5388	0.095
11	datafix	0.5731	0.0301	0.5414	0.111
12	rfoob	0.6843	0.1165	0.7251	0.1174
13	mlpoob	0.6282	0.1578	0.6095	0.0143
14	ard	0.4571	0.1741	0.5361	0.1239

Table 26: Time Profiling Comparison w.r.t. Sample-sizes

n	drpermmean	drpermstd	grfmean	grfst	knockoffmean	knockoffstd	l1misspecmean	l1misspecstd	locommean	locostd	mmdmean	mmdstd	permucatemean	permucatestd	rfmean	rfstd	shapmean	shapstd	walltotalmean	walltotalstd	locorriskmean	locorriskstd
200	21.108	3.965	23.831	12.755	9.693	1.419	5.71	0.81	3.69	0.39	1.115	0.315	32.077	10.723	1.267	0.829	3.469	1.263	92.155	23.294	3.106	2.341
300	34.798	6.308	4.371	1.214	11.992	1.648	11.59	0.82	14.2	2.2	2.148	0.613	52.135	9.848	0.823	0.334	3.78	0.864	112.727	15.816	12.684	3.835
500	35.92	4.886	9.408	1.325	25.587	11.257	11.59	1.82	12.48	2.84	5.465	1.093	52.092	7.098	0.95	0.203	5.299	0.842	128.329	12.327	16.412	2.934
750	46.4	4.691	10.887	1.391	38.117	5.116	10.86	1.36	22.8	3.74	9.128	1.225	48.578	7.501	1.027	0.208	6.823	1.496	144.457	14.339	18.834	4.072
1000	55.497	8.733	14.071	3.503	35.059	8.77	26.66	4.99	34.21	6.84	16.634	2.287	60.771	10.703	0.948	0.229	8.412	1.117	182.075	24.771	22.037	5.097
1500	100.609	14.274	24.966	5.693	22.16	2.3	35.19	4.02	38.23	6.4	21.006	4.923	94.789	16.942	1.766	0.464	16.223	3.114	302.051	45.736	35.085	7.408
2000	93.15	31.83	21.53	6.604	36.444	37.504	52.01	11.48	56.58	16.31	12.781	4.452	75.932	23.157	1.349	0.421	14.678	5.892	274.069	73.085	28.277	10.795
3000	69.154	41.886	43.589	11.409	256.954	102.464	67.37	14.15	75.01	16.31	14.694	4.727	145.383	42.229	2.87	0.975	29.649	9.653	689.306	171.061	18.002	11.802

budget; that is the next-round streaming result the shares are designed to prioritize.

To summary, this attribution can serve as the guidance of the next round/stage continuous training procedure. Another opportunity lies in the fact that this FSDS procedure can enhance the continuous time LLM inference - the **Attribution-Guided Online Distillation (AGOD)**. At online step t , AGOD treats the modality-specific gap as a clever covariate rather than a post-hoc diagnostic: given reference batch $\mathcal{D}_0$ and stream batch $\mathcal{D}_t$, it estimates $g_m^{(t)} = \text{Normalize}(\text{AUC}_m^{(t)} \cdot \text{VIMP}_m^{(t)} + \gamma \cdot \text{PO-risk}_m^{(t)})$ for $m \in \{\text{audio, image, text}\}$ via an RF domain classifier and a CFPerm meta-learner (with RF-Domain-only fallback when overlap < 0.3), maps $\alpha^{(t)} = \text{Softmax}(\mathbf{g}^{(t)}/\tau)$ through a gated EMA smoother, and updates the student by $\mathcal{L}^{(t)} = \mathcal{L}_{\text{global}}^{(t)} + \sum_m \alpha_m^{(t)} \mathcal{L}_m^{(t)} + \lambda \Omega(\alpha^{(t)})$. **Opportunities.** The control signal already routes capacity toward concept drift, but task gains remain small; the natural next steps are (i) replacing the linear teacher/student with frozen CLIP/CLAP/EmbeddingGemma on true ChronoBerg windows, (ii) allocating *update budget* (steps or tokens) by α with a global-loss floor so stable modalities are not starved, (iii) calibrating γ, τ, λ and using residualized CATE heterogeneity so strong covariate shift does not leak PO-risk across views, and (iv) extending alignment from modality-level MSE to slice/patch-token and sample-wise weights, evaluated with multi-seed retrieval, forgetting curves, and FLOPs when only one modality drifts.

Table 27: Time Profiling Comparison w.r.t. Sample-sizes - with neural network based methods, n = 2000

p	20	50	150	200
datafixmean	37.71	54.61	70.62	101.93
datafixsd	11.11	18.3	22.49	27.93
rfoobmean	10.57	25.68	46.87	93.67
rfoobsd	1.65	9.1	22.1	41.38
fslnetmean	9.02	15.83	24.11	59.87
fslnetsd	3.33	4.13	7.64	26.95

Table 28: Overlap Sensitivity for Concept Drift

overlap	method	nreps	F1mean	F1std	precisionmean	precisionstd	recallmean	recallstd	selectionAUCmean	selectionAUCstd
0	0.01	datafix	2	0.0	0.0	0.0	0.0	0.0	0.564	0.038
13	0.1	datafix	2	0.0	0.0	0.0	0.0	0.0	0.58	0.211
26	0.4	datafix	2	0.0	0.0	0.0	0.0	0.0	0.567	0.292
1	0.01	deltashap	2	0.8	0.094	0.8	0.094	0.8	0.094	0.862
14	0.1	deltashap	2	0.767	0.141	0.767	0.141	0.767	0.141	0.889
27	0.4	deltashap	2	0.833	0.047	0.833	0.047	0.833	0.047	0.9
2	0.01	drperm	2	0.667	0.094	0.667	0.094	0.667	0.094	0.749
15	0.1	drperm	2	0.633	0.047	0.633	0.047	0.633	0.047	0.707
28	0.4	drperm	2	0.733	0.0	0.733	0.0	0.733	0.0	0.744
3	0.01	fslnet	2	0.059	0.083	0.25	0.354	0.033	0.047	0.584
16	0.1	fslnet	2	0.111	0.157	0.333	0.471	0.067	0.094	0.482
29	0.4	fslnet	2	0.0	0.0	0.0	0.0	0.0	0.464	0.053
4	0.01	grf	2	0.767	0.047	0.767	0.047	0.767	0.047	0.869
17	0.1	grf	2	0.733	0.094	0.733	0.094	0.733	0.094	0.789
30	0.4	grf	2	0.767	0.047	0.767	0.047	0.767	0.047	0.896
5	0.01	knockoff	2	0.9	0.047	0.9	0.047	0.9	0.047	0.93
18	0.1	knockoff	2	0.933	0.0	0.933	0.0	0.933	0.0	0.989
31	0.4	knockoff	2	0.933	0.0	0.933	0.0	0.933	0.0	0.948
6	0.01	llmisspec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.916
19	0.1	llmisspec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.927
32	0.4	llmisspec	2	0.867	0.0	0.867	0.0	0.867	0.0	0.902
7	0.01	llspec	2	0.9	0.047	0.9	0.047	0.9	0.047	0.967
20	0.1	llspec	2	0.933	0.0	0.933	0.0	0.933	0.0	0.984
33	0.4	llspec	2	0.933	0.0	0.933	0.0	0.933	0.0	0.962
8	0.01	locohsic	2	0.5	0.047	0.5	0.047	0.5	0.047	0.489
21	0.1	locohsic	2	0.533	0.0	0.533	0.0	0.533	0.0	0.593
34	0.4	locohsic	2	0.533	0.094	0.533	0.094	0.533	0.094	0.576
9	0.01	locommd	2	0.5	0.141	0.5	0.141	0.5	0.141	0.513
22	0.1	locommd	2	0.433	0.047	0.433	0.047	0.433	0.047	0.367
35	0.4	locommd	2	0.4	0.0	0.4	0.0	0.4	0.0	0.418
10	0.01	locorrisk	2	0.7	0.047	0.7	0.047	0.7	0.047	0.802
23	0.1	locorrisk	2	0.867	0.0	0.867	0.0	0.867	0.0	0.918
36	0.4	locorrisk	2	0.933	0.0	0.933	0.0	0.933	0.0	0.96
11	0.01	permucate	2	0.733	0.094	0.733	0.094	0.733	0.094	0.824
24	0.1	permucate	2	0.767	0.141	0.767	0.141	0.767	0.141	0.824
37	0.4	permucate	2	0.767	0.047	0.767	0.047	0.767	0.047	0.836
12	0.01	rfdomain	2	0.533	0.094	0.533	0.094	0.533	0.094	0.482
25	0.1	rfdomain	2	0.4	0.0	0.4	0.0	0.4	0.0	0.393
38	0.4	rfdomain	2	0.4	0.283	0.4	0.283	0.4	0.283	0.349

Table 29: Sensitivity Analysis of the choice of the propensity score model for the Meta-Learner Based methods in terms of the feature selection efficiency - Synthetic Datasets

	shift	method	model	avgAUC	stdAUC	avgF1	stdF1
0	linearconceptdrift	drperm	logistic	1.0	0.0	1.0	0.0
1	linearconceptdrift	drperm	rf	1.0	0.0	1.0	0.0
2	linearconceptdrift	drperm	xgb	1.0	0.0	1.0	0.0
3	linearconceptdrift	permucate	logistic	0.9969	0.007	0.95	0.1118
4	linearconceptdrift	permucate	rf	0.95	0.1118	0.95	0.1118
5	linearconceptdrift	permucate	xgb	0.9	0.1369	0.9	0.1369
6	linearcovariateshift	drperm	logistic	1.0	0.0	1.0	0.0
7	linearcovariateshift	drperm	rf	1.0	0.0	1.0	0.0
8	linearcovariateshift	drperm	xgb	1.0	0.0	1.0	0.0
9	linearcovariateshift	permucate	logistic	0.4156	0.189	0.35	0.2236
10	linearcovariateshift	permucate	rf	1.0	0.0	1.0	0.0
11	linearcovariateshift	permucate	xgb	0.1437	0.2029	0.1	0.1369
12	nonlinearconceptdrift	drperm	logistic	1.0	0.0	1.0	0.0
13	nonlinearconceptdrift	drperm	rf	1.0	0.0	1.0	0.0
14	nonlinearconceptdrift	drperm	xgb	1.0	0.0	1.0	0.0
15	nonlinearconceptdrift	permucate	logistic	0.9875	0.028	0.95	0.1118
16	nonlinearconceptdrift	permucate	rf	1.0	0.0	1.0	0.0
17	nonlinearconceptdrift	permucate	xgb	0.9187	0.1118	0.9	0.1369
18	nonlinearcovariateshift	drperm	logistic	1.0	0.0	1.0	0.0
19	nonlinearcovariateshift	drperm	rf	1.0	0.0	1.0	0.0
20	nonlinearcovariateshift	drperm	xgb	1.0	0.0	1.0	0.0
21	nonlinearcovariateshift	permucate	logistic	1.0	0.0	1.0	0.0
22	nonlinearcovariateshift	permucate	rf	1.0	0.0	1.0	0.0
23	nonlinearcovariateshift	permucate	xgb	0.6219	0.1823	0.6	0.2236

Table 30: Sensitivity Analysis of the choice of the propensity score model for the Meta-Learner Based methods in terms of the feature selection efficiency - Real-Datasets

	scenario	method	estimator	avgAUC	stdAUC	avgF1	stdF1
0	chronoberg	drperm	logistic	0.4633	0.0802	0.3022	0.099
1	chronoberg	drperm	rf	0.8027	0.0988	0.6222	0.1283
2	chronoberg	drperm	xgb	0.6635	0.0644	0.5067	0.0854
3	chronoberg	permucate	logistic	0.5173	0.1281	0.3422	0.1395
4	chronoberg	permucate	rf	0.4686	0.1103	0.2666	0.0962
5	chronoberg	permucate	xgb	0.495	0.0807	0.2755	0.0793
6	tabular	drperm	logistic	0.5894	0.2998	0.5545	0.2864
7	tabular	drperm	rf	0.6877	0.2853	0.6536	0.2754
8	tabular	drperm	xgb	0.6824	0.2684	0.6458	0.2653
9	tabular	permucate	logistic	0.7397	0.276	0.7075	0.2622
10	tabular	permucate	rf	0.7947	0.2262	0.7464	0.236
11	tabular	permucate	xgb	0.7555	0.2654	0.7074	0.2621

17 Attribution as the model performance adaptation

We leverage the following procedure: based on model performance, we then switch back to target the modality that contributes most to the distribution shift—as measured by both MMD and PO-risk—and adjust the learning rate on that sub-model. This further enhances the next-step distribution shift adaptation. We refer to this as the **attribution-**

Table 31: Sensitivity Analysis of the choice of the outcome model the Meta-Learner Based methods in terms of the feature selection efficiency: Synthetic Data

	shift _{type}	method	estimator	avgAUC	stdAUC	avgF1	stdF1
0	jitter	drperm	rf	0.3756	0.1497	0.1667	0.0786
1	jitter	permucate	rf	0.3915	0.1795	0.1667	0.0786
2	linear	drperm	mlp	0.6001	0.2853	0.5723	0.2678
3	linear	drperm	rf	0.6583	0.2877	0.6233	0.267
4	linear	drperm	ridge	0.6568	0.2914	0.6181	0.2748
5	linear	drperm	xgb	0.6744	0.3152	0.621	0.2959
6	linear	permucate	mlp	0.6818	0.3089	0.6317	0.2829
7	linear	permucate	rf	0.7733	0.2397	0.7198	0.2365
8	linear	permucate	ridge	0.7742	0.2634	0.7119	0.2597
9	linear	permucate	xgb	0.7238	0.3028	0.6766	0.2816
10	linearconcept	drperm	mlp	0.9469	0.0506	0.775	0.2189
11	linearconcept	drperm	rf	1.0	0.0	1.0	0.0
12	linearconcept	drperm	ridge	1.0	0.0	1.0	0.0
13	linearconcept	drperm	xgb	1.0	0.0	1.0	0.0
14	linearconcept	permucate	mlp	1.0	0.0	1.0	0.0
15	linearconcept	permucate	rf	0.9953	0.0148	0.975	0.0791
16	linearconcept	permucate	ridge	1.0	0.0	1.0	0.0
17	linearconcept	permucate	xgb	1.0	0.0	1.0	0.0
18	linearcovariate	drperm	mlp	0.9578	0.0683	0.85	0.2415
19	linearcovariate	drperm	rf	1.0	0.0	1.0	0.0
20	linearcovariate	drperm	ridge	1.0	0.0	1.0	0.0
21	linearcovariate	drperm	xgb	1.0	0.0	1.0	0.0
22	linearcovariate	permucate	mlp	0.5	0.1622	0.275	0.1419
23	linearcovariate	permucate	rf	0.4953	0.1828	0.4	0.2415
24	linearcovariate	permucate	ridge	0.5094	0.1485	0.2	0.1581
25	linearcovariate	permucate	xgb	0.586	0.2585	0.475	0.2486
26	miFGSM	drperm	rf	0.4815	0.1715	0.2592	0.1283
27	miFGSM	permucate	rf	0.4656	0.2378	0.2592	0.1697
28	nonlinearconcept	drperm	mlp	1.0	0.0	1.0	0.0
29	nonlinearconcept	drperm	rf	1.0	0.0	1.0	0.0
30	nonlinearconcept	drperm	ridge	1.0	0.0	1.0	0.0
31	nonlinearconcept	drperm	xgb	1.0	0.0	1.0	0.0
32	nonlinearconcept	permucate	mlp	0.8641	0.1527	0.775	0.1845
33	nonlinearconcept	permucate	rf	0.9812	0.0593	0.975	0.0791
34	nonlinearconcept	permucate	ridge	0.8016	0.2105	0.675	0.2372
35	nonlinearconcept	permucate	xgb	1.0	0.0	1.0	0.0
36	nonlinearcovariate	drperm	mlp	0.5798	0.3141	0.5358	0.2882
37	nonlinearcovariate	drperm	rf	0.6081	0.3087	0.5843	0.3036
38	nonlinearcovariate	drperm	ridge	0.6253	0.3132	0.6053	0.3014
39	nonlinearcovariate	drperm	xgb	0.569	0.3304	0.5677	0.3126
40	nonlinearcovariate	permucate	mlp	0.7091	0.255	0.6501	0.272
41	nonlinearcovariate	permucate	rf	0.8234	0.2271	0.7687	0.2376
42	nonlinearcovariate	permucate	ridge	0.6887	0.2594	0.6132	0.2801
43	nonlinearcovariate	permucate	xgb	0.7112	0.2911	0.6765	0.2894
44	nonlinearinteraction	drperm	mlp	0.6462	0.2747	0.6145	0.2623
45	nonlinearinteraction	drperm	rf	0.6653	0.3093	0.6387	0.2927
46	nonlinearinteraction	drperm	ridge	0.6515	0.2895	0.6274	0.2733
47	nonlinearinteraction	drperm	xgb	0.7452	0.3464	0.7063	0.3347
48	nonlinearinteraction	permucate	mlp	0.8017	0.2557	0.7685	0.2425
49	nonlinearinteraction	permucate	rf	0.8754	0.1602	0.8379	0.1858
50	nonlinearinteraction	permucate	ridge	0.8472	0.1648	0.8121	0.1929
51	nonlinearinteraction	permucate	xgb	0.8463	0.1927	0.81	0.2019
52	rFGSM	drperm	rf	0.4074	0.1123	0.2222	0.1571
53	rFGSM	permucate	rf	0.3783	0.0486	0.1111	0.0

as-learning-rate-adaptor procedure. We evaluate the same control plane—MMD² covariate / PO concept sensors, EMA Softmax routing state α , param-group LR actuator, and an optional hard-gate adaptor—on Amazon Reviews,

Table 32: Random Forest Binary Classifier in comparison with other procedures - Boruta, ShadowVIMP and VITA-PIMP for synthetic data and tabular data

	shift _{type}	method	scenario	avgAUC	stdAUC	avgF1	stdF1
14	beta=0.2	boruta	synthetic	0.4305	0.137	0.175	0.1687
15	beta=0.2	dccondperm	synthetic	0.4922	0.1494	0.175	0.1208
16	beta=0.2	dcloco	synthetic	0.5734	0.2035	0.325	0.2372
17	beta=0.2	dcoob	synthetic	0.4625	0.172	0.2	0.1972
18	beta=0.2	rfvanilla	synthetic	0.4531	0.2233	0.175	0.2648
19	beta=0.2	shadowVIMP	synthetic	0.4984	0.157	0.175	0.1687
20	beta=0.2	vitaPIMP	synthetic	0.4625	0.217	0.175	0.2058
21	gamma=0.2	boruta	synthetic	0.9828	0.029	0.9	0.1291
22	gamma=0.2	dccondperm	synthetic	0.8953	0.1192	0.775	0.2486
23	gamma=0.2	dcloco	synthetic	0.8	0.1441	0.6	0.1748
24	gamma=0.2	dcoob	synthetic	0.9359	0.0753	0.75	0.1179
25	gamma=0.2	rfvanilla	synthetic	0.9422	0.074	0.8	0.1581
26	gamma=0.2	shadowVIMP	synthetic	0.9172	0.0807	0.775	0.1845
27	gamma=0.2	vitaPIMP	synthetic	0.8547	0.1162	0.675	0.2648
28	interaction	boruta	tabular	0.7233	0.2149	0.6953	0.2211
29	interaction	dccondperm	tabular	0.9966	0.0107	0.9846	0.0436
30	interaction	dcloco	tabular	0.8421	0.1649	0.7961	0.1891
31	interaction	dcoob	tabular	0.9971	0.01	0.9864	0.0377
32	interaction	rfvanilla	tabular	0.9993	0.0032	0.9948	0.0255
33	interaction	shadowVIMP	tabular	1.0	0.0	1.0	0.0
34	interaction	vitaPIMP	tabular	0.9948	0.0134	0.9763	0.0493
49	linear	boruta	tabular	0.4721	0.2216	0.4519	0.2255
50	linear	dccondperm	tabular	0.5009	0.241	0.4801	0.2258
51	linear	dcloco	tabular	0.5246	0.2425	0.4998	0.2358
52	linear	dcoob	tabular	0.5682	0.2336	0.5784	0.2207
53	linear	rfvanilla	tabular	0.4623	0.2251	0.468	0.2379
54	linear	shadowVIMP	tabular	0.5101	0.2138	0.5011	0.2059
55	linear	vitaPIMP	tabular	0.4879	0.2564	0.5024	0.2394
84	shift	boruta	tabular	0.7416	0.2254	0.7153	0.2691
85	shift	dccondperm	tabular	0.989	0.0237	0.9733	0.0542
86	shift	dcloco	tabular	0.8475	0.1562	0.8088	0.1754
87	shift	dcoob	tabular	0.9887	0.0236	0.9691	0.0559
88	shift	rfvanilla	tabular	0.9964	0.0103	0.9802	0.0454
89	shift	shadowVIMP	tabular	0.9957	0.0132	0.9802	0.0454
90	shift	vitaPIMP	tabular	0.9848	0.0325	0.9565	0.0698

MSR-VTT, and multiple image-text packs (COCO splits, MM train/test, Fashion-IQ, Indiana CXR, Microscopy CLIP). **B1** is equal routing / equal LR with all modalities updated ($\text{flops}_{\text{rel}} = 1$), not a no-adapt freeze. **B5** uses MMD-cov + PO scores; **B5g** additionally skips modality proj. BWD when $\alpha_m < \theta$ (structured adapt-dropout; FWD still paid); **B5r** replaces the gate by random shutdown ($p_{\text{keep}} = 0.67$). Across datasets, online Acc lift is broadly consistent with equal-LR adaptation, while B5g reliably reduces adapt FLOPs to about $0.75\text{--}0.82\times$ (Table 49). Amazon remains the clearest Acc win for B5 over B1 ($\approx +0.03$); on video/image-text smokes B5 is typically on par with B1, and B5g trades a small Acc lift for lower update cost. Random shutdown (B5r) is a useful FLOPs baseline but is not equivalent to attribution-guided gating: B5g beats B5r on several image-text packs, while on others random keeps more Acc at higher FLOPs (Table 50). The results make complete sense: attribution-guided gating (B5g) achieves a better accuracy-FLOPs

Table 33: Random Forest Binary Classifier in comparison with other procedures - Boruta, ShadowVIMP and VITA-PIMP for Chronoberg and FMoW data

	shift _{type}	method	scenario	avgAUC	stdAUC	avgF1	stdF1
0	FGSM	boruta	chronoberg	0.5545	0.0418	0.0667	0.0574
1	FGSM	boruta	fmow	0.569	0.0769	0.1495	0.1125
2	FGSM	dccondperm	chronoberg	0.664	0.0858	0.4222	0.1148
3	FGSM	dccondperm	fmow	0.6642	0.049	0.391	0.0753
4	FGSM	dcloco	chronoberg	0.554	0.1315	0.3333	0.1283
5	FGSM	dcloco	fmow	0.5538	0.0667	0.3468	0.1146
6	FGSM	dcoob	chronoberg	0.6667	0.1044	0.4333	0.1223
7	FGSM	dcoob	fmow	0.6231	0.0416	0.3744	0.0916
8	FGSM	rfvanilla	chronoberg	0.6577	0.1184	0.4222	0.1463
9	FGSM	rfvanilla	fmow	0.6353	0.0596	0.366	0.0742
10	FGSM	shadowVIMP	chronoberg	0.6704	0.0841	0.4667	0.1463
11	FGSM	shadowVIMP	fmow	0.6483	0.0837	0.3458	0.099
12	FGSM	vitaPIMP	chronoberg	0.6635	0.1192	0.4667	0.1262
13	FGSM	vitaPIMP	fmow	0.645	0.0936	0.3897	0.0749
35	jitter	boruta	chronoberg	0.5952	0.0445	0.1111	0.0
36	jitter	boruta	fmow	0.7037	0.1642	0.0991	0.0476
37	jitter	dccondperm	chronoberg	0.982	0.0346	0.9333	0.0994
38	jitter	dccondperm	fmow	0.8768	0.1924	0.7179	0.3109
39	jitter	dcloco	chronoberg	0.7619	0.0571	0.6	0.0609
40	jitter	dcloco	fmow	0.7276	0.1642	0.5778	0.1401
41	jitter	dcoob	chronoberg	0.9767	0.039	0.9111	0.1449
42	jitter	dcoob	fmow	0.8462	0.2189	0.6923	0.3525
43	jitter	rfvanilla	chronoberg	0.9714	0.0481	0.9111	0.1449
44	jitter	rfvanilla	fmow	0.8412	0.2415	0.718	0.3635
45	jitter	shadowVIMP	chronoberg	0.9841	0.0254	0.9111	0.1449
46	jitter	shadowVIMP	fmow	0.9098	0.1519	0.7692	0.3353
47	jitter	vitaPIMP	chronoberg	0.9852	0.0252	0.9333	0.0994
48	jitter	vitaPIMP	fmow	0.9041	0.1556	0.7692	0.3353
56	miFGSM	boruta	chronoberg	0.6	0.0516	0.1111	0.0
57	miFGSM	boruta	fmow	0.6858	0.1335	0.1169	0.074
58	miFGSM	dccondperm	chronoberg	0.9778	0.0468	0.9333	0.0994
59	miFGSM	dccondperm	fmow	0.8948	0.1641	0.7928	0.2544
60	miFGSM	dcloco	chronoberg	0.7619	0.0958	0.5556	0.1361
61	miFGSM	dcloco	fmow	0.7575	0.2067	0.6072	0.2321
62	miFGSM	dcoob	chronoberg	0.9873	0.0229	0.9333	0.0994
63	miFGSM	dcoob	fmow	0.9073	0.1496	0.7795	0.2423
64	miFGSM	rfvanilla	chronoberg	0.9767	0.037	0.9111	0.1449
65	miFGSM	rfvanilla	fmow	0.8939	0.1794	0.7795	0.2423
66	miFGSM	shadowVIMP	chronoberg	0.9884	0.0206	0.9333	0.0994
67	miFGSM	shadowVIMP	fmow	0.9408	0.102	0.8195	0.2536
68	miFGSM	vitaPIMP	chronoberg	0.9862	0.021	0.9556	0.0609
69	miFGSM	vitaPIMP	fmow	0.9127	0.1411	0.8062	0.2522
70	rFGSM	boruta	chronoberg	0.6048	0.0361	0.1111	0.0
71	rFGSM	boruta	fmow	0.7091	0.1687	0.0991	0.0476
72	rFGSM	dccondperm	chronoberg	0.9831	0.0268	0.8889	0.1361
73	rFGSM	dccondperm	fmow	0.871	0.2086	0.7436	0.3795
74	rFGSM	dcloco	chronoberg	0.7566	0.0639	0.5556	0.0786
75	rFGSM	dcloco	fmow	0.7468	0.1537	0.6513	0.1344
76	rFGSM	dcoob	chronoberg	0.9746	0.0377	0.8889	0.1361
77	rFGSM	dcoob	fmow	0.8594	0.235	0.718	0.3635
78	rFGSM	rfvanilla	chronoberg	0.9714	0.0442	0.9111	0.1449
79	rFGSM	rfvanilla	fmow	0.8495	0.2437	0.718	0.3635
80	rFGSM	shadowVIMP	chronoberg	0.9852	0.0232	0.9333	0.0994
81	rFGSM	shadowVIMP	fmow	0.9082	0.159	0.7949	0.3553
82	rFGSM	vitaPIMP	chronoberg	0.9778	0.036	0.8889	0.1361
83	rFGSM	vitaPIMP	fmow	0.8966	0.1769	0.7436	0.3795

trade-off than random shutdown (B5r) in most cases, because it selectively skips updates for modalities that contribute little to the shift.

We want to mention that it will serve as a promising future exploration direction - Beyond equal-LR(learning rate) online updates or randomly shut down some parts of the update neural networks, we treat per-modality attribution scores (concept vs. covariate telemetry $\rightarrow$ routing mass α_m) as a *next-stage adapter signal*: they do not replace the backbone,

Table 34: Covariate-shift ranking on AFFEC. RF Domain Classifier VIMP for $W \sim X$, with $k = 8$. Batch: early runs $\{0, 1\}$ vs late runs $\{2, 3\}$. Domain AUC 0.999.

Modality	Ranked indices	Channel names (same order)
eye_tracking	[6, 15, 12, 11, 4, 5, 0, 1]	fixation_duration, validity, pupil_y_left, pupil_x_left, gaze_x_right, gaze_y_right, fixation_x, fixation_y
pupil	[5, 4, 13, 11, 12, 15, 17, 10]	pupil_latency, pupil_peak, eye_pos_y_raw, eye_pos_y, eye_pos_x_raw, eye_pos_velocity_y, eye_pos_range_x, eye_pos_x
cursor	[1, 0, 2, 3]	cursor_y, cursor_x, cursor_velocity, cursor_state
gsr_eda	[15, 32, 31, 4, 3, 35, 30, 14]	acc_y, gsr_feat_32, gsr_feat_31, gsr_scr_count, gsr_tonic, gsr_feat_35, gsr_feat_30, acc_x
eeg	[50, 54, 23, 2, 56, 3, 47, 15]	EEG_50, EEG_54, EEG_23, F3, EEG_56, F4, EEG_47, EEG_15

Table 35: Concept-drift ranking on AFFEC. PO-risk path RF VIMP on pseudo-outcome (blended with within-batch $Y \sim X$ importance gap), ordered with $k = 8$. Observed PO-risk 0.0000.

Modality	Ranked indices	Channel names (same order)
eye_tracking	[15, 9, 6, 0, 2, 1, 4, 3]	validity, eye_open_left, fixation_duration, fixation_x, gaze_x_left, fixation_y, gaze_x_right, gaze_y_left
pupil	[2, 0, 5, 4, 13, 10, 11, 15]	pupil_diameter_filt, pupil_diameter, pupil_latency, pupil_peak, eye_pos_y_raw, eye_pos_x, eye_pos_y, eye_pos_velocity_y
cursor	[0, 1, 2, 3]	cursor_x, cursor_y, cursor_velocity, cursor_state
gsr_eda	[0, 29, 2, 31, 1, 34, 14, 33]	gsr_raw, gsr_feat_29, gsr_phasic, gsr_feat_31, gsr_filtered, gsr_feat_34, acc_x, gsr_feat_33
eeg	[62, 13, 46, 12, 52, 44, 27, 59]	EEG_62, EEG_13, EEG_46, EEG_12, EEG_52, EEG_44, EEG_27, EEG_59

but decide *what to adapt next*—which modality param-groups receive larger LR, and (optionally) which projection BWD paths are gated. In this view, AGOD is less a new architecture than a portable control plane for downstream training: sensor $\rightarrow$ EMA state $\alpha \rightarrow$ LR / hard-gate actuator. Our multi-dataset smoke (Amazon, MSR-VTT, image-text packs) is intended as preliminary evidence that such attribution-conditioned adaptation is transferable and can trade Acc lift against adapt FLOPs (B5g), with random shutdown (B5r) as a non-attribution sparsity baseline. We leave stronger claims—longer real drift, encoder-side adapt cost, and coupling α to distillation / adapter selection—to future work.

17.1 Adapter design logic.

We cast attribution not as an explanation head, but as a *next-stage adapter policy* over an already-trained multimodal encoder. Let s_m be a per-modality shift score (ship rule: concept $\uparrow$, covariate $\downarrow$), and let $\alpha = \text{EMA}(\text{Softmax}(s/\tau))$ be the routing state. The adapter has two coupled levers on modality projection params θ_m :

1. Soft adapter (LR scheduler).

$$\text{LR}_m = \text{lr}_0(\beta + (1 - \beta)\alpha_m|M|).$$

High- α modalities take larger steps; low- α modalities remain updated but shrunk. This is continuous, always-on adaptation.

2. Hard adapter (adapt-gate). If $\alpha_m < \theta$, zero $\partial L / \partial \theta_m$ for that window (structured adapt-dropout). Forward inference is unchanged; only projection BWD—and thus adapt FLOPs—is skipped. At least one modality stays active.

Thus the design target is *what to train next under a shift*, not *what to infer*: soft adapter reallocates gradient budget by LR; hard adapter sparsifies that budget when routing mass collapses. Random shutdown (B5r) keeps the same sparsity interface without using α , and serves only as a non-attribution control. Extending this adapter logic later means replacing the LR map by a richer scheduler, or replacing the binary gate by rank/depth/LoRA selection under the same α prior.

17.2 Towards Contributing to the Continuous via the Feature Selection

Borrowing ideas from Mixture of Experts (MoE, [47]), we leverage modality-specific attribution proportions for a KL loss—analogueous to the balance loss used to ensure and recognize each expert’s contribution—to demonstrate the next-stage procedure. Specifically, we apply a KL loss to the weights output from each modality, $KL(w_i || \alpha_i)$ represents the modality-specific contribution of PO-risk (the leave-one-group-out VIMP via pseudo-outcome risk, primarily targeting

Table 36: Post-hoc LOBO bbox attribution on COCO early/late. Named mechanisms f^* selected by PO-risk VIMP; images located by $\hat{\tau}^2$ and f^* ; each box is removed with its text, features recalculated, and Δ PO contribution ranked.

Step	Operation
Form	{caption, bbox_text: box + text_description}
Select f^*	PO / RF named VIMP on image-level bbox features
Locate images	high $\hat{\tau}(X)^2$ and $ f^* $
LOBO	remain = boxes \ {k}; recalculate_bbox_features
Score	$\Delta_k^{\text{rel}} = (\hat{\tau}(X)^2 - \hat{\tau}(X^{(-k)})^2) / \hat{\tau}(X)^2$
Y	rank-quantile mapped to [0, 1]
PO-risk (fit)	0.0018

Table 37: Selected mechanisms f^* and top PO named ranks.

Rank	Feature	PO VIMP
1	nn_center_dist_mean *	0.06811
2	cx_std *	0.05569
3	cy_mean *	0.05416
4	cx_mean	0.05326
5	aspect_std	0.05113
6	aspect_mean	0.04726
7	cy_std	0.04579
8	largest_area_frac *	0.04425
9	h_mean	0.04362
10	nn_center_dist_min	0.04349

concept drift). This yields a portable, parsimonious product use case that highlights the potential of the feature selection procedure for continuous training of deployed models—somewhat akin to online stacking procedures([9]). As demonstrated in Tables 51 (original) and 52 (the portable product use case procedure), this highly portable and parsimonious approach already shows potential for boosting production ML systems. It serves as a first step toward integrating Meta Learners and Pseudo-outcome Risk into Continuous Learning.

17.3 FallBack Procedure

In real-life data applications, there are quite a few possibilities that the procedure with the best empirical performance won’t give people concise or the optimal procedure/performance. We present the following two circumstances.

- **When significant distribution shift is detected but data quality is suspect:** If data integrity failures are observed (e.g., sudden missing rate spikes, schema changes, or out-of-bound values), the fallback mechanism prioritizes root-cause diagnosis of data quality issues before any ML-based feature attribution. All downstream distribution shift characterization procedures are halted until the Data Integrity Health Score recovers above the safety threshold in terms of the data quality issues (e.g., $\mathcal{H} \geq 0.6$).
- **When the batch sample size is too small:** If the sample size $n < 50$ or the sample-to-feature ratio $\gamma = n/p < 2$, the system directly falls back to univariate marginal statistics (e.g., PSI and KL divergence), abandoning joint distribution modeling. If $50 \leq n < 100$ and $\gamma \geq 2$, the RF-Domain classifier with OOB-VIMP is permitted, but its output is flagged as “low confidence” and only the top-5 ranked features are recommended for manual inspection. Otherwise, the standard general pipeline (L0/L1/L2) proceeds.

Table 38: Category counter of LOBO top boxes (n=40 located images).

Category	# as top box	mean $ \Delta \text{ PO rel} $
person	4	0.9696
car	3	0.9631
traffic light	3	0.9635
bench	2	0.6768
bed	2	0.9933
toilet	2	0.9903
surfboard	2	0.9963
chair	2	0.9708

Table 39: Example images: which bounding box has largest leave-one-out relative $\Delta \text{ PO}$ contribution $(\hat{\tau}^2 - \hat{\tau}_{(-k)}^2)/\hat{\tau}^2$.

Image	Batch	Box	Category	$\Delta \text{ PO rel}$
500859	0	bbx_1	bench	0.9694
330531	0	bbx_2	pizza	0.9069
230976	0	bbx_2	car	0.9334
270918	0	bbx_2	person	0.9207
235788	0	bbx_2	bus	0.9036
391735	0	bbx_2	person	0.9588
234147	0	bbx_1	carrot	1.0000
109869	0	bbx_1	bench	-0.3841
558577	0	bbx_1	bed	0.9912
520437	0	bbx_1	suitcase	0.9158

Table 40: RF Domain Classifier on Fashion-IQ CLIP ($n_1 = 2823$, $n_2 = 1211$, $d_{\text{img}} = d_{\text{txt}} = 512$). Batch: train vs test. Domain AUC 0.725; modality share image 0.697 / text 0.303.

Modality	Ranked local indices (high→low)
image	[381, 316, 496, 218, 124, 265, 226, 18, 379, 151, 393, 253, 492, 426, 459, 47, 108, 46, 355, 118]
text	[326, 177, 212, 198, 209, 294, 53, 64, 35, 369, 415, 271, 119, 108, 0, 216, 295, 179, 55, 106]

Table 41: Coordinate-wise 1-D MMD ranking on Fashion-IQ CLIP (max_n= 100; full leave-one-out MMD is numerically flat at $p = 1024$). Modality share image 0.550 / text 0.450.

Modality	Ranked local indices (high→low)
image	[190, 496, 426, 84, 493, 331, 338, 133, 95, 490, 313, 47, 264, 362, 224, 507, 363, 176, 151, 20]
text	[233, 186, 104, 211, 175, 378, 359, 201, 50, 10, 368, 271, 364, 90, 219, 375, 406, 60, 75, 471]

Table 42: PO-risk RF VIMP on Fashion-IQ CLIP. Observed PO-risk 10.845; modality share image 0.280 / text 0.720.

Modality	Ranked local indices (high→low)
image	[39, 471, 489, 200, 499, 102, 358, 504, 195, 501, 370, 146, 71, 181, 317, 27, 345, 19, 239, 160]
text	[315, 442, 443, 67, 117, 179, 369, 351, 111, 448, 15, 64, 115, 198, 23, 181, 57, 100, 159, 82]

Table 43: Covariate-shift ranking on Indiana/OpenI CXR ($n=7426$, $d_{\text{img}}=d_{\text{txt}}=512$). RF Domain Classifier VIMP for $W \sim X$. Batch: **projection** Frontal vs Lateral. Domain AUC 1.000; modality VIMP share image 0.996 / text 0.004.

Modality	Ranked local indices (high→low)
image	[330, 105, 235, 145, 278, 335, 494, 117, 293, 82, 111, 410, 154, 21, 115, 68, 0, 77, 431, 171]
text	[138, 187, 447, 38, 246, 423, 436, 476, 34, 426, 288, 115, 171, 444, 55, 310, 427, 364, 375, 210]

Table 44: Concept-drift ranking on Indiana/OpenI CXR. PO-risk RF VIMP (blended with within-batch $Y \sim X$ importance gap), Observed PO-risk 0.054; modality VIMP share image 0.591 / text 0.409.

Modality	Ranked local indices (high→low)
image	[133, 249, 146, 338, 57, 301, 170, 4, 370, 30, 511, 122, 248, 99, 28, 197, 199, 437, 268, 165]
text	[302, 241, 461, 489, 506, 196, 3, 257, 304, 256, 60, 72, 423, 428, 420, 268, 127, 361, 249, 0]

Table 45: Covariate-shift ranking on multimodal train/test set ($n=4034$, $d_{\text{img}}=d_{\text{txt}}=512$). RF Domain Classifier VIMP for $W \sim X$. Batch: `split` train vs test. Domain AUC 0.727; modality VIMP share image 0.606 / text 0.394.

Modality	Ranked local indices (high→low)
image	[265, 377, 147, 124, 317, 108, 84, 316, 496, 459, 218, 348, 47, 18, 227, 38, 11, 114, 379, 479]
text	[261, 116, 414, 63, 244, 146, 504, 67, 505, 96, 469, 186, 164, 379, 412, 0, 259, 189, 211, 73]

Table 46: Concept-drift ranking on multimodal train/test set. PO-risk RF VIMP (blended with within-batch $Y \sim X$ importance gap), ordered by `np.argsort(-VIMP)[:k]` with $k=20$. Observed PO-risk 16.214; modality VIMP share image 0.255 / text 0.745.

Modality	Ranked local indices (high→low)
image	[370, 39, 216, 175, 321, 304, 315, 489, 195, 360, 95, 28, 357, 336, 102, 66, 173, 209, 324, 501]
text	[115, 300, 501, 442, 113, 100, 23, 179, 96, 57, 348, 137, 453, 25, 1, 111, 419, 206, 67, 335]

Table 47: Modality contribution to within-video covariate shift $P(X | W)$. Batch W is the early vs. late window split inside each video ($n = 320$ windows, 16 videos; column-standardized X). RF-Domain, coordinate-MMD, and PO-risk VIMP are L1-normalized shares. Mean $|d|$ is Cohen’s d on region-mean activation, averaged over 16 videos $\times$ 8 embedding bins. Holm cells count video $\times$ region tests with Holm $p < 0.05$ (within video).

Modality	RF-Domain	MMD	PO-risk	mean $ d $	Holm cells	Role in W -shift
Video (768-d)	0.772	0.897	0.813	0.787	10	primary
Audio (512-d)	0.191	0.101	0.174	0.518	2	secondary
Text (768-d)	0.037	0.002	0.013	0.000	0	structural null [†]

Friedman test of equal per-video RF shares: $\chi^2 = 30.13$, $p = 2.87 \times 10^{-7}$. Wilcoxon signed-rank (Holm) rejects video=audio, video=text, and audio=text at $p < 10^{-4}$. [†]Clip-level caption (window-invariant on this extract), hence a structural zero for W -shift, not a claim that language is uninformative for retrieval.

Table 48: Modality contribution to within-video covariate shift $P(X | W)$. Batch W is the early vs. late window split inside each video ($n = 320$ windows, 16 videos; column-standardized X). RF-Domain, coordinate-MMD, and PO-risk VIMP are L1-normalized shares. Mean $|d|$ is Cohen’s d on region-mean activation, averaged over 16 videos $\times$ 8 embedding bins. Holm cells count video $\times$ region tests with Holm $p < 0.05$ (within video).

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Table 49: AGOD portable prototype smoke across datasets. Policies: B1 equal LR; B5 MMD-cov+PO; B5g = B5 + α -gate; B5r = B5 + random adapt-shutdown ($p_{\text{keep}}=0.67$). Induced $P(Y)$ drift, $T=6$ windows. FLOPs are projection-head adapt cost (features pre-extracted). FWD inference always runs.

Dataset	Mods	B1	B5	B5g	B5r	B5g FLOPs	B5r FLOPs
Amazon Reviews	img, txt	+0.058	+0.087	—	—	—	—
MSR-VTT packed	video, txt, audio	+0.070	+0.063	+0.048	+0.057	0.746	0.841
COCO outdoor/indoor	img, txt, bbox	+0.034	+0.019	+0.031	+0.018	0.810	0.841
COCO time order	img, txt, bbox	+0.012	+0.028	+0.007	+0.001	0.778	0.841
COCO center split	img, txt, bbox	+0.012	+0.028	+0.007	+0.001	0.778	0.841
MM train/test	img, txt	−0.004	−0.022	+0.010	−0.007	0.778	0.911
Fashion-IQ	img, txt	−0.004	−0.022	+0.010	−0.007	0.778	0.911
Indiana CXR	img, txt	+0.024	+0.016	+0.007	+0.025	0.778	0.911
Microscopy CLIP	img, txt	+0.040	+0.027	+0.018	+0.039	0.822	0.911

Table 50: Attribution gate (B5g) vs random adapt-shutdown (B5r).

Dataset	B5g Acc $\uparrow$	B5r Acc $\uparrow$	Δ Acc	FLOPs (g / r)
MSR-VTT packed	+0.048	+0.057	−0.010	0.75 / 0.84
COCO outdoor/indoor	+0.031	+0.018	+0.013	0.81 / 0.84
COCO time order	+0.007	+0.001	+0.006	0.78 / 0.84
COCO center split	+0.007	+0.001	+0.006	0.78 / 0.84
MM train/test	+0.010	−0.007	+0.018	0.78 / 0.91
Fashion-IQ	+0.010	−0.007	+0.018	0.78 / 0.91
Indiana CXR	+0.007	+0.025	−0.018	0.78 / 0.91
Microscopy CLIP	+0.018	+0.039	−0.021	0.82 / 0.91

Table 51: Attribution-guided online stacking. `stack_alpha` = CE + λ KL($w||\alpha$).

Dataset	Variant	MSE drop $\uparrow$	Acc $\uparrow$	Acc post
MSR-VTT	mean_ce	−0.0080	+0.000	0.651
MSR-VTT	stack_ce	+0.0076	+0.008	0.655
MSR-VTT	stack_alpha	+0.0097	+0.008	0.655
Amazon	mean_ce	+0.0339	+0.049	0.543
Amazon	stack_ce	+0.0357	+0.033	0.539
Amazon	stack_alpha	+0.0383	+0.069	0.571
COCO	mean_ce	+0.0236	+0.034	0.860
COCO	stack_ce	+0.0251	+0.033	0.865
COCO	stack_alpha	+0.0240	+0.036	0.860
Affec	mean_ce	−0.0501	−0.055	0.560
Affec	stack_ce	+0.0180	+0.019	0.629
Affec	stack_alpha	+0.0369	+0.039	0.622

Table 52: Lift of `stack_alpha` over baselines (same towers/windows).

Dataset	vs. baseline	Δ MSE drop	Δ Acc
MSR-VTT	mean_ce	+0.0177	+0.008
MSR-VTT	stack_ce	+0.0021	+0.000
Amazon	mean_ce	+0.0044	+0.020
Amazon	stack_ce	+0.0026	+0.036
COCO	mean_ce	+0.0004	+0.002
COCO	stack_ce	−0.0011	+0.003
Affec	mean_ce	+0.0870	+0.094
Affec	stack_ce	+0.0189	+0.020

Table 53: Alternate weight sockets. Same towers; only the prior / temperature changes.

Dataset	Variant	MSE drop $\uparrow$	Acc $\uparrow$	Acc post
COCO	mean_ce	+0.0236	+0.034	0.860
COCO	stack_ce	+0.0251	+0.033	0.865
COCO	stack_alpha	+0.0240	+0.036	0.860
COCO	stack_uniform	+0.0247	+0.036	0.862
COCO	stack_fixed	+0.0213	+0.025	0.857
COCO	stack_temp	+0.0251	+0.037	0.872
COCO	stack_erank	+0.0246	+0.037	0.854
Affec	mean_ce	−0.0501	−0.055	0.560
Affec	stack_ce	+0.0180	+0.019	0.629
Affec	stack_alpha	+0.0369	+0.039	0.622
Affec	stack_uniform	+0.0225	+0.024	0.652
Affec	stack_fixed	+0.0093	+0.009	0.634
Affec	stack_temp	+0.0742	+0.071	0.625
Affec	stack_erank	+0.0379	+0.037	0.626

Table 54: Lift vs. `mean_ce` / `stack_ce` on Affec (the informative stream).

Variant	Δ MSE vs mean	Δ Acc vs mean	Δ MSE vs stack	Δ Acc vs stack
<code>stack_ce</code>	+0.0681	+0.074	—	—
<code>stack_alpha</code>	+0.0870	+0.094	+0.0190	+0.019
<code>stack_uniform</code>	+0.0726	+0.079	+0.0045	+0.004
<code>stack_fixed</code>	+0.0594	+0.064	−0.0087	−0.010
<code>stack_temp</code>	+0.1243	+0.126	+0.0563	+0.052
<code>stack_erank</code>	+0.0880	+0.092	+0.0199	+0.018

Table 55: Arize train-tail vs production-head monitors. RFPerm tests $Y \mid X$; RF-domain accuracy and eigen-shift (SAS misalignment, RIR, PET) read embedding covariate shift. PET state from locked train-tail covariance.

Dataset	cut	RFPerm p	RF-dom. acc	SAS mis	RIR	PET _{u}	PET
xtreme	prod_head	0.463	0.777	0.722	1.36	0.166	CAUTION
ecomm early EN	prod_head	0.195	0.944	0.662	1.43	0.195	CAUTION
ecomm Spanish	Spanish-window head	0.463	0.913	0.643	1.42	0.189	CAUTION

Table 56: Eigen-shift geometry panel, scaler locked on the train tail. Spectrum TV is total variation between eigenvalue masses; cov rel-F is $\|\Sigma_1 - \Sigma_0\|_F / \|\Sigma_0\|_F$.

Dataset	SAS mis	RIR	eig TV	cov rel-F	$\ \Delta\mu\ _2$
xtreme	0.722	1.36	0.043	0.536	0.551
ecomm early EN	0.662	1.43	0.016	0.504	0.520
ecomm Spanish	0.643	1.42	0.028	0.563	0.928

Table 57: FSDS covariate-shift feature list, ranked by RF-domain importance. CFPerm p is a confirm statistic on the screened set, not the CS rank. $\Delta = \text{mean}^{\text{new}} - \text{mean}^{\text{train}}$.

Dataset	rank	feature	RF-dom. imp	Δ	CFPerm p
xtreme	1	<code>rate_0</code>	0.049	+0.034	0.871
	5	<code>rate_PER</code>	0.033	−0.019	0.226
ecomm early EN	5	<code>product</code>	0.015	+0.005	0.774
ecomm Spanish	1	<code>is_spanish</code>	0.102	+0.258	0.806
	2	<code>de</code>	0.053	+0.046	0.710
	4	<code>que</code>	0.036	+0.028	0.710